MEDICAL EXAM REVISION SERIES

BASIC HISTOLOGY

The Exam Companion

A structured summary of Junqueira's Basic Histology, Text & Atlas, 17th edition, written for medical students preparing for examinations.

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23 chapters condensed

100 exam questions

10 master comparison tables

1 rapid-review chapter

Condensed from the 17th edition · For study use · Not a substitute for the atlas plates

Contents

How to Use This Guide 4

The Twelve Laws of Histology 5

Exam Technique 6

About the 17th Edition 7

Source Map 8

PART I — CELLS AND TECHNIQUES 9

1 · Histology & Its Methods of Study 10

2 · The Cytoplasm 14

3 · The Nucleus 17

PART II — THE FOUR BASIC TISSUES 20

4 · Epithelial Tissue 21

5 · Connective Tissue 25

6 · Adipose Tissue 28

7 · Cartilage 30

8 · Bone 32

9 · Nerve Tissue & the Nervous System 35

10 · Muscle Tissue 37

PART III — TRANSPORT, DEFENSE, AND BLOOD 40

11 · The Circulatory System 41

12 · Blood 43

13 · Hemopoiesis 45

14 · The Immune System & Lymphoid Organs 47

PART IV — DIGESTION AND RESPIRATION 51

15 · Digestive Tract 52

16 · Organs Associated with the Digestive Tract 56

17 · The Respiratory System 59

18 · Skin 62

PART V — EXCRETION AND REPRODUCTION 65

19 · The Urinary System 66
20 · Endocrine Glands 69
21 · The Male Reproductive System 72
22 · The Female Reproductive System 74
23 · The Eye & Ear: Special Sense Organs 77

PART VI — RAPID REVIEW AND REFERENCE 79

24 · The Last-24-Hours Rapid Review 80
Appendix A · Stains and What They Prove 86
Appendix B · The Vocabulary Examiners Expect 87

PART VII — ASSESS YOUR KNOWLEDGE 88

100 Examination Questions 89
Answer Key with Explanations 100
Scoring and What to Do Next 103

How to Use This Guide

Histology is the one preclinical subject where you cannot bluff. Everything is either there on the slide or it is not. But histology is also the most efficient subject in medicine, because a small number of structural rules generate almost every fact you will be asked.

This guide condenses **Junqueira's Basic Histology: Text & Atlas, 17th edition (Anthony L. Mescher, McGraw Hill, 2024; ISBN 978-1-264-93039-5)** into the material that decides exam marks. It is built on the book's own contents page: **every chapter follows the same section sequence, in the same order, using the book's own section titles**, with printed page numbers so you can read a section here and then turn straight to the corresponding pages in your copy.

How each chapter is built

Block	What it gives you
Chapter line	Printed page range and the book's own section titles
The core idea	The one organizing principle that makes the chapter cohere
Sections	The book's sections, in the book's order, condensed
Comparison tables	The highest-yield format in histology — learn the rows, not the prose
CLINICAL	The disease link that turns a fact into a mark
MICRO	How to recognize the tissue on an unknown slide or EM image
TIP / MNEMONIC	Traps, confusions, and memory hooks

At the end of every chapter in the book there is a **Summary of Key Points** and an **Assess Your Knowledge** question set. This guide is designed to be read before those, so that when you reach them you already know the answers.

Three questions answer 80% of slide questions

1. **What tissue is this?** — epithelium, connective tissue, muscle, nervous tissue, or a fluid.
2. **What organ or region?** — the tissue arrangement identifies the organ, not the cell type.
3. **What is it doing here?** — every structure you can see exists because of a function the cell must perform. Name the function and the marks follow.

EXAM TIP

Never memorize a slide as a picture. Memorize the rule that generates the picture ("stratified squamous with keratin on top = surface exposed to abrasion and drying"). The book's atlas plates teach you the appearance; this guide teaches you the rule behind it.

Three study plans

If you have six weeks. One part of this guide per week. Read the corresponding chapter, then this section, then cover the tables and redraw them from memory. Do the book's "Assess Your Knowledge" questions and this guide's MCQs at the end of the week. Reserve the final week for the Rapid Review chapter only.

If you have ten days. Two to three chapters per day, tables only on the first pass. Spend 60% of your time on the six comparison tables that carry the most marks: epithelium types, connective tissue fibers, cartilage, muscle, leukocytes, and GI tract wall.

If you have 48 hours. Read Rapid Review (Chapter 24) end to end, then the 100 MCQs, then the six tables. Do not attempt prose. Histology rewards pattern recognition under time pressure, and patterns live in tables.

The Twelve Laws of Histology

These twelve statements are true across the whole book. When you are stuck in an exam, reason from them.

1. **Structure follows function, and function follows structure.** Every visible feature is a solution to a physical problem — diffusion distance, mechanical stress, surface area, water loss, conduction speed.
2. **Form is set by the apical surface, the basal surface, and the junctions between cells.** Look at all three before naming a tissue.
3. **Epithelium sits on something and is never crossed by a blood vessel.** Avascularity plus a basement membrane defines it.
4. **Connective tissue is defined by its extracellular matrix, not its cells.** The matrix tells you the mechanical job: tension, elasticity, compression, or filtration.
5. **Cells that make protein look basophilic; cells that store or secrete protein look eosinophilic.** Rough ER is blue-purple with hematoxylin; secretory product and cytoplasm are pink with eosin.
6. **Cells that make steroid hormone are full of smooth ER and lipid droplets** — hence foamy, pale, or vacuolated cytoplasm (adrenal cortex, Leydig cells, corpus luteum).
7. **Muscle is classified by nucleus position and striation, not by shape alone.** Peripheral multiple nuclei = skeletal; central single nucleus with striation = cardiac; central single nucleus without striation = smooth.
8. **The number of cell layers tells you the mechanical job.** One layer = exchange, absorption, secretion, or a thin barrier. Many layers = protection from abrasion.
9. **Surface area is bought with folds.** Microvilli, villi, plicae circulares, basal infoldings, and cristae all exist for the same arithmetic reason.
10. **Tubes are built in concentric layers, and each layer has one job.** In blood vessels: intima for exchange, media for tone, adventitia for anchorage. In the gut: mucosa for digestion, submucosa for supply, muscularis for movement.
11. **A capsule and a hilum identify a solid organ as lymph node, spleen, thymus, or gland.** The route of fluid flow tells you which one.
12. **Every histology fact has a clinical twin.** If you cannot name the disease, you do not yet own the fact.

Exam Technique

Answering a slide or image question

Never begin with a guess. Use this sequence every single time:

1. **Magnification and staining.** H&E at low power versus PAS versus silver tells you what the examiner thinks is important.
2. **The boundary.** Is there a free surface? A lumen? A capsule? Blood vessels inside the tissue? This immediately separates epithelium-lined organs from solid organs.
3. **The dominant tissue.** Note the arrangement — cords, acini, tubules, follicles, layers, bundles.
4. **The one diagnostic cell.** Parietal cell, Paneth cell, podocyte, Purkinje fiber, Clara cell, Sertoli cell, odontoblast. Examiners choose images that contain one.
5. **Name it, then justify it in the same sentence.** "This is thyroid gland because the parenchyma is arranged in follicles of colloid lined by a single layer of cuboidal cells." Justification earns the marks; the name alone earns half.

Answering a "describe and give the function of" essay

Use a fixed skeleton — it works for any organ:

Capsule → cortex → medulla → parenchyma arrangement → cell types → blood supply → ducts or outflow → function → clinical correlation.

Ten headings, ten marks, almost regardless of the organ. Write the headings first, then fill them. An examiner marking 200 scripts recognizes structure instantly and rewards it.

Ten phrases that lose marks

- "It looks like…" without saying why it looks that way.
- Naming a cell without its product or function.
- Confusing "basophilic" (blue, acidic structure) with "eosinophilic" (pink, basic protein).
- Saying "simple squamous" for endothelium without noting it is a type of simple squamous epithelium.
- Using "mucous" for "mucosa" — different words, different things.
- Describing the adrenal medulla and cortex as one tissue; they are embryologically unrelated.
- Calling transitional epithelium "stratified squamous"; the umbrella cells are unique.
- Forgetting that the thymus has **no** follicles and the spleen has **no** afferent lymphatics.
- Mixing up the two adrenal cortex zones that produce cortisol and aldosterone.
- Writing "bone is dead tissue" — osteocytes are metabolically active and connected by canaliculi.

HIGH-YIELD PEARL

In a viva or OSCE-style histology station, the examiner is usually testing whether you can connect a structure to a function and a disease. Practise every fact in a three-part sentence: "Structurally X, which allows Y, and when it fails you get Z."

About the 17th Edition

The seventeenth edition (2024) retains the two features that made the book a standard for over fifty years: a **concise, straightforward text** with no unnecessary detail, and an **exceptionally clear set of photomicrographs and labeled illustrations**. For the student revising for an examination, three features matter most:

- **Summary of Key Points** at the end of every chapter — the author's own list of what matters.
- **Summary Tables** in each chapter — condensed comparisons in exactly the form examiners like.
- **Medical Applications boxes** — the clinical relevance of each structure.

The online version on AccessMedicine adds **500 multiple-choice questions** across the chapters, a **Lab Guide** with microscopic directions for each laboratory session, and links to a collection of **virtual light microscope slides**. If your university library subscribes, that Lab Guide is the closest thing to sitting at a microscope with a demonstrator beside you.

EXAM TIP

Where this guide and the book's Summary of Key Points disagree in emphasis, trust the book: it is the examiner's source. Use this guide for structure and recall, and the book for the final word.

Source Map

Every chapter section here corresponds to the printed pages below, so the three books — this guide, the textbook, and your uploaded PDF — can be read side by side. **Chapters 1–7 are checked line by line against** `MyDocument_part_1.pdf`**, Chapters 8–14 against part 2, and Chapters 15–19 plus the pituitary, adrenal, islet and thyroid sections of Chapter 20 against part 3.** Part 3 ends part-way through Chapter 20 (at the thyroid follicles), so the parathyroid and pineal sections of Chapter 20 and all of Chapters 21–23 will be checked when part 4 is uploaded. The PDF page column is written `p1`, `p2`, `p3` for the part file, then the page inside that file.

Ch	Title	Book pages	Your PDF page	Part of this guide
1	Histology & Its Methods of Study	1–16	p1 · 13	I
2	The Cytoplasm	17–52	p1 · 40	I
3	The Nucleus	53–70	p1 · 92	I
4	Epithelial Tissue	71–95	p1 · 122	II
5	Connective Tissue	96–121	p1 · 162	II
6	Adipose Tissue	122–128	p1 · 197	II
7	Cartilage	129–137	p1 · 208	II
8	Bone	138–160	p2 · 3	II
9	Nerve Tissue & the Nervous System	161–192	p2 · 37	II
10	Muscle Tissue	193–214	p2 · 87	II
11	The Circulatory System	215–236	p2 · 120	III
12	Blood	237–253	p2 · 156	III
13	Hemopoiesis	254–266	p2 · 181	III
14	The Immune System & Lymphoid Organs	267–294	p2 · 199 + p3 · 1	III
15	Digestive Tract	295–328	p3 · 23	IV
16	Organs Associated with the Digestive Tract	329–348	p3 · 71	IV
17	The Respiratory System	349–370	p3 · 100	IV
18	Skin	371–392	p3 · 132	IV
19	The Urinary System	393–412	p3 · 161	V
20	Endocrine Glands	413–438	p3 · 191	V
21	The Male Reproductive System	439–459	part 4	V
22	The Female Reproductive System	460–489	part 4	V
23	The Eye & Ear: Special Sense Organs	490–524	part 4	V
App.	Light Microscopy Stains	525–526	part 4	Appendix A

PART I — CELLS AND TECHNIQUES

Chapters 1–3 · microscopy, the cytoplasm, and the nucleus

1 · Histology & Its Methods of Study

BOOK PAGES 1–16 | Part 1, pages 13–39 · Preparation of Tissues for Study · Light Microscopy · Electron Microscopy · Autoradiography · Cell & Tissue Culture · Enzyme Histochemistry · Visualizing Specific Molecules · Interpretation of Structures in Tissue Sections

The core idea

Histology explores the body's tissues and how they are arranged to produce functional organs. Tissues have two interacting components: **cells and extracellular matrix (ECM)**. The ECM is not packing material — it supports the cells, carries the fluid that brings nutrients and removes waste, and its molecules bind specific cell-surface receptors connected to structural components inside the cell, forming a continuum in which cells and matrix function together.

Organs form by an orderly combination of these tissues. Because cells and matrix components are small, histology depends on microscopes and molecular methods — and every image you will interpret has been fixed, dehydrated, cleared, embedded, sectioned, stained and mounted. Knowing what each step does tells you which appearances are real and which are artefacts.

Preparation of Tissues for Study

Step	What the book says it does
Fixation	Pieces of organ are placed as soon as possible in **stabilizing or cross-linking compounds**. **Formalin** — a buffered isotonic solution of 37% formaldehyde — is widely used for light microscopy; **glutaraldehyde** for electron microscopy. Both react with the **amine groups (NH_2) of proteins**, preventing degradation by proteases; glutaraldehyde also **cross-links adjacent proteins**. Fixative must diffuse fully through the tissue, so organs are cut into small fragments, or fixed by **vascular perfusion**
Dehydration	Water extracted gradually through increasing **ethanol** solutions, ending in 100% ethanol
Clearing	Ethanol replaced by an organic solvent miscible with both alcohol and the embedding medium; the tissue becomes **translucent**
Infiltration	Tissue placed in melted **paraffin at 52–60 °C** in an oven, which evaporates the clearing solvent
Embedding	Paraffin allowed to harden in a small container at room temperature
Sectioning	Block trimmed and cut on a **microtome**. Paraffin sections are typically **3–10 µm**; EM requires sections **less than 1 µm**

The units the book expects you to handle: 1 µm = 1/1000 mm; 1 nm = 0.001 µm = 10^{-6} mm; 1 Å = 0.1 nm = 10^{-4} µm.

Plastic resins are used for both light and electron microscopy. They are infiltrated after dehydration and harden when cross-linking polymerisers are added; plastic embedding **avoids the higher temperatures of paraffin** and therefore avoids tissue distortion.

CLINICAL CORRELATION

Biopsies are tissue samples removed during surgery or routine procedures, fixed in vials of formalin. When the result is needed before the procedure ends — typically to know whether a growth is malignant before the patient is closed — the biopsy is **frozen in liquid nitrogen**, which preserves structure and hardens the tissue, and sectioned on a **cryostat** in a subfreezing cabinet. Freezing is also used for histochemistry of **sensitive enzymes or small molecules**, because freezing, unlike fixation, does not inactivate most enzymes; and because clearing solvents dissolve lipids, **frozen sections are used whenever lipid-containing structures must be studied**.

Staining

Most cells and extracellular material are completely colourless. Dyes behave like acidic or basic compounds and form **electrostatic (salt) linkages** with ionisable radicals of tissue macromolecules.

Term	Meaning in the book	Examples
Basophilic	Anionic (net negative) components with affinity for **basic dyes** — nucleic acids, glycosaminoglycans	Toluidine blue, alcian blue, methylene blue, **hematoxylin**
Acidophilic	Cationic components with many ionised amino groups, staining with **acidic dyes** — mitochondria, secretory granules, collagen	**Eosin**, orange G, acid fuchsin

H&E is the most common method. **Hematoxylin** stains DNA in the nucleus, RNA-rich cytoplasm, and cartilage matrix dark blue-purple; **eosin** stains other cytoplasmic structures and collagen pink and acts as the **counterstain**. **Trichrome** stains (eg, Masson) allow greater distinction among extracellular components.

The PAS reaction uses the hexose rings of polysaccharides and other carbohydrate-rich structures and stains them distinctly **purple or magenta** — the basis for demonstrating glycogen, mucin, and basement membranes. The **Feulgen reaction** is a modification of PAS that stains **DNA specifically**.

Enzyme digestion identifies what a stain is binding: pretreating a section with an enzyme that digests one substrate. Pretreatment with **ribonuclease** greatly reduces cytoplasmic basophilia with little effect on the nucleus, proving that the cytoplasmic staining depended on RNA.

Lipid-soluble dyes such as **Sudan black** reveal lipid-rich structures — provided the processing steps that remove lipids (heat and organic solvents) are avoided. **Metal impregnation** with silver salts visualizes certain ECM fibers and specific elements in nervous tissue. The book's Appendix lists the staining procedures used for most of its light micrographs.

Slide preparation, from fixation to observation, takes **12 hours to 2½ days**, depending on tissue size, embedding medium, and staining method.

Light Microscopy

The **bright-field microscope** has three lens sets:

- **Condenser** — collects and focuses a cone of light onto the slide.
- **Objective** — enlarges and projects the image; routine objectives are **×4** (large field), **×10** (medium), **×40** (detail).
- **Eyepiece (ocular)** — magnifies a further **×10**, giving total magnifications of **×40, ×100, ×400**. Total magnification = objective × ocular.

Resolving power — the smallest distance at which two structures are seen as separate — is the critical factor, and its maximum for a light microscope is about **0.2 µm**, permitting clear images at **1000–1500×**. Anything smaller or thinner than 0.2 µm (a single ribosome, a microfilament) cannot be distinguished, and two mitochondria closer than 0.2 µm appear as one. Resolution depends mainly on the **objective lens**; the ocular only enlarges and does **not** improve resolution. Magnification is of value only when accompanied by resolution.

Virtual microscopy converts a stained slide into high-resolution digital images captured in a grid at multiple magnifications, stored so a browser can navigate the original slide — now replacing glass slide collections in teaching laboratories.

Method	Principle	Use
Fluorescence	Fluorophores emit longer-wavelength light when irradiated; bright on a dark background	**DAPI** and **Hoechst** bind DNA (blue); **acridine orange** binds DNA and RNA and emits different colors, localizing them separately; fluorescein-coupled probes and labeled antibodies
Phase-contrast	Light changes speed through structures of different refractive index, producing contrast in transparent objects	**Living, unstained cultured cells**; **DIC (Nomarski optics)** gives a more three-dimensional image

Method	Principle	Use
Confocal	A laser point source and a **pinhole aperture** optically conjugated to the focal plane block out-of-focus light; the beam scans the specimen	Thin **optical sections**, localization with great precision, and 3-D reconstruction
Polarizing	A polarizing filter passes light vibrating in one direction; oriented macromolecules rotate it and appear bright against a dark background — **birefringence**	Collagen, cellulose, microtubules, actin filaments; only material with repetitive periodic structure is seen

Digital cameras extend all of these methods: images can be quantified, enhanced, and analysed by software for features not visible through the eyepieces.

Electron Microscopy

The wavelength of an electron beam is far shorter than that of light, allowing a **1000-fold increase in resolution**.

Transmission EM (TEM) resolves to about **3 nm** — enough to view isolated particles magnified as much as **400,000×**; tissue sections of **40–90 nm** embedded in resin are routinely studied at up to about **120,000×**. Electrons pass through the section to produce an image in black, white, and gray: areas that transmitted electrons appear **electron lucent** (bright), while denser areas or those binding heavy metal ions appear **electron dense** (dark). The column is configured like an upside-down light microscope, with electromagnetic lenses and a tungsten filament cathode accelerating electrons at 60–120 kV.

Scanning EM (SEM) images the surface in three dimensions.

Autoradiography

A **radiolabelled precursor** is incorporated into newly synthesized macromolecules and detected by a photographic emulsion applied over the section — so the silver grains mark the **cellular sites where those macromolecules were made**. The classic application is **^{3}H-thymidine** labeling of cells in S phase.

Cell & Tissue Culture

Living cells maintained outside the body, observed with phase-contrast or DIC microscopy, allow behavior, differentiation, and drug responses to be studied directly — without fixation, which is precisely the advantage.

Enzyme Histochemistry

A substrate is converted by a specific enzyme into an **insoluble, coloured product** deposited at the site of the enzyme, localizing enzyme activity within cells. **Acid phosphatase** for lysosomes is the standard example.

Visualizing Specific Molecules

Immunohistochemistry (IHC) depends on the specific reaction between an antigen and an antibody labeled with a visible marker — a fluorescent compound or **peroxidase** for light microscopy, **gold particles** for TEM.

Method	How it works	Why it is used
Direct IHC	A labeled **primary antibody** binds the antigen directly	Simple, fewer steps
Indirect IHC	An unlabelled primary antibody is detected by labeled **secondary antibodies**	**More commonly used** — the extra level of binding **amplifies the signal** and gives greater technical flexibility

In situ hybridization (ISH) uses labeled **cDNA probes** to detect specific gene sequences or mRNAs microscopically.

Interpretation of Structures in Tissue Sections

Two rules from the book's own section:

1. Many processing, preparation, and staining steps introduce **artefacts** — spaces and precipitates not present in living tissue — and these must be recognized for what they are.
2. Sections are essentially **two-dimensional planes through three-dimensional structures**, so what you see is a slice, not the whole object.

UNDER THE MICROSCOPE

Before identifying anything, decide whether it could be an artefact. A clear space around every cell is shrinkage; a straight edge through the tissue is a knife mark; granular debris is precipitated stain. True structure has irregular, biological boundaries.

EXAM TIP

Three numbers carry the chapter: **0.2 µm** (light microscope resolution), **3–10 µm** (paraffin section thickness), **about 3 nm** (TEM resolution). If a question asks why the electron microscope reveals organelles that the light microscope cannot, the answer is **the shorter wavelength of the electron beam**, not the magnification.

HIGH-YIELD PEARL

The examiner's favourite Chapter 1 chain is: **dehydration precedes clearing**, clearing makes the tissue translucent, and **freezing avoids both** — which is why frozen sections are used for rapid diagnosis, for enzymes, and for lipids.

2 · The Cytoplasm

BOOK PAGES 17–52 | Part 1, pages 40–91 · Cell Differentiation · The Plasma Membrane · Cytoplasmic Organelles · The Cytoskeleton · Inclusions

The core idea

A cell is a compartmentalization machine. Each organelle is a membrane-bound solution to a specific logistics problem: making protein, making lipid, generating ATP, degrading waste, or moving cargo. Learn the job, and the morphology follows automatically.

Cell Differentiation

The chapter opens with differentiation, because it explains why one genome produces hundreds of cell shapes. Cells become specialized by **differential gene expression** — the same DNA, different genes switched on. Differentiation is accompanied by changes in the amount and type of organelles, which is exactly what you recognize down a microscope: an erythroblast loses its nucleus, a plasma cell fills with rough ER, a muscle cell fills with contractile filaments.

Stem cells retain the capacity to divide and to produce differentiated progeny. They divide **asymmetrically**: one daughter renews the stem-cell pool, the other commits to differentiation. This mechanism explains tissue renewal in the gut, epidermis, and bone marrow (Chapter 3).

The Plasma Membrane

A **7.5–10 nm phospholipid bilayer** with embedded proteins — the **fluid mosaic model**. Amphipathic phospholipids form the bilayer spontaneously; cholesterol modulates fluidity; glycolipids and glycoproteins project outward as the **glycocalyx**, which mediates cell recognition, adhesion, and immune identity.

Protein class	Position	Function
Integral (transmembrane)	Span the bilayer	Channels, pumps, receptors, adhesion
Peripheral	Attached to one face	Signaling, cytoskeletal anchoring, enzymes

Lipid rafts are cholesterol- and sphingolipid-enriched microdomains that concentrate signaling molecules. **Caveolae** are small flask-shaped invaginations involved in uptake and signaling.

Transport across the membrane

- **Simple diffusion** — small, non-polar molecules down a gradient.
- **Osmosis** — water movement through the bilayer and aquaporins.
- **Facilitated diffusion** — carrier or channel, down the gradient, no energy.
- **Active transport** — against the gradient, ATP-dependent. The **Na^+/K^+-ATPase** exchanges 3 Na^+ out for 2 K^+ in and consumes a large fraction of the cell's resting energy budget.
- **Endocytosis** — **phagocytosis** (large particles, actin-driven), **pinocytosis** (fluid), **receptor-mediated endocytosis** (clathrin-coated pits; the classic example is LDL uptake, and its failure causes familial hypercholesterolaemia).
- **Exocytosis** — vesicle fusion with the plasma membrane; requires SNARE proteins and calcium.
- **Transcytosis** — endocytosis on one face plus exocytosis on the other; used by endothelial cells and antibodies crossing epithelia.

Cytoplasmic Organelles

Mitochondria. Double membrane; the inner membrane folds into **cristae** carrying the electron transport chain and ATP synthase. The matrix contains circular mitochondrial DNA, mitochondrial ribosomes, and calcium granules. Mitochondria are **maternally inherited** and semi-autonomous.

Cells with huge energy demand (cardiac muscle, brown fat, proximal tubule, sperm midpiece) are packed with them. **Cells that synthesize steroid hormone have mitochondria with tubular rather than lamellar cristae** — a favourite EM question.

Ribosomes. 15–20 nm particles of rRNA plus protein, made of a large and small subunit. Free ribosomes make cytosolic protein for internal use; ribosomes bound to the rough ER make protein for secretion, membranes, or lysosomes. **Polyribosomes (polysomes)** are strings of ribosomes on one mRNA, and a **polyribosome** is the functional unit of translation.

Endoplasmic reticulum (ER). An anastomosing network of intercommunicating channels and cisternae, continuous with the outer nuclear membrane. **Rough ER (RER)** — flattened cisternae studded with ribosomes. Sites of synthesis of secretory proteins, integral membrane proteins, and lysosomal enzymes. It is the structure responsible for **basophilia** — and in neurons, stacked RER plus polysomes form the **Nissl bodies**.

Smooth ER (SER) — tubular and ribosome-free. Three jobs: **lipid and steroid synthesis**, **detoxification** (hepatic cytochrome P450), and **calcium storage** (sarcoplasmic reticulum of muscle). Hepatocytes, adrenal cortical cells, Leydig cells, and the corpus luteum are rich in SER — which is why steroid-secreting cells look pale and foamy rather than blue.

Golgi apparatus. Vesicles leave the ER in coats marked by **COPII** and fuse with the cis (forming) face; transport onward is mediated by **COPI**-coated vesicles. It is a stack of 4–8 flattened cisternae with a **cis (forming) face** receiving vesicles from the ER and a **trans (maturing) face** releasing vesicles. Functions: modification (glycosylation, sulfation, phosphorylation), sorting, and packaging. The **trans-Golgi network** is the dispatch center routing proteins to secretory granules, lysosomes, or the membrane. Cells specialized for secretion (plasma cells, pancreatic acinar cells, goblet cells) have an obvious, often supra-nuclear Golgi.

Lysosomes. 0.05–0.5 µm vesicles of acid hydrolases at pH ≈ 5 maintained by a proton pump. **Primary lysosomes** are newly formed; **secondary lysosomes** are the product of fusion with a phagosome or autophagosome; **residual bodies** remain after digestion and accumulate as **lipofuscin**, the "wear-and-tear" pigment of long-lived cells such as cardiac myocytes and neurons.

Peroxisomes. Small, membrane-bound oxidative organelles containing **catalase** and oxidases. They perform β-oxidation of very long-chain fatty acids, detoxify ethanol, and destroy H_2O_2.

Proteasomes. Cytosolic barrel-shaped complexes that degrade ubiquitin-tagged proteins.

The Cytoskeleton

Element	Diameter	Protein	Main roles
Microfilament	6–8 nm	**Actin**	Microvilli core, cell cortex, stress fibers, cleavage furrow, muscle contraction
Intermediate filament	10 nm	**Keratin, vimentin, desmin, GFAP, neurofilament, lamin**	Mechanical strength; tissue-specific identity
Microtubule	25 nm	**Tubulin (α/β dimer)**	Spindle, cilia and flagella, intracellular transport

Intermediate filaments are the histologist's best friend, because each cell type keeps its own: **keratin** in epithelium, **vimentin** in mesenchyme, **desmin** in muscle, **GFAP** in astrocytes, **neurofilament** in neurons, **lamins** in the nucleus. This is why immunohistochemistry can identify the origin of a tumor that looks undifferentiated.

Microtubules are polar. Kinesin walks cargo toward the plus end (anterograde, away from the cell body); **dynein** walks toward the minus end (retrograde, back to the cell body). Rabies and herpes viruses exploit dynein to travel to the neuronal soma.

Cilia and flagella have a **9 + 2** arrangement with dynein arms; the **basal body** has **9 + 0**. Defective dynein causes **primary ciliary dyskinesia**, of which **Kartagener syndrome** (situs inversus,

bronchiectasis, sinusitis, male infertility) is the classic form.

Inclusions

Non-membranous accumulations of metabolites or pigments: **glycogen** (PAS-positive, liver and muscle), **lipid droplets** (Sudan-positive), **melanin**, **lipofuscin**, **hemosiderin**, and crystals.

CLINICAL CORRELATION

Lysosomal storage diseases are the direct consequence of a missing acid hydrolase. In **Tay–Sachs disease** hexosaminidase A is absent, so GM2 ganglioside accumulates in neurons, producing developmental regression, a cherry-red macular spot, and death in early childhood. In **I-cell disease** the enzyme that tags hydrolases with mannose-6-phosphate is defective, so the enzymes are secreted instead of delivered to lysosomes.

UNDER THE MICROSCOPE

Learn to read a cell's job from its EM. Abundant RER and a large Golgi = protein secretion. Abundant SER and lipid droplets, mitochondria with tubular cristae = steroid secretion. Many mitochondria and few other organelles = energy transduction. Numerous lysosomes and residual bodies = phagocytosis or long life. Bundles of filaments = mechanical work.

MEMORY HOOK

"Keratins Keep Epithelium; Vimentin is Versatile mesenchyme; Desmin Does muscle; GFAP Glues astrocytes; Neurofilaments are Neuronal."

EXAM TIP

Two traps repeat every year. The Golgi **modifies and sorts** proteins; the RER makes them. And the mitochondrion is the powerhouse; the lysosome is the incinerator.

3 · The Nucleus

BOOK PAGES 53–70 | Part 1, pages 92–121 · Components of the Nucleus · The Cell Cycle · Mitosis · Stem Cells & Tissue Renewal · Meiosis · Apoptosis

The core idea

The nucleus stores the genome, transcribes it, and duplicates it before division. Its morphology is a readout of transcriptional activity: **dark, condensed chromatin means silent genes; pale, dispersed chromatin means active genes.**

Components of the Nucleus

Nuclear envelope. Two membranes separated by a **perinuclear cisterna**, continuous with the rough ER. Beneath the inner membrane lies the **nuclear lamina**, a meshwork of **lamins A, B, and C** that supports the envelope and organizes chromatin. **Mutations in lamin A cause Hutchinson–Gilford progeria.**

Centrosomes. Each cell has a pair of **centrioles** — nine microtubular triplets — surrounded by pericentriolar material: the **centrosome**, the microtubule-organizing centre that builds the mitotic spindle, and the **basal body** of a cilium.

Nuclear pores. 3,000–4,000 per nucleus, each a **nuclear pore complex** regulating traffic. Small molecules diffuse; proteins and RNA require **importins and exportins** and a Ran-GTP gradient.

Chromatin. Each human cell holds about **2 metres of DNA — 3.2 billion base pairs** — which must be packaged into a nucleus a few micrometres across. The DNA wraps around a core of **eight histones** (two copies each of H2A, H2B, H3, H4) with **about 150 bp** per turn, plus a larger **H1** histone on the outside: that unit is the **nucleosome**, which in EM looks like **10-nm beads on a string**. Further folding into 30-nm fibers and beyond involves nonhistone scaffold proteins and is less well understood. **Heterochromatin** is condensed and transcriptionally inactive; **euchromatin** is dispersed and active. **Constitutive** heterochromatin is permanently silent; **facultative** heterochromatin can be reactivated. The **Barr body** is a facultatively heterochromatic, inactivated X chromosome — one per somatic cell in a female.

Nucleolus. Not membrane-bound; a factory for ribosomal subunits, with a **pars fibrosa** (rRNA transcription) and a **pars granulosa** (ribosome assembly), organized around **nucleolar organizer regions** on the acrocentric chromosomes **13, 14, 15, 21, and 22**. Cells that make large amounts of protein have prominent nucleoli, and malignant cells often do too.

The Cell Cycle

Phase	What happens	Duration (typical cycling cell)
G1	Cell grows, checks nutrients and DNA integrity; restriction point	~10 h
S	DNA replication; histones synthesized	~8 h
G2	Repair check, prepare for mitosis	~4 h
M	Mitosis and cytokinesis	~1 h

Progression is driven by **cyclin–CDK complexes**. **p53** halts the cycle at G1 to allow repair and triggers apoptosis if repair fails — hence "guardian of the genome", and hence its mutation in more than half of human cancers. **Rb** is the G1/S brake lost in retinoblastoma. **G0** is the quiescent state of cells that have left the cycle.

Mitosis

- **Prophase** — chromosomes condense, centrioles move to opposite poles, the nuclear envelope begins to break down.

- **Metaphase** — chromosomes align at the equatorial plate; the spindle attaches to kinetochores.
- **Anaphase** — sister chromatids separate and are pulled poleward.
- **Telophase** — nuclear envelopes re-form, chromosomes decondense.
- **Cytokinesis** — an **actin** contractile ring pinches the cell in two.

Stem Cells & Tissue Renewal

- **Totipotent** — can form the whole organism (zygote, early blastomeres).
- **Pluripotent** — can form all three germ layers (inner cell mass; embryonic stem cells; induced pluripotent stem cells reprogrammed with the Yamanaka factors Oct4, Sox2, Klf4, c-Myc).
- **Multipotent** — form a restricted set of cell types (hematopoietic stem cell).
- **Unipotent** — one cell type.

Tissues are **labile** (continuous turnover — epidermis, gut, marrow), **stable** (low turnover, can be triggered — liver, kidney, fibroblasts), or **permanent** (neurons, cardiac myocytes, lens fibers).

Meiosis

Two divisions produce **four haploid cells** from one diploid cell, with **crossing over** between homologous chromosomes during **pachytene of prophase I**, held together by the **synaptonemal complex**. Meiosis generates genetic diversity and halves the chromosome number; **non-disjunction** produces aneuploid gametes, the mechanism behind trisomy 21.

Apoptosis

Programmed cell death: **caspases** are activated, **cytochrome c** escapes mitochondria, the **Bcl-2 family** decides survival versus death, chromatin condenses, the cell shrinks and buds into **apoptotic bodies** that are phagocytosed **without inflammation**.

Feature	Apoptosis	Necrosis
Trigger	Physiological or programmed	Injury, ischemia, toxin
Energy	ATP-dependent	Passive
Cell size	Shrinks	Swells
Nucleus	Condenses, fragments	Pyknosis, karyorrhexis, karyolysis
Membrane	Intact, blebbing	Ruptures
Inflammation	**Absent**	**Present**
Spread	Single cells	Contiguous sheets

CLINICAL CORRELATION

Lamin A mutation causes progeria with premature aging. Changes in nuclear shape and chromatin texture are also one of the oldest cancer-screening tools: the irregular, hyperchromatic nucleus of a dysplastic cervical cell is what a Pap smear is looking for.

UNDER THE MICROSCOPE

An active cell has a pale nucleus with a prominent, reticulated nucleolus and lots of euchromatin. A quiescent cell — a small lymphocyte, a spermatozoon, a superficial keratinocyte — has dense heterochromatin and little visible nucleolus. **Nuclear texture is a functional readout.**

EXAM TIP

One Barr body in a normal female somatic cell, **zero** in a male, one in Klinefelter syndrome (47, XXY), none in Turner syndrome (45, X). Count Barr bodies and you have counted X chromosomes minus one.

Chapter 3 — one-line answers

- Nuclear lamina = lamins; lamin A mutation = progeria.

- Nuclear envelope = two membranes + perinuclear cisterna, continuous with rough ER.
- Nuclear pores = 3,000–4,000 per nucleus; importins and exportins, Ran-GTP gradient.
- Nucleosome = eight histones + about 150 bp of DNA; H1 sits outside; 10-nm beads on a string.
- Heterochromatin = condensed and silent; euchromatin = dispersed and active.
- Barr body = facultative heterochromatin of one inactivated X chromosome.
- Nucleolus = rRNA synthesis and ribosome assembly; NORs on 13, 14, 15, 21, 22.
- Cell cycle = G1 → S → G2 → M; DNA replicates in S.
- Cyclin–CDK complexes drive the cycle; p53 arrests at G1; Rb is the G1/S brake.
- p53 = G1 checkpoint and apoptosis; mutated in most cancers.
- Mitosis order = prophase, metaphase, anaphase, telophase; actin ring divides the cell.
- Meiosis = four haploid cells, crossing over in pachytene, held by the synaptonemal complex.
- Non-disjunction = aneuploid gametes, the mechanism of trisomy 21.
- Apoptosis = caspases, cytochrome c, Bcl-2 family, apoptotic bodies, no inflammation.
- Necrosis = swelling, membrane rupture, inflammation.
- Labile tissue turns over constantly; stable tissue can be triggered; permanent tissue cannot regenerate.

PART II — THE FOUR BASIC TISSUES

Chapters 4–10 · the four basic tissues and their specializations

4 · Epithelial Tissue

BOOK PAGES 71–95 | Part 1, pages 122–161 · Characteristic Features of Epithelial Cells · Specializations of the Apical Cell Surface · Types of Epithelia · Transport Across Epithelia · Renewal of Epithelial Cells

The core idea

Epithelial tissues are composed of closely aggregated polyhedral cells adhering strongly to one another and to a thin layer of ECM, forming cellular sheets that line the cavities of organs and cover the body surface. Epithelia (Gr. epi, upon + thele, nipple) line all external and internal surfaces of the body, and **all substances that enter or leave an organ must cross this type of tissue**. Everything else — the number of layers, the shape of the cells, the specializations on the apex — is a solution to one of four problems: **protection, absorption, secretion, or transport**.

The four basic tissues (the book's Table 4–1, and the framework for the whole subject):

Tissue	Cells	Extracellular matrix	Main functions
Epithelial	Aggregated polyhedral cells	**Small amount**	Lining of surface or body cavities; glandular secretion
Connective	Several types of fixed and wandering cells	**Abundant amount**	Support and protection of tissues/organs
Muscle	Elongated contractile cells	Moderate amount	Strong contraction; body movements
Nervous	Elongated cells with extremely fine processes	**Very small amount**	Transmission of nerve impulses

Most organs are divided into the **parenchyma** — the cells responsible for the organ's specialized functions — and the **stroma**, which plays a supporting role. **Except in the brain and spinal cord, the stroma always consists of connective tissue.** This single sentence answers a common question about why the CNS looks different from every other organ.

Characteristic Features of Epithelial Cells

The principal functions of epithelial tissues are **covering, lining, and protecting surfaces** (eg, epidermis), **absorption** (eg, the intestinal lining), and **secretion** (eg, parenchymal cells of glands). Specific cells of certain epithelia may be contractile (**myoepithelial cells**) or specialized sensory cells, such as those of taste buds or the olfactory epithelium.

Cell shape and nuclear shape correspond: **columnar** cells have elongated nuclei, **squamous** cells flattened nuclei, and **cuboidal or pyramidal** cells spherical nuclei. This is the fastest identification rule in the whole subject.

The defining features:

1. Cells are closely apposed with minimal intercellular space.
2. There is a **free (apical) surface** facing air, a lumen, or a cavity.
3. Cells rest on a **basement membrane**.
4. The tissue is **avascular** — nourishment arrives by diffusion from vessels in the underlying connective tissue.
5. Cells are **polarized**: apical, lateral, and basal domains differ in structure and function.

The basement membrane, visible with PAS or silver in the light microscope and as two layers in EM:

Layer	Composition	Made by
Lamina lucida	Laminin, entactin, perlecan	Epithelial cells

Layer	Composition	Made by
Lamina densa	**Type IV collagen**, heparan sulfate proteoglycan	Epithelial cells
Lamina reticularis	**Type III (reticular) collagen**	Fibroblasts of underlying connective tissue

Cell junctions

Junction	Transmembrane protein	Cytoskeletal link	Function
Tight junction (zonula occludens)	Claudins, occludins	Actin	Seals the intercellular space; creates the **blood–brain barrier** and apical/basal polarity
Adherent junction (zonula adherens)	**E-cadherin** (Ca^{2+}-dependent)	Actin	Mechanical adhesion; the belt that holds sheets together
Desmosome (macula adherens)	**Desmoglein, desmocollin**	**Keratin**	Spot-welds resisting shear; autoantibodies cause **pemphigus vulgaris**
Gap junction (nexus)	**Connexins**	None	Direct cytoplasmic communication
Hemidesmosome	**Integrins**	Keratin → laminin	Anchors basal cells; antibodies cause **bullous pemphigoid**

The pattern is worth memorizing: **tight seals, adherens belts to actin, desmosomes to keratin, gap junctions for talk, hemidesmosomes for feet.**

Specializations of the Apical Cell Surface

Microvilli. Finger-like extensions with a core of **actin** filaments cross-linked by villin and fimbrin and anchored in a **terminal web**. They multiply surface area for absorption. In the proximal tubule and small intestine they form a **brush border** visible as a fuzzy eosinophilic rim. In the epididymis and inner ear, unusually long, stiff microvilli are called **stereocilia** — a misnomer, because they contain no microtubules.

Cilia. Motile, **9 + 2** microtubule axoneme with dynein arms, anchored to a **basal body (9 + 0)**. They move mucus, cerebrospinal fluid, and the ovum. A **primary cilium** is a single non-motile 9 + 0 cilium acting as a sensory antenna — its failure underlies polycystic kidney disease.

Basal infoldings. Deep folds of the basal membrane packed with mitochondria — the "striated" appearance in the ducts of salivary glands and the distal tubule of the kidney. They are the morphological signature of **active ion transport**.

Types of Epithelia

Type	Structure	Location	Function
Simple squamous	One flat layer, central nucleus	Alveoli, endothelium, mesothelium, Bowman capsule	Diffusion, filtration, lubrication
Simple cuboidal	One layer, nuclei round and central	Kidney tubules, small ducts, thyroid follicles, surface of ovary	Secretion, absorption, conduit
Simple columnar	One layer, tall cells	Stomach, intestine, gallbladder, uterine tube	Absorption, secretion, protection
Pseudostratified ciliated columnar	Single layer with nuclei at different heights; cilia and goblet cells	Trachea, bronchi, nasal cavity	Mucus clearance ("respiratory epithelium")
Stratified squamous, keratinized	Many layers, outermost anucleate keratin	Epidermis	Protection against abrasion and dehydration

Type	Structure	Location	Function
Stratified squamous, non-keratinized	Many layers, outermost live nucleated cells	Oral cavity, esophagus, vagina, anal canal, cornea	Protection in a wet environment
Stratified cuboidal / columnar	Two or more layers	Ducts of sweat glands, conjunctiva	Conduit with protection
Transitional (urothelium)	Stratified, with dome-shaped **umbrella cells** and uroplakins	Renal pelvis, ureter, bladder, proximal urethra	Stretchable barrier to urine
Epithelioid	Cells in clusters with no free surface	Adrenal cortex, islets of Langerhans, liver cords	Endocrine secretion

Two naming traps decide many marks. **Endothelium and mesothelium are both simple squamous epithelium** — one lines blood vessels, the other body cavities; they are not separate tissue types. And **pseudostratified** epithelium is truly a **single layer** in which every cell touches the basement membrane, even though the nuclei sit at different levels.

Transport Across Epithelia

An epithelium that moves material from one side to the other is built like a machine: **entry at the apical face, exit at the basal face**, with the tight junction preventing back-leak and mitochondrial-rich basal infoldings powering the pumps. In the intestinal enterocyte, glucose enters apically via **Na^+-glucose cotransport (SGLT1)** and exits basolaterally via **GLUT2**; the Na^+ gradient is maintained by basolateral Na^+/K^+-ATPase. This is **transcellular transport**. **Paracellular** transport passes between cells and is limited by tight junctions.

Renewal of Epithelial Cells

Epithelia wear out. They are replaced from **stem cells** whose location varies with the tissue: the **basal layer** of stratified squamous epithelium, the **crypt base** of the intestine, the **isthmus** of the gastric gland, and the basal layer of the epidermis. Renewal rate is a diagnostic index: the intestinal epithelium turns over in days, the epidermis in weeks. Because the stem cells rest on the basement membrane, an epithelium can regenerate completely after superficial injury — but not after the basement membrane itself is destroyed.

Glands

- **Exocrine** — connected to a surface by a duct. **Endocrine** — ductless, secreting into the blood.
- **Unicellular** — the goblet cell (mucin; empty cup-shaped apex after secretion).
- **Simple** — one unbranched duct (simple tubular = intestinal crypt; simple coiled tubular = eccrine sweat gland; simple branched acinar = sebaceous gland).
- **Compound** — branched duct (salivary glands, pancreas, mammary gland).
- **Mechanism: merocrine** (exocytosis — pancreas, sweat glands), **apocrine** (membrane and cytoplasm bud off — mammary gland), **holocrine** (whole cell disintegrates — sebaceous gland).

Myoepithelial cells sit between secretory cells and the basement membrane and contract to expel product — visible in mammary and salivary glands.

CLINICAL CORRELATION

Epithelial junctions are the target of autoimmune blistering. In **pemphigus vulgaris**, antibodies against **desmoglein** break desmosomes, so the epidermis detaches within the epithelium and thin-walled blisters form. In **bullous pemphigoid**, antibodies against **hemidesmosomal** antigens split the epidermis from the basement membrane and the blister is thick-walled and tense. The level of the split is dictated by which junction fails.

UNDER THE MICROSCOPE
Count nuclei, not cells. More than one nuclear level with a flat continuous surface = stratified. Then the top layer: **flat and anucleate** = keratinized; **flat with nuclei** = non-keratinized; **dome-shaped and large** = transitional; **tall with cilia** = respiratory.

MEMORY HOOK
"Simple for exchange, stratified for stress, transitional for stretch." For cilia, remember **9 + 2 moves, 9 + 0 anchors.**

EXAM TIP
Where is ciliated simple columnar epithelium found? The **uterine tube** — cilia move the ovum — and parts of the respiratory tract, where the epithelium is pseudostratified. Students often wrongly assign all ciliated epithelium to the lung.

Chapter 4 — one-line answers

- Epithelium sits on a basal lamina, is avascular, and is always adjacent to connective tissue.
- Simple = one layer (exchange, absorption, secretion); stratified = protection; pseudostratified = all cells touch the basement membrane.
- Squamous = flat; cuboidal = height ≈ width; columnar = height > width.
- Keratinized stratified squamous = dry, abrasive surfaces; nonkeratinized = wet surfaces (mouth, esophagus, vagina).
- Urothelium (transitional) = dome-shaped umbrella cells; stretches.
- Tight junction (zonula occludens) = seal and fence; it creates apical–basal polarity.
- Zonula adherens anchors actin; desmosomes anchor intermediate filaments; gap junctions couple cells.
- Hemidesmosomes attach the basal cell to the basal lamina.
- Basal lamina = lamina lucida + lamina densa (type IV collagen, laminin); the reticular lamina (type III) belongs to connective tissue.
- Microvilli = actin core, increase surface; stereocilia = long, stiff microvilli; cilia = 9 + 2 microtubules, motile.
- Basal bodies anchor cilia and have a 9 + 0 arrangement.
- Keratin 5/14 mark the basal layer; keratin 1/10 mark suprabasal layers.
- Apical specializations serve absorption or movement; basal specializations anchor and receive signals.
- Epithelia renew from stem cells: labile (gut, skin) versus stable (liver, kidney).

5 · Connective Tissue

BOOK PAGES 96–121 | Part 1, pages 162–196 · Cells of Connective Tissue · Fibers · Ground Substance · Types of Connective Tissue

The core idea

Connective tissue is the tissue whose defining feature is **abundance of extracellular matrix**. The cells are widely separated because the matrix — not the cells — does the mechanical work. Name the matrix and you have named the tissue's job: collagen for tension, elastin for recoil, ground substance for compression and diffusion.

Cells of Connective Tissue

Cell	Origin	Function	Identifying feature
Fibroblast / fibrocyte	Mesenchyme	Synthesises collagen, elastin, GAGs; **fibrocyte** is the quiescent form	Elongated, pale, abundant RER; **myofibroblast** also has actin for wound contraction
Macrophage	Monocyte	Phagocytosis, antigen presentation, cytokines	Irregular, abundant lysosomes; **Kupffer cell** in liver, **microglia** in CNS, **Langerhans cell** in epidermis, **osteoclast** in bone
Mast cell	Bone marrow	Releases **histamine** and **heparin**; IgE-mediated allergy	Round, many dark granules; degranulates on IgE cross-linking
Plasma cell	B lymphocyte	Secretes antibody	**Clock-face nucleus**, eccentric, perinuclear clear zone (Golgi)
Adipocyte	Mesenchyme	Store fat	Signet-ring when unilocular
Pericyte	Mesenchyme	Contractile support of capillaries	On the abluminal surface of capillaries

Fibers

Fiber	Protein	Properties	Where
Collagen	Tropocollagen (types I–XXVIII)	High tensile strength, inelastic	Everywhere; the most abundant protein in the body
Reticular	**Type III collagen**	Fine, branched; silver-staining	Lymphoid organs, bone marrow, liver, endocrine glands
Elastic	**Elastin** core + **fibrillin** microfibrils	Stretch and recoil	Aorta, skin, lung, elastic cartilage, ligaments

Type	Form	Main location	Disease if defective
I	Thick fibrils	Bone, dermis, tendon, cornea	**Osteogenesis imperfecta**
II	Fibrils	Hyaline and elastic cartilage, vitreous	Chondrodysplasias
III	Reticular fibers	Lymph node, spleen, bone marrow, skin	**Ehlers–Danlos vascular type**
IV	Network	**Basement membrane**	Alport syndrome
V	Fibrils with type I	Cornea, bone, placenta	Ehlers–Danlos

Collagen synthesis in five steps — a reliable structured-question answer:

1. **Translation** on the RER as preprocollagen.
2. **Hydroxylation** of proline and lysine — **requires vitamin C**.

3. **Glycosylation** in the ER and Golgi.
4. **Procollagen** is secreted and the propeptides cleaved → **tropocollagen**.
5. **Fibril assembly** with cross-links formed by **lysyl oxidase**, which requires **copper**.

This pathway explains two diseases. Without vitamin C, hydroxylation fails and the result is **scurvy** — bleeding gums, poor wound healing, perifollicular hemorrhage. Without copper, cross-linking fails, as in **Menkes disease**. In **Ehlers-Danlos syndrome** the defect is in collagen processing or type; in **Marfan syndrome** the defect is in **fibrillin**, so the aorta dilates and the lens dislocates.

Ground Substance

An amorphous, hydrated gel of **glycosaminoglycans (GAGs)**, **proteoglycans**, and **glycoproteins**, which resists compression, lubricates, and controls diffusion.

- **GAGs** (also called mucopolysaccharides) are long polymers of repeating disaccharide units, usually with many negative charges that attract water: **hyaluronic acid** (the only **unsulfated** one, not attached to protein, and the backbone of the gel) and the **sulfated GAGs** — chondroitin sulfate, keratan sulfate, dermatan sulfate, and heparan sulfate.
- **Proteoglycans** are a core protein with GAG side chains — the classic is **aggrecan**, a very large molecule with a **250-kDa core protein** heavily bound with chondroitin and keratan sulfate and joined to hyaluronan by a **link protein**. Its water-binding is why cartilage resists compression.
- **Perlecan** is the key proteoglycan of **all basal laminae** — the one name that links ground substance to the basement membrane.
- **Glycoproteins** — **fibronectin** (binds cells to matrix via integrins, important in wound healing) and **laminin** (basement membrane).

Types of Connective Tissue

- **Loose (areolar)** — abundant ground substance, scattered fibers, many cells. Fills spaces, surrounds vessels, underlies epithelia.
- **Dense irregular** — thick collagen bundles in many directions; dermis, organ capsules. Resists stress from all directions.
- **Dense regular** — parallel collagen bundles; **tendon** and ligament. Resists stress in one direction.
- **Elastic** — many elastic fibers; elastic ligament, vocal cord, large arteries.
- **Reticular** — type III collagen network; lymphoid organs and marrow.
- **Mucous (Wharton's jelly)** — gelatinous, in the umbilical cord.
- **Embryonic** — mesenchyme.

Besides fibroblasts, the resident population includes **adipocytes**, **macrophages**, mast cells, and the **leukocytes** that migrate in (especially **lymphocytes** and plasma cells in chronic inflammation) — which is why the book lists several types of fixed and wandering cells.

Repair. Inflammation → **granulation tissue** (proliferating capillaries, fibroblasts, macrophages) → collagen deposition → **scar**. **Primary intention** occurs in clean apposed wounds; **secondary intention** in larger, open wounds with more granulation and contraction.

CLINICAL CORRELATION

Scurvy is the purest histology-to-clinic link in the chapter: inadequate hydroxylation produces defective collagen, which produces fragile capillaries (bleeding gums, petechiae) and failed wound healing. **Scleroderma** is the opposite failure — excess collagen in skin and internal organs. **Keloid** is an overgrowth of dense collagen beyond the original wound margin, common in darker skin.

UNDER THE MICROSCOPE

Collagen is pink, wavy, and in bundles; elastin is sparse, often darker, and best shown by orcein. A plasma cell has an **eccentric clock-face nucleus with a pale perinuclear halo**; a mast cell's granules stain metachromatically; the eosinophil's bilobed nucleus is unmistakable.

MEMORY HOOK

"1 bone, 2 cartilage, 3 reticulin, 4 basement membrane."

EXAM TIP

The most examined distinction here is **dense regular versus dense irregular**, and the answer is one word: **direction**. Regular = parallel = tendon = one axis of force. Irregular = random = dermis = many axes.

Chapter 5 — one-line answers

- Connective tissue = cells + fibers + ground substance; it is vascular except in cartilage.
- Type I collagen = bone, tendon, dermis, scar. Type II = hyaline and elastic cartilage. Type III = reticulin. Type IV = basal lamina.
- Collagen synthesis needs vitamin C for proline and lysine hydroxylation; deficiency = scurvy.
- Elastic fibers = elastin core + fibrillin microfibrils; orcein stains them.
- Ground substance = GAGs and proteoglycans; **hyaluronan is the only unsulfated GAG**.
- Fibroblast makes collagen; myofibroblast contracts (wound closure).
- Plasma cell = eccentric clock-face nucleus, perinuclear Golgi halo, antibody factory.
- Mast cell = heparin, histamine, proteases; metachromatic granules; IgE receptor.
- Macrophage = monocyte-derived phagocyte; foreign-body and Langhans giant cells are fused macrophages.
- Adipocytes, mast cells, and leukocytes are resident connective tissue cells.
- Loose (areolar) tissue = cells and vessels; dense irregular = dermis; dense regular = tendon.
- Reticular tissue = type III collagen framework of lymphoid organs and marrow.
- Edema = excess fluid in ground substance; it follows increased pressure, leaky vessels, or low albumin.

6 · Adipose Tissue

BOOK PAGES 122–128 | Part 1, pages 197–207 · White Adipose Tissue · Brown Adipose Tissue

The core idea

Adipose tissue is an endocrine and metabolic organ disguised as a filling tissue. White fat stores energy; brown fat burns it for heat. Both are built from the same cell with a completely different internal architecture.

White Adipose Tissue

Unilocular. Each white adipocyte is a very large cell, **50–150 µm** in diameter, containing one enormous lipid droplet that displaces the cytoplasm to a thin rim and flattens the nucleus against the plasmalemma — the **signet-ring** appearance. Adipose tissue is normally **15–20% of body weight in men** and somewhat more in women. The lipid dissolves during routine processing, so on an H&E slide the cell appears as an empty circle with a thin border.

Functions: energy storage, insulation, cushioning of organs (retro-orbital, perirenal, mesenteric), and **endocrine signaling**.

Where the stored fat comes from — three sources, straight from the book: dietary fat carried as **chylomicrons**; triglycerides made in the **liver** and carried as **VLDLs**; and fatty acids and glycerol **synthesized by the adipocytes themselves**. Mobilization is by **hormone-sensitive lipase**, activated by **norepinephrine** from the adrenal gland and sympathetic nerves, and by growth hormone; **insulin inhibits** that lipase and stimulates lipid synthesis. The cells are supported by **reticular fibers**, with connective tissue septa dividing the tissue into lobules.

Adipokines — the exam list:

- **Leptin** — from adipocytes, acts on the hypothalamus to reduce appetite.
- **Adiponectin** — increases insulin sensitivity; low in obesity.
- **Resistin, TNF-α, IL-6** — pro-inflammatory, part of why obesity is a chronic inflammatory state.
- **Aromatase** — converts androgens to estrogens; a peripheral estrogen source in obesity.

Brown Adipose Tissue

Multilocular. Many small lipid droplets, a **central round nucleus**, and — the diagnostic feature — **numerous mitochondria with abundant cristae** containing **uncoupling protein 1 (UCP1, thermogenin)**, which short-circuits the proton gradient so energy is released as heat instead of ATP.

Found in the fetus and newborn (interscapular, axillary, perirenal, along the great vessels), much reduced in adults. Sympathetic innervation drives it. **"Beige" adipocytes** appear within white fat after cold exposure.

Feature	White	Brown
Lipid droplets	One (unilocular)	Many (multilocular)
Nucleus	Peripheral	Central
Mitochondria	Few	**Many, cristae-rich**
UCP1	Absent	**Present**
Main role	Energy storage, endocrine	**Non-shivering thermogenesis**
Sympathetic innervation	Sparse	Rich

CLINICAL CORRELATION

In **lipodystrophy** fat is absent or redistributed, and the metabolic consequences are paradoxical: severe insulin resistance with hypertriglyceridaemia, because the remaining fat cannot store lipid safely. In **obesity** the same storage limit is approached, so lipid spills into liver (steatosis), muscle, and pancreas — **lipotoxicity**. Brown fat is a therapeutic target: activating it increases energy expenditure, and PET imaging shows active depots in adults.

UNDER THE MICROSCOPE

One big hole per cell with a flattened nucleus at the rim = white. Many small holes with a round central nucleus and granular eosinophilic cytoplasm = brown.

EXAM TIP

Do not confuse adipose tissue with **steatosis**. Adipocytes are a normal tissue with a capsule and vessels; steatosis is lipid accumulating inside hepatocytes or myocytes as vacuoles.

Chapter 6 — one-line answers

- White adipocyte: **one large lipid droplet**, nucleus flattened at the rim, few mitochondria.
- Brown adipocyte: **many small droplets**, round central nucleus, many cristae-rich mitochondria.
- White fat = energy storage plus an **endocrine organ**: leptin, adiponectin, and other adipokines.
- Brown fat = **non-shivering thermogenesis** through **UCP1**, which uncouples the proton gradient.
- Brown fat is richest in the newborn (neck, back, mediastinum) and persists in adult neck and supraclavicular depots.
- Beige/brite adipocytes are an inducible, brown-like population within white fat.
- Adipose tissue is vascular and innervated; white fat's sympathetic supply is sparse, brown fat's is rich.
- Obesity is expansion by hypertrophy and hyperplasia; the storage limit produces lipotoxicity.
- Lipodystrophy = absent or redistributed fat with severe insulin resistance.
- Leptin signals energy stores to the hypothalamus; adiponectin increases insulin sensitivity.

7 · Cartilage

BOOK PAGES 129–137 | Part 1, pages 208–217 · Hyaline Cartilage · Elastic Cartilage · Fibrocartilage · Cartilage Formation, Growth, & Repair

The core idea

Cartilage is a **tough, resilient** connective tissue whose **ECM** — collagen plus proteoglycan and water — is the bulk of the tissue; **fibroblasts** of the perichondrium, not chondrocytes, produce its outer fibrous layer. that structurally supports soft tissues — notably in the respiratory tract — and provides **cushioned, low-friction surfaces in joints**. Chondrocytes are a small percentage of the tissue's mass, which is mainly a flexible extracellular matrix of collagen plus abundant proteoglycans, notably **aggrecan**, which binds a large amount of water. **Cartilage always lacks blood vessels, lymphatics, and nerves**, but usually has a vascularized **perichondrium** of dense connective tissue. It is **avascular and aneural**, which explains both its durability under load and its disastrously poor healing.

The shared architecture

- Cells: **chondroblasts** (immature, at the surface) and **chondrocytes** (mature, in **lacunae**).
- Matrix: **territorial matrix** immediately around the lacuna (rich in proteoglycan, basophilic) and **interterritorial matrix** between cells (more collagen, eosinophilic).
- **Isogenous groups** — clusters of 2–8 chondrocytes from one parent cell, indicating recent **interstitial growth**.
- **Perichondrium** — fibrous outer layer plus chondrogenic inner layer; present on most cartilage, **absent on fibrocartilage and on articular surfaces**.

Hyaline Cartilage

The commonest type and the template for most of the fetal skeleton. Glassy, homogeneous matrix; **collagen makes up 40% of hyaline cartilage's dry weight**, all of it embedded in a hydrated gel of proteoglycans and structural glycoproteins, with more proteoglycan and less collagen immediately around the lacunae (the **territorial matrix**). Perichondrium is present except on **articular surfaces** and the epiphyses of growing long bones, where it forms the smooth, low-friction bearing surface of joints. Found in the **trachea, bronchi, nose, ribs, epiphyseal plate, and articular surfaces**.

Elastic Cartilage

Type II collagen plus a network of **elastic fibers**, giving flexibility with recoil. Found in the **epiglottis, pinna, external auditory canal, and Eustachian tube** — that is, everywhere something must bend and spring back.

Fibrocartilage

Type I and type II collagen in coarse bundles, with chondrocytes arranged in **parallel rows** between them. **No perichondrium.** Found where compressive and tensile forces meet: **intervertebral disc, symphysis pubis, menisci, temporomandibular joint**. The disc is the perfect example, with an inner **nucleus pulposus** (notochord remnant, rich in hyaluronic acid) and an outer **annulus fibrosus** of concentric fibrocartilage lamellae.

Feature	Hyaline	Elastic	Fibrocartilage
Collagen	**Type II**	Type II + **elastic fibers**	**Type I + II**
Cells	Isogenous groups	More numerous	**Rows** between bundles
Perichondrium	Present (not on articular surfaces)	Present	**Absent**

Feature	Hyaline	Elastic	Fibrocartilage
Locations	Trachea, bronchi, nose, ribs, epiphyseal plate, joints	Epiglottis, pinna, auditory canal	Disc, symphysis pubis, menisci
Function	Smooth bearing surface, template for bone	Flexible support	Shock absorption, tensile strength

Cartilage Formation, Growth, & Repair

- **Appositional growth** — new cartilage added at the periphery by chondroblasts of the perichondrium. Increases width.
- **Interstitial growth** — chondrocytes divide within lacunae and secrete matrix, pushing cells apart. Increases length; possible only in young cartilage.

Because cartilage is avascular, nutrition depends on diffusion from the perichondrium or synovial fluid. This limits repair: defects fill with **fibrocartilage** rather than hyaline cartilage, and that patch is mechanically inferior.

CLINICAL CORRELATION

Osteoarthritis is a failure of the chondrocyte to maintain matrix: proteoglycans are depleted, water content rises, collagen fibrils fragment, and the surface fissures. Because hyaline cartilage has no vessels and no nerves, pain comes late and from the underlying bone, capsule, and synovium. **Rheumatoid arthritis** is different: an autoimmune synovitis destroys cartilage from the surface inward, producing a pannus.

UNDER THE MICROSCOPE

Identify cartilage by the **lacuna**: a cell alone in a clear space within a solid matrix. Then the matrix: **glassy purple with cells in clusters = hyaline; wavy dark fibers = elastic; thick parallel pink bundles with cells in rows = fibrocartilage.**

EXAM TIP

Two marks are lost repeatedly: **hyaline cartilage contains type II collagen, not type I**, and **fibrocartilage has no perichondrium**, which is why disc injuries persist.

Chapter 7 — one-line answers

- Cartilage = chondrocytes in lacunae + solid matrix; no blood vessels, no nerves.
- Hyaline cartilage = **type II collagen** + aggrecan; glassy; perichondrium present except at articular surfaces.
- Elastic cartilage = type II collagen + **elastic fibers**; perichondrium present; epiglottis, auricle, auditory tube.
- Fibrocartilage = **type I + type II collagen**, rows of chondrocytes, **no perichondrium**; discs, menisci, pubic symphysis.
- Territorial matrix surrounds each lacuna; interterritorial matrix lies between groups.
- Isogenous groups = daughter cells from recent division, still in one lacuna.
- Appositional growth = from perichondrium, adds width.
- Interstitial growth = division inside lacunae, adds length, young cartilage only.
- Articular cartilage is hyaline **without perichondrium** and heals poorly.
- Cartilage repair fills with fibrocartilage, not hyaline.
- Osteoarthritis = matrix failure; rheumatoid arthritis = synovial inflammation with pannus.
- Chondroblasts make matrix; chondrocytes maintain it.

8 · Bone

BOOK PAGES 138–160 | Part 2, pages 3–36 · Bone Cells · Bone Matrix · Periosteum & Endosteum · Types of Bone · Osteogenesis · Bone Remodeling & Repair · Metabolic Role of Bone · Joints

The core idea

Bone is the main constituent of the adult skeleton: it gives **solid support**, protects vital organs of the cranial and thoracic cavities, encloses **medullary cavities** where blood cells form, and acts as a **reservoir of calcium and phosphate**, storing or releasing them to keep body fluid concentrations constant. It also forms a system of **levers** that multiply the force of muscle contraction into movement. Its stiffness comes from mineral, its toughness from collagen, and its capacity to adapt from four cell types that continuously demolish and rebuild it — **every skeletal disease is a failure of one of those cells, or of the balance between them.**

Bone Cells

Cell	Origin	Function	Marker / feature
Osteoprogenitor	Mesenchyme	Divide and differentiate into osteoblasts	Periosteum (inner layer), endosteum
Osteoblast	Osteoprogenitor (periosteum, endosteum)	Secrete **osteoid**, rich in **type I collagen** plus proteoglycans; mineralize it; regulate osteoclasts via **RANKL**	Cuboidal, basophilic, cover existing bone surfaces; **alkaline phosphatase** high
Osteocyte	Osteoblast that becomes embedded	Maintain matrix; sense mechanical load	In **lacunae**, connected by **canaliculi** via gap junctions
Osteoclast	**Monocyte/macrophage lineage**	**Resorb bone**	Large, multinucleated, **ruffled border**, resorption bay (**Howship lacuna**); **TRAP**, carbonic anhydrase II, H^+-ATPase

As osteoid calcifies and hardens it **entraps some osteoblasts**, which then differentiate into osteocytes sitting in **lacunae** between the lamellae, with cytoplasmic processes in **canaliculi**. Osteocytes maintain the calcified matrix and are fed from microvasculature in the **central canals of osteons** through those canaliculi.

The osteoclast is the exception examiners always test. Every other bone cell is mesenchymal; the osteoclast is **monocyte-derived** and is the cell required for bone remodeling. It dissolves mineral by pumping H^+ and digests collagen with cathepsin K.

The RANK–RANKL–OPG axis: osteoblasts express **RANKL**, which binds **RANK** on osteoclast precursors and drives differentiation; they also secrete **osteoprotegerin (OPG)**, a decoy receptor that blocks RANKL. **PTH** shifts the balance toward resorption; **estrogen** suppresses it.

Bone Matrix

- **Inorganic (~65%)** — calcium phosphate as **hydroxyapatite** crystals; compressive strength.
- **Organic (~35%)** — **type I collagen** (90% of the organic fraction), osteocalcin, osteonectin, osteopontin, proteoglycans, growth factors; tensile strength.

Remove the mineral and bone becomes flexible; remove the collagen and it becomes brittle. **Bone is strong because it is both at once.**

Periosteum & Endosteum

Periosteum — dense connective tissue with a mainly **fibrous layer** over a more **cellular** layer; small vessels penetrate the matrix from it. Absent on articular surfaces and at tendon insertions.

Endosteum covers all trabeculae around the marrow cavities — not merely the shaft. Both are sources of osteoprogenitor cells for growth and repair.

Types of Bone

Compact (cortical) bone — dense, on the surface; built of **osteons (Haversian systems)**: concentric lamellae around a central **Haversian canal** containing vessels and nerves, with **osteocytes** in lacunae linked by canaliculi, and **Volkmann canals** running transversely.

Spongy (cancellous) bone — a network of **trabeculae** with marrow between them, oriented along lines of mechanical stress. No true osteons.

Woven versus lamellar. Woven bone is disorganized and produced rapidly in the fetus, fracture callus, and disease (Paget, osteosarcoma). Lamellar bone is ordered and mechanically superior; adult bone is lamellar but always being remodelled through **basic multicellular units (BMUs)**.

Osteogenesis

Intramembranous ossification — mesenchyme condenses and differentiates directly into osteoblasts, depositing bone in a fibrous membrane. Produces the **flat bones of the skull, mandible, and clavicle**. No cartilage template.

Endochondral ossification — a hyaline cartilage model is replaced by bone. This is how **long bones** form and grow in length. The **epiphyseal plate** has **five zones**:

Zone (from epiphysis toward diaphysis)	What happens
1. Reserve (resting)	Small, inactive chondrocytes
2. Proliferative	Chondrocytes divide in columns — the actual growth in length
3. Hypertrophic	Chondrocytes enlarge and the matrix calcifies
4. Calcified cartilage	Chondrocytes die; matrix calcifies
5. Ossification	Osteoblasts deposit bone on the calcified spicules

Growth in **width** is **appositional**, by periosteal osteoblasts. The **primary ossification center** appears in the diaphysis; **secondary centers** in the epiphyses. When the plate ossifies, an **epiphyseal line** remains and length growth stops.

Bone Remodeling & Repair

Fracture → **hematoma** → inflammation → **soft callus** (fibrocartilage) → **hard callus** (woven bone) → **remodelling** to lamellar bone. Immobilization and blood supply decide the outcome. **Wolff law** — bone is laid down where it is loaded and resorbed where it is not — explains both trabecular architecture and the bone loss of bed rest.

Metabolic Role of Bone

Bone is the body's calcium and phosphate reservoir.

Hormone	Source	Effect on bone	Net serum calcium
PTH	Parathyroid chief cells	Increases resorption, increases renal reabsorption, activates vitamin D	**Raises**
Calcitriol (1,25-$(OH)_2$D)	Kidney	Increases intestinal absorption and bone resorption	**Raises**
Calcitonin	Thyroid **C cells**	Inhibits osteoclasts	**Lowers**

Joints

Fibrous joints (synarthroses) — bones united by dense connective tissue, no cavity (sutures of the skull, gomphosis of a tooth).

Cartilaginous joints (amphiarthroses) — bones united by cartilage: **primary** by hyaline cartilage (epiphyseal plate, first sternocostal joint); **secondary** by fibrocartilage (intervertebral disc, symphysis pubis).

Synovial joints (diarthroses) — a joint cavity between **hyaline articular cartilage** on each bone, enclosed by a **capsule** of dense connective tissue lined by **synovial membrane**, which secretes **synovial fluid** for lubrication and nutrition of the avascular cartilage. Some joints add a **meniscus** (fibrocartilage), a **labrum**, or a **fat pad**. Because articular cartilage has no perichondrium and no vessels, it heals hardly at all — the anatomical basis of degenerative joint disease.

CLINICAL CORRELATION

Six to learn as a set. **Osteoporosis** — reduced bone mass with normal mineralization; estrogen deficiency drives it, trabeculae are lost first, fragility fractures follow. **Rickets/osteomalacia** — unmineralized osteoid; in children the growth plate widens and deforms (rachitic rosary, bowed legs). **Paget disease** — chaotic excessive remodelling with mosaic woven bone. **Osteopetrosis** — osteoclast failure, dense but brittle bone with a crowded marrow cavity. **Achondroplasia** — failure of chondrocyte proliferation in the plate, short limbs with normal trunk. **Osteogenesis imperfecta** — defective type I collagen, brittle bones, blue sclerae.

UNDER THE MICROSCOPE

On a ground section, **concentric lamellae around a central canal** = an osteon = compact bone. On a decalcified H&E section, osteoblasts are the row of cuboidal cells on the surface and osteocytes sit in lacunae. A **multinucleated cell in a scalloped pit** is an osteoclast resorbing.

MEMORY HOOK

"OsteoBlast Builds, OsteoClast Chews, OsteoCytes Communicate." For the epiphyseal plate, **"Rest, Proliferate, Hypertrophy, Calcify, Ossify"** — read from the joint outward.

EXAM TIP

The commonest error is saying osteoclasts are mesenchymal. They are **hematopoietic**, from the monocyte–macrophage line. The second is reversing the epiphyseal zones: growth happens at the **epiphyseal** side, ossification at the **diaphyseal** side.

9 · Nerve Tissue & the Nervous System

BOOK PAGES 161–192 | Part 2, pages 37–86 · Development of Nerve Tissue · Neurons · Glial Cells & Neuronal Activity · Central Nervous System · Peripheral Nervous System · Neural Plasticity & Regeneration

The core idea

Nervous tissue is built for **rapid signaling over distance**. The neuron is a polarized cell with a receiving end (dendrites and soma), a conducting cable (axon), and a transmitting end (synapse). Everything else — myelin, glia, barriers — exists to make that signaling faster, safer, and better supplied.

Development of Nerve Tissue

The nervous system arises from the **neural plate**, which folds into the **neural tube** (giving the CNS) while **neural crest** cells migrate away to form the peripheral neurons, Schwann cells, satellite cells, adrenal medulla, melanocytes, and much of the facial skeleton. This embryology explains three exam facts: the CNS is lined by **ependyma** derived from the tube; Schwann cells are neural crest; and the adrenal medulla is neural crest while the cortex is mesoderm.

Neurons

- **Perikaryon (soma)** — large, pale nucleus with a prominent nucleolus and abundant **Nissl bodies** (stacked RER and polysomes). Nissl substance is absent from the **axon hillock** — a useful landmark.
- **Dendrites** — short, branched, receptive; contain Nissl bodies.
- **Axon** — one per neuron, arises at the axon hillock, contains no Nissl bodies, specialized for conduction.
- **Axonal transport** — **kinesin** anterograde; **dynein** retrograde. Rabies, herpes, and tetanus toxin exploit retrograde transport.

Classification: multipolar (most CNS neurons and motor neurons), bipolar (retina, olfactory epithelium), pseudounipolar (sensory ganglia), anaxonic. Also by function — sensory, motor, interneuron.

The synapse. A chemical synapse has a presynaptic terminal with **synaptic vesicles** and mitochondria, a **synaptic cleft** (~20–30 nm), and a postsynaptic membrane with receptors. Transmission is unidirectional and modifiable. **Electrical synapses (gap junctions)** are fast and bidirectional.

Glial Cells & Neuronal Activity

Cell	Location	Function	Exam hook
Astrocyte	CNS	**The most numerous cell of the CNS**; hundreds of processes enclose neuronal perikarya, synapses, and capillaries, providing regulated microenvironments	**GFAP**; foot processes on capillaries; glial scar
Oligodendrocyte	CNS	Wraps flattened processes around portions of axons, forming **myelin sheaths**	One cell, many segments
Microglia	CNS	Resident phagocyte, immune surveillance	**Mesodermal** origin — the odd one out
Ependymal cell	CNS	Lines ventricles; cilia move CSF; choroid plexus	Produces CSF
Schwann cell (neurolemmocyte)	PNS	**Myelinates one segment of one axon**	Neural crest origin
Satellite cell	PNS ganglia	Supports sensory and autonomic neurons	—

The book counts **six major glial types**: oligodendrocytes, astrocytes, ependymal cells, and microglia in the CNS; Schwann cells and satellite cells in the PNS. **Myelination:** in the PNS a Schwann cell wraps one axon segment; in the CNS an oligodendrocyte wraps several. The gaps are **nodes of Ranvier**, where voltage-gated Na^+ channels cluster, so the impulse jumps by **saltatory conduction**. Velocity rises with axon diameter and myelin thickness.

Central Nervous System

- **Cerebral cortex** — six layers; **pyramidal cells** are the output neurons; gray matter outside.
- **Cerebellar cortex** — three layers: **molecular**, **Purkinje** (large flask-shaped cells, the only output), and **granular**.
- **Spinal cord** — gray matter in an H-shaped core, white matter outside; **anterior horn** motor, posterior horn sensory. This is the reverse of the cerebral cortex.
- **Meninges** — dura, arachnoid, pia; CSF in the subarachnoid space.
- **Choroid plexus** — produces CSF.
- **Blood–brain barrier** — tight junctions between endothelial cells, reinforced by astrocyte end-feet and pericytes. Absent in the circumventricular organs (area postrema, median eminence, pineal, posterior pituitary), which is how circulating hormones reach the brain.

Peripheral Nervous System

- **Sensory ganglia** — pseudounipolar neurons with satellite cells; **no synapses inside the ganglion**.
- **Autonomic ganglia** — multipolar neurons, synapses present.
- **Peripheral nerve** — axons in fascicles surrounded by **perineurium**, the whole nerve by **epineurium**, individual fibers by **endoneurium**.

Neural Plasticity & Regeneration

After axotomy the distal stump undergoes **Wallerian degeneration**; the soma shows **central chromatolysis**; Schwann cells form **bands of Büngner** to guide regrowth; regeneration proceeds at about **1 mm/day**. CNS axons regenerate poorly because of inhibitory molecules (Nogo, MAG) and glial scarring. **Plasticity** — the remodelling of synapses with use — is the basis of learning and of rehabilitation after injury.

CLINICAL CORRELATION

Multiple sclerosis is autoimmune demyelination in the CNS. **Guillain–Barré syndrome** is demyelination in the PNS, ascending from the legs. **Parkinson disease** is loss of dopaminergic neurons in the substantia nigra with Lewy bodies. **Alzheimer disease** shows neurofibrillary tangles and amyloid plaques. **Poliomyelitis** destroys anterior horn motor neurons and produces flaccid paralysis.

UNDER THE MICROSCOPE

Identify neurons by the **large pale nucleus, prominent nucleolus, and clumped Nissl bodies**. The axon hillock is the pale cone where Nissl substance stops. In peripheral nerve, look for **wavy bundles of axons with myelin rings** and connective tissue sheaths. In the cerebellar cortex, the **single row of enormous Purkinje cells** is unmistakable.

EXAM TIP

Oligodendrocyte = CNS, several axons. Schwann cell = PNS, one segment of one axon. And **microglia are the only glia of mesodermal origin.**

10 · Muscle Tissue

BOOK PAGES 193–214 | Part 2, pages 87–119 · Skeletal Muscle · Cardiac Muscle · Smooth Muscle · Regeneration of Muscle Tissue

The core idea

Muscle tissue consists of cells that optimize the universal cell property of **contractility**. Essentially **all muscle cells originate from embryonic mesoderm** and differentiate by elongation and abundant synthesis of the myofibrillar proteins **actin and myosin**. All three types use the same actin–myosin interaction but arrange it differently: **striated and voluntary** (skeletal), **striated and involuntary** (cardiac), **unstriated and involuntary** (smooth).

Skeletal Muscle

Organization, large to small: muscle → fascicles → fibers → myofibrils → myofilaments. Each fiber is a long, cylindrical, **multinucleated** syncytium with **peripheral nuclei**, produced by fusion of myoblasts. Satellite cells persist beneath the basal lamina and mediate limited regeneration.

The sarcomere, bounded by two **Z lines**:

Band	Contents	Behavior in contraction
A band	Thick filaments (**myosin**)	Constant length
I band	Thin filaments (**actin**) only	Shortens
H band	Thick filaments only, no overlap	Shortens
Z line	Anchor for actin	Moves closer
M line	Centre of the A band	—

Sliding filament mechanism: myosin heads bind actin, hydrolyse ATP, and pivot, pulling thin filaments toward the M line. **The A band never changes length.**

Control. The **sarcoplasmic reticulum** stores Ca^{2+}; **T-tubules** are invaginations of the sarcolemma that encircle each myofibril **near the aligned A- and I-band boundaries**, where each T-tubule associates with **two terminal cisternae** — a **triad**. Depolarization spreads along the T-tubules to trigger simultaneous Ca^{2+} release throughout the fiber. An action potential releases Ca^{2+}, which binds **troponin C**, moves **tropomyosin** off the myosin-binding site, and contraction begins. Relaxation requires Ca^{2+} reuptake.

Innervation. The **motor end plate** — acetylcholine binds nicotinic receptors, depolarises the sarcolemma, and triggers contraction. One motor neuron plus all the fibers it supplies is a **motor unit**.

Fiber types:

Type	Speed	Metabolism	Fatigue	Color
I (slow, red)	Slow	**Oxidative**	Resistant	Red
IIa (fast, oxidative-glycolytic)	Fast	Mixed	Intermediate	Intermediate
IIx/IIb (fast, glycolytic)	Fast	**Glycolytic**	Fast fatigue	White

Sensory receptors. Muscle spindles (intrafusal fibers, Ia afferents, gamma motor neurons) sense length and stretch; **Golgi tendon organs** sense tension.

Cardiac Muscle

- **Striated**, branched fibers, **one or two central nuclei**.

- **Intercalated discs** — the diagnostic feature — contain **fascia adherens** (anchors actin), **desmosomes** (mechanical strength), and **gap junctions** (electrical coupling), making the heart a functional syncytium.
- **Diads** at the **Z line** (one T-tubule + one cisterna) rather than **triads** of the **transverse tubule system** at the A–I junction.
- Abundant mitochondria and lipid droplets; highly oxidative; resistant to fatigue.
- **Purkinje fibers** — specialized conducting cells, larger and paler, with fewer myofibrils; they conduct rapidly through the ventricles.
- **Regeneration is essentially absent**; damage is repaired by fibrous scar.

Smooth Muscle

- **Spindle-shaped** cells with a single **central** nucleus and no striations.
- Contractile apparatus: **dense bodies** (analogous to Z lines) linked by desmin and actin.
- **No troponin.** Calcium binds **calmodulin**, which activates **myosin light-chain kinase (MLCK)**.
- **Caveolae** replace T-tubules; the SR is sparse and Ca^{2+} entry across the membrane matters more.
- **Unitary (single-unit)** smooth muscle — gap junctions, spontaneous rhythmicity, in gut, uterus, bladder. **Multiunit** — each cell separately innervated, in the iris and vas deferens.
- Capable of **hypertrophy and hyperplasia** (uterine myometrium in pregnancy).

Regeneration of Muscle Tissue

Skeletal muscle regenerates weakly, from **satellite cells**, and only if the basal lamina survives; otherwise the defect fills with fibrous tissue. **Cardiac muscle does not regenerate** — infarcted myocardium is replaced by scar. **Smooth muscle** can both hypertrophy and, in the uterus, proliferate.

Feature	Skeletal	Cardiac	Smooth
Striation	**Yes**	**Yes**	No
Nuclei	**Many, peripheral**	1–2, **central**	1, **central**
Cell shape	Long cylinder	Branched	Spindle
T-tubules	A–I junction, **triads**	Z line, **diads**	None (caveolae)
Ca^{2+} sensor	**Troponin C**	**Troponin C**	**Calmodulin**
Junctions	None between fibers	**Intercalated discs**	Gap junctions (unitary)
Control	Somatic, voluntary	Autonomic, involuntary	Autonomic, involuntary
Regeneration	Satellite cells (limited)	None	Hypertrophy/hyperplasia

CLINICAL CORRELATION

Duchenne muscular dystrophy is an X-linked defect of **dystrophin**, which links the sarcolemma to the extracellular matrix; without it, contraction tears the membrane, fibers die, and fat and fibrous tissue replace muscle (pseudohypertrophy). **Myasthenia gravis** is autoantibody blockade of the nicotinic acetylcholine receptor, producing fatigable weakness. **Cardiac hypertrophy** is an adaptation to pressure overload; beyond a point the myocytes outgrow their blood supply and fail.

UNDER THE MICROSCOPE

Look first for **striations**; if present, look at the **nuclei**. **Many nuclei at the edge = skeletal. One or two central nuclei with dark transverse bands = cardiac (intercalated discs).** No striations and central nuclei in spindle-shaped cells = smooth.

MEMORY HOOK

"Skeletal = Surface nuclei. Cardiac = Central nucleus, Cross-striations, interCalated discs. Smooth = Spindle, Single central nucleus, no Striations."

EXAM TIP
The A band does not shorten. This is asked more often than any other muscle fact: I and H shorten, A stays constant, Z lines approach.

Chapter 10 — one-line answers

- Skeletal muscle: long cylinders, **many peripheral nuclei**, triads at the **A–I junction**.
- Cardiac muscle: branched cells, **one or two central nuclei**, **intercalated discs**, diads at the **Z line**.
- Smooth muscle: spindle cells, **one central nucleus**, no striations, no T-tubules (caveolae).
- Sarcomere = Z line to Z line; **A band constant**, I band and H zone shorten.
- Thin filament = actin + troponin + tropomyosin; thick filament = myosin.
- Ca^{2+} binds **troponin C** in striated muscle and **calmodulin** in smooth muscle.
- Smooth muscle contraction uses **myosin light-chain kinase**; it has no troponin.
- Intercalated discs contain **fasciae adherentes, desmosomes, and gap junctions** — not tight junctions.
- Skeletal muscle regenerates from **satellite cells**, and only if the basal lamina survives.
- Cardiac muscle does not regenerate; infarction heals by scar.
- Type I fibers = slow, oxidative, fatigue-resistant. Type II = fast, glycolytic.
- Duchenne dystrophy = **dystrophin** defect; myasthenia gravis = antibody to the nicotinic receptor.
- Muscle spindle = stretch receptor; Golgi tendon organ = tension receptor.

PART III — TRANSPORT, DEFENSE, AND BLOOD

Chapters 11–14 · circulation, blood, hemopoiesis, and immunity

11 · The Circulatory System

BOOK PAGES 215–236 | Part 2, pages 120–155 · Heart · Tissues of the Vascular Wall · Vasculature · Lymphatic Vascular System

The core idea

The circulatory system pumps and directs blood cells and substances to all tissues, and in an adult the total length of its vessels is estimated at **100,000–150,000 km**. Every vessel in the body is a variation on one three-layered tube. What changes between an aorta and a capillary is **how much muscle and elastic tissue each layer contains** — decided entirely by the pressure the vessel must withstand and the job it must do.

Heart

- **Endocardium** — endothelium + subendothelial connective tissue + a layer of Purkinje fibers; continuous with the intima of vessels.
- **Myocardium** — the thickest layer; cardiac muscle in complex spirals, atria thinner than ventricles, left ventricle thickest.
- **Epicardium** — connective tissue containing many **adipocytes**, covered by mesothelium; the visceral layer of the pericardium.
- **Cardiac skeleton** — dense irregular connective tissue rings that anchor the valves and electrically insulate atria from ventricles.
- **Valves** — endocardium folded over a core of dense connective tissue; **avascular**. The mitral and tricuspid have chordae tendineae and papillary muscles.

Tissues of the Vascular Wall

Layer	Contents	Function
Intima	Endothelium, basal lamina, subendothelial connective tissue, **internal elastic lamina** (in arteries)	Non-thrombogenic surface, exchange, barrier
Media	**Smooth muscle** circularly arranged, elastic lamellae (in elastic arteries), **external elastic lamina**	Tone and compliance
Adventitia	Loose connective tissue, **vasa vasorum**, **nervi vascularis**	Anchorage and supply of the wall

The rule of the wall: the thicker the media, the higher the pressure the vessel handles. Arteries have a thick media; veins a thin media and a thick adventitia. That single ratio lets you identify a vessel on any slide.

Vasculature

Vessel	Intima	Media	Adventitia	Distinguishing feature
Elastic artery (aorta, pulmonary, common carotid)	Endothelium + subendothelial	**Many elastic lamellae** alternating with smooth muscle	Thin, with vasa vasorum	**Elastic lamellae**
Muscular artery	Prominent **internal elastic lamina**	Thick, 3–40 layers of smooth muscle	Moderate	**Well-defined IEL**
Arteriole	Endothelium + thin subendothelium	1–2 layers of smooth muscle	Minimal	**The resistance vessel**
Capillary	Endothelium + basal lamina only	None	None	One cell thick

Vessel	Intima	Media	Adventitia	Distinguishing feature
Venule	Endothelium	None or sparse pericytes	Thin	Site of **leukocyte diapedesis**
Small/medium vein	Endothelium	Thin	**Thickest layer**	Often collapsed on section
Large vein (vena cava)	Endothelium	Thin	**Thick, with longitudinal smooth muscle**	Valve-like intimal folds

Capillaries:

Type	Structure	Location	Function
Continuous	Uninterrupted endothelium, tight junctions	Muscle, skin, lung, CNS	Regulated exchange; the **blood–brain barrier** is this type plus tight junctions and astrocyte feet
Fenestrated	Pores with diaphragms	Kidney glomerulus, intestine, endocrine glands	Rapid fluid and solute exchange
Sinusoidal	Discontinuous, large gaps, incomplete basal lamina	**Liver, spleen, bone marrow**	Free passage of cells and large molecules

Microcirculation. Arteriole → metarteriole → capillary bed → venule, with **precapillary sphincters** regulating flow. **Arteriovenous anastomoses** shunt blood directly, important in skin thermoregulation.

Lymphatic Vascular System

Begins as blind-ended **lymphatic capillaries** with overlapping endothelial flaps and **anchoring filaments**, which open when interstitial pressure rises. They converge into collecting vessels with valves, then nodes, trunks, and the **thoracic duct** and **right lymphatic duct**. Lymphatics have **no continuous basal lamina** and no fenestrations — which is exactly why tumor cells and bacteria enter them so easily.

CLINICAL CORRELATION

Atherosclerosis begins with endothelial dysfunction: LDL crosses the endothelium, is oxidized, and provokes monocyte recruitment. Macrophages become **foam cells**, a fatty streak forms, smooth muscle migrates into the intima, and a fibrous cap covers a necrotic core. Rupture of a thin cap exposes thrombogenic material and produces the acute coronary syndrome. **Aneurysm** follows destruction of medial elastic fibers. **Varicose veins** follow valve failure. **Lymphedema** follows lymphatic obstruction — classically after axillary node clearance.

UNDER THE MICROSCOPE

Arteries: **round lumen, thick wall, corrugated internal elastic lamina**. Veins: **irregular collapsed lumen, thin media**. In elastic arteries the media shows wavy pink bands with no orderly muscle coat; in muscular arteries the media is uniform smooth muscle.

EXAM TIP

The three capillaries are distinguished by **continuity, not size**: continuous (tight), fenestrated (pores), sinusoidal (gaps wide enough for a red cell). Liver, spleen, marrow = sinusoidal. Kidney glomerulus and intestine = fenestrated. Brain = continuous with tight junctions.

12 · Blood

BOOK PAGES 237–253 | Part 2, pages 156–180 · Composition of Plasma · Blood Cells

The core idea

Blood is connective tissue with a liquid matrix. Its cells all derive from one stem cell, and their morphology is a direct readout of their job: **red for oxygen, granular white for killing, agranular white for immunity, and fragments for clotting.**

Composition of Plasma

About 55% of blood volume. Water plus plasma proteins — **albumin, α- and β-globulins, complement proteins, and fibrinogen**, all secreted in the **liver**, plus the **immunoglobulins** — and electrolytes, nutrients, hormones, and waste. On clotting, some proteins are consumed and others are released from platelets, leaving **serum**, whose biological properties differ from plasma. **Serum is plasma minus fibrinogen** (i.e., after clotting).

Blood Cells

Erythrocytes

- **Biconcave disc**, **7.5 µm** in diameter, **2.6 µm** thick at the rim but only **0.75 µm** at the center — flexible enough to pass small capillaries; **no nucleus, no organelles**, so the cell is packed with hemoglobin.
- Lifespan about **120 days**; removed by splenic macrophages; the globin is recycled and the hem converted to bilirubin.
- Shape maintained by **spectrin and ankyrin**; defects produce hereditary spherocytosis.
- **Hemoglobin** — four globin chains, each with a hem group carrying Fe^{2+}. Adult HbA ($\alpha_2\beta_2$) and fetal HbF ($\alpha_2\gamma_2$, higher affinity for O_2).
- **Reticulocytes** are young red cells with residual RNA, appearing polychromatic; a raised count signals active marrow output.
- **Hematocrit** — the sedimented erythrocyte column — is about **44–45% of total blood volume** in healthy adults; the thin layer between cells and plasma is the **buffy coat** of leukocytes and platelets. An average adult has **5–6 L** of blood.

The leukocytes — broadly grouped as **granulocytes** (neutrophils, eosinophils, basophils) and **agranulocytes** (lymphocytes, monocytes)

Cell	% of WBC	Size	Nucleus	Cytoplasm	Main function
Neutrophil	**60–70%**	10–14 µm	**3–5 lobes**, dense	Pale; fine granules	**Phagocytosis of bacteria**; the acute responder
Eosinophil	**2–4%**	10–14 µm	**Bilobed**	**Bright orange-red** granules	**Parasites and allergy**; major basic protein
Basophil	**0.5–1%**	10–14 µm	**S-shaped**, obscured	**Large dark blue** granules	**Histamine and heparin**
Lymphocyte	**20–30%**	7–12 µm	Round, dense	Pale rim	**T cells**, **B cells**, **NK cells**
Monocyte	**3–8%**	**14–20 µm** (largest)	**Kidney-shaped**	Abundant blue-gray	Become **macrophages**

Neutrophil granules: azurophilic (primary) granules contain myeloperoxidase and defensins; **specific (secondary)** granules contain lysozyme and lactoferrin. The neutrophil dies after phagocytosis; the accumulation of dead neutrophils and debris is **pus**. Each **phagocyte** kills ingested

bacteria inside a phagolysosome.

Platelets are 2–3 µm anucleate fragments of **megakaryocyte** cytoplasm, with a peripheral clear zone (hyalomere) and a central granular zone (granulomere). They adhere, activate, and aggregate to form the primary hemostatic plug; **von Willebrand factor** links them to collagen.

CLINICAL CORRELATION

The differential count is a diagnostic instrument. **Neutrophilia** suggests bacterial infection or steroid use; **eosinophilia** parasites or allergy; **lymphocytosis** viral infection or chronic lymphocytic leukemia; **pancytopenia** marrow failure. In **sickle cell disease** a single glutamic acid → valine substitution in β-globin allows deoxygenated HbS to polymerize; the cell sickles, occludes small vessels, and is removed by the spleen.

UNDER THE MICROSCOPE

Count lobes, not granules. **Neutrophil: 3–5 lobes. Eosinophil: exactly two, orange granules. Basophil: granules so dark the nucleus is hidden. Monocyte: kidney-shaped nucleus, largest cell. Lymphocyte: round dense nucleus, almost no cytoplasm.**

MEMORY HOOK

"Never Let Monkeys Eat Bananas" = Neutrophil, **L**ymphocyte, **M**onocyte, **E**osinophil, **B**asophil — in descending order of abundance.

EXAM TIP

Eosinophil granules stain with eosin because they are **basic proteins** — the naming is the reverse of what students expect. The basophil's granules contain **heparin and histamine**, so the "basic" cell is the one full of allergy mediators.

13 · Hemopoiesis

BOOK PAGES 254–266 | Part 2, pages 181–198 · Stem Cells, Growth Factors, & Differentiation · Bone Marrow · Maturation of Erythrocytes · Maturation of Granulocytes · Maturation of Agranulocytes · Origin of Platelets

The core idea

All blood cells descend from one **hematopoietic stem cell**, and the pathway from that cell to a mature erythrocyte, granulocyte, or platelet is a sequence of named stages you can recognize by **decreasing nucleus, increasing cytoplasm, and changing color**. It is the most examinable lineage in histology because every stage has a name.

Sites of hemopoiesis

Period	Site
First trimester	Yolk sac (mesoblastic phase)
Second trimester	**Liver and spleen** (hepatic phase)
Third trimester	**Bone marrow** — marrow becomes the major organ in the third trimester as skeletal elements ossify

In the adult, active **red marrow** persists in the **sternum, ribs, vertebrae, pelvis, skull, and proximal femur**. The rest is **yellow (fatty) marrow**, which can revert to red marrow under demand — why chronic anemia produces marrow expansion and frontal bossing.

Stem Cells, Growth Factors, & Differentiation

The **hematopoietic stem cell** is CD34⁺, self-renewing, and pluripotent, sitting in a **niche** of stromal cells, other cells, and ECM. Committed **progenitor cells** — also called **CFUs** (colony-forming units) — proliferate and differentiate under growth factors, which are also called **CSFs** (colony-stimulating factors) or cytokines. Differentiation is driven by **growth factors**, examinable by name:

Factor	Target
Erythropoietin (EPO)	Erythroid progenitors (kidney, some liver)
Thrombopoietin (TPO)	Megakaryocytes (liver)
G-CSF	Granulocyte production
GM-CSF, M-CSF	Granulocyte–monocyte and monocyte lineages
IL-3	Early multipotent progenitors
IL-5	Eosinophils
SCF (stem cell factor)	Stem cell maintenance

Bone Marrow

Between the trabeculae lie **hemopoietic cords** and **sinusoids** with a discontinuous endothelium, so mature cells can enter the circulation. The **myeloid:erythroid (M:E) ratio** is normally **2:1 to 4:1**; it shifts toward erythroid in hemolytic anemia and toward myeloid in infection or leukemia.

Maturation of Erythrocytes

Proerythroblast → basophilic → polychromatophilic → orthochromatophilic erythroblast → reticulocyte → erythrocyte, developing in **erythropoietic islands or cords**. The names track the cytoplasm's transition from **RNA-rich to hemoglobin-filled**; at the last stage the nucleus is **extruded**, producing a **reticulocyte** that still contains some polyribosomes when released into the circulation.

The cell **shrinks**, the nucleus **condenses and is extruded**, hemoglobin accumulates so the cytoplasm shifts from blue to pink, and the nucleolus disappears. The sequence takes about **5–7 days**. EPO drives it; iron, vitamin B_{12}, and folate are required.

Maturation of Granulocytes

Myeloblast → promyelocyte → myelocyte → metamyelocyte → band cell → segmented neutrophil (or eosinophil/basophil).

The suffixes carry the meaning: **-blast** = dividing precursor (**myeloblast**, **lymphoblast**, **proerythroblast**), **-cyte** = the later, non-dividing cell (**myelocyte**, **lymphocyte**, **reticulocyte**); **thrombocyte** is another name for the platelet.

Primary (azurophilic) granules appear at the promyelocyte stage, **specific granules** at the myelocyte stage, the nucleus indents at the metamyelocyte stage, and becomes a **C-shaped band** before segmenting. A **left shift** — more bands and younger forms in the peripheral blood — indicates accelerated production: immature neutrophilic metamyelocytes called **band (stab) cells** are released prematurely when the circulating neutrophil compartment is depleted by bacterial infection.

Maturation of Agranulocytes

Lymphopoiesis begins in the marrow for B cells and in the thymus for T cells. **Monopoiesis** produces monocytes that become tissue macrophages. Both pass through stages in which the nucleus becomes less lobated, the cytoplasm more abundant, and the chromatin pattern characteristic of the mature cell.

Origin of Platelets

Megakaryoblasts → megakaryocytes, which become polyploid and develop **demarcation membranes** that delineate future platelet territories; platelets bud from the cytoplasm under TPO control.

CLINICAL CORRELATION

Aplastic anemia is marrow failure with an empty, fatty marrow and pancytopenia. **Leukemia** is the opposite: marrow packed with malignant blasts, crowding out normal lineages and producing anemia, infection, and bleeding. **Megaloblastic anemia** (B_{12} or folate deficiency) shows nuclear–cytoplasmic asynchrony — the nucleus matures more slowly than the cytoplasm, so cells are large and hypersegmented. **Iron deficiency** produces microcytic hypochromic cells because hemoglobin synthesis lags behind cell division.

UNDER THE MICROSCOPE

Identify the lineage by **nucleus-to-cytoplasm ratio and granule color**. Early cells have a large nucleus, a thin rim of basophilic cytoplasm, and a nucleolus. As cells mature, the ratio falls, granules appear, and the nucleus condenses. **Maturation moves in one direction: nucleus shrinks, cytoplasm grows, color shifts from blue to pink or from blue to granular.**

EXAM TIP

Two facts carry disproportionate marks: **EPO comes from the kidney** (peritubular interstitial cells), and **TPO comes mostly from the liver**. Students routinely invert these.

14 · The Immune System & Lymphoid Organs

BOOK PAGES 267–294 | Part 2, pages 199–217 + part 3, pages 1–22 · Innate & Adaptive Immunity · Cytokines · Antigens & Antibodies · Antigen Presentation · Cells of Adaptive Immunity · Thymus · Mucosa-Associated Lymphoid Tissue · Lymph Nodes · Spleen

The core idea

The immune system is a set of organs arranged to **bring antigen to lymphocytes**. Each organ is a different solution to the same problem: how to filter a fluid (blood, lymph, or air/food), trap antigen, and present it to the right cell.

Innate & Adaptive Immunity

Feature	Innate	Adaptive
Timing	**Present from birth**, acts within minutes	**Develops more slowly**, improves with exposure
Specificity	Broad molecular patterns	**Exact antigen**
Memory	None	**Yes**
Cells	**Granulocytes** (mainly neutrophils), macrophages, dendritic cells, NK cells	**T and B lymphocytes**
Mediators	**Defensins, complement, lysozyme, interferons**, cytokines	Antibodies, cytotoxic granules

The book's one-line definition is worth memorizing: innate immunity is present from birth and uses leukocytes (mainly granulocytes) plus soluble proteins; adaptive immunity develops more slowly and is **based on antigen presentation to lymphocytes**.

Cytokines

Polypeptide hormones of the immune system — the book's phrase — secreted by immune cells to regulate one another. The ones worth naming: **interleukins** (IL-1, IL-2 for T-cell proliferation, IL-4 for IgE and Th2 responses, IL-5 for eosinophils, IL-6 for acute-phase response), **interferons** (antiviral; IFN-γ activates macrophages), **TNF-α** (inflammation, fever, cachexia), **chemokines** (directed leukocyte migration), and **colony-stimulating factors** (hemopoiesis, Chapter 13). Cytokines act locally (paracrine) or on the secreting cell (autocrine), and their Storm-like excess produces the systemic inflammatory response.

Antigens & Antibodies

Antigens are molecules recognized by the adaptive immune system; **epitopes** are the specific parts bound by a receptor. **Antibodies** are immunoglobulins:

Class	Structure	Main role
IgG	Monomer	Most abundant; crosses the placenta; opsonizes
IgM	Pentamer	First responder; complement activation
IgA	Dimer with secretory component	Mucosal secretions
IgE	Monomer	Mast-cell binding; allergy and parasites
IgD	Monomer	B-cell receptor

Antigen Presentation

T cells cannot see free antigen. They recognize peptide displayed on **MHC** molecules:

- **MHC class I** — on the surface of **all nucleated cells**, bearing fragments of that cell's own proteins (and viral proteins if infected); recognized by **CD8⁺ cytotoxic T cells**.
- **MHC class II** — only on **antigen-presenting cells (APCs)**, mostly derived from monocytes; presents fragments of **endocytosed foreign** proteins to **CD4⁺ helper T cells**. This is the defining difference: MHC I says "here is what I am making"; MHC II says "here is what I have eaten".

This pairing — **CD4/MHC II, CD8/MHC I** — is the most examined immunology fact in the chapter.

Cells of Adaptive Immunity

T lymphocytes mature in the thymus and are responsible for cell-mediated immunity. Their **T-cell receptor (TCR)** is usually an **αβ heterodimer**; each T cell also carries one CD marker that fixes its class:

T-cell class	Marker	Job
Helper T cell	**CD4⁺**	Recognizes MHC II; secretes cytokines that activate B cells and macrophages
Cytotoxic T cell	**CD8⁺**	Recognizes MHC I; kills infected or tumor cells
Regulatory T cell	**CD4⁺ CD25⁺**	Suppresses immune responses and prevents autoimmunity
γδ T cell	γδ TCR	Mainly in **epithelia**; early defense at body surfaces

B lymphocytes mature in bone marrow and carry a **B-cell receptor (BCR)** — surface **IgM or IgD** — that binds antigen directly, without MHC. On activation they become **plasma cells** (antibody factories) and **memory cells**. **NK cells** kill without prior sensitization and without MHC restriction.

How a Lymphoid Nodule Forms

This sequence explains every secondary lymphoid organ in the chapter:

1. Lymphocytes are packed in a meshwork of **reticulin** made by **fibroblastic reticular cells**; most APCs among them are **dendritic cells** with many processes.
2. In a secondary organ, **follicular dendritic cells (FDCs)** display antigen–antibody complexes on their large surface. A B cell whose BCR binds that antigen is held there.
3. With cytokines from helper T cells, that B cell **proliferates clonally**, producing a large transient nodule with a pale **germinal center**.
4. The cells disperse as **plasma cells, various T cells, and B and T memory cells** — which is why the same nodule can look "active" (large germinal center) or "resting" (dense primary follicle).

Thymus

- **Capsule** with trabeculae dividing it into lobules, each with a **cortex** and a **medulla**.
- **No follicles.** The cortex is densely packed with developing **thymocytes** and epithelial reticular cells; the medulla is paler and contains **Hassall corpuscles** — concentric whorls of epithelial cells, unique to the thymus.
- **Blood–thymus barrier** — formed by **squamous thymic epithelial cells** joined by desmosomes and occluding junctions, together with vascular endothelium and pericytes — prevents unregulated exposure of thymocytes to antigens.
- **Cortex** is packed with **T lymphoblasts (thymocytes)**, macrophages, and **thymic epithelial cells (TECs)**. Stellate TECs form a **cytoreticulum** to which lymphocytes attach instead of reticulin fibers; these TECs are **antigen-presenting cells** expressing **MHC class II** and secreting cytokines — which is why the thymus is also counted among endocrine glands.
- **Hassall corpuscles** — aggregates of medullary TECs, concentrically arranged, **up to 100 µm**, **unique to the medulla**; their cells secrete cytokines that control local dendritic cells and promote regulatory T cells for peripheral tolerance.

- Function: **T-cell maturation and selection**. **Positive selection occurs in the cortex**: only thymocytes whose TCR recognizes the individual's own MHC molecules survive. **Negative selection occurs in the medulla**: thymocytes that bind self-antigen too tightly are deleted, leaving tolerance. Medullary TECs express **Aire**, which lets them display a wide sample of tissue self-antigens for this test.
- **Involutes** after puberty, replaced by adipose tissue.

Mucosa-Associated Lymphoid Tissue

Tonsils (palatine, lingual, pharyngeal/adenoid, tubal) — no capsule, no afferent lymphatics, epithelium invaginates into crypts. **Peyer's patches** in the ileum; **appendix**; **MALT** throughout the gut, respiratory tract, and genitourinary tract.

Lymph Nodes

Region	Contents	Function
Cortex	**Lymphoid follicles** — primary (dense) or secondary (with pale **germinal center** and dark mantle)	**B-cell** zone
Paracortex	Diffuse lymphoid tissue, **high endothelial venules (HEV)**	**T-cell** zone; lymphocytes enter from blood here
Medulla	Cords and sinuses, macrophages and plasma cells	Filtration; antibody secretion

Flow: afferent lymphatics → subcapsular sinus → trabecular sinuses → medullary sinuses → **efferent lymphatics at the hilum**. The one-way flow explains why nodal metastases lodge first in the subcapsular sinus.

Spleen

- **No afferent lymphatics.** It filters **blood**.
- **White pulp** — lymphocytes around a central arteriole: the **periarteriolar lymphoid sheath (PALS)** is the T-cell zone; follicles at its periphery are the B-cell zone.
- **Red pulp** — **cords of Billroth** and **sinusoids** whose elongated endothelial **stave cells** are separated by narrow slits.
- **Two routes through the red pulp.** In the **closed circulation**, capillaries from the penicillar arterioles connect directly to sinusoids. In the **open circulation** (about half of the penicillar arterioles), capillaries are open-ended and dump blood into the splenic cords, so plasma and cells must squeeze between stave cells to re-enter the vasculature. Platelets, motile leukocytes, and thin flexible erythrocytes pass; **stiff or swollen erythrocytes at the end of their 120-day life are blocked and removed by macrophages**.
- **Iron recycling.** Iron released from **hemoglobin** during erythrocyte degradation is stored by macrophages in **ferritin**, released bound to **transferrin**, and reused for erythropoiesis. Iron-free heme binds **hemopexin** or is converted to **bilirubin** and excreted in bile.
- **Functions:** blood filtration, removal of old erythrocytes, immune response to blood-borne antigen, and **extramedullary hemopoiesis** in fetal life and marrow failure.

Organ	Capsule	Afferent lymphatics	Follicles	Unique feature
Thymus	Yes	None	**No**	**Hassall corpuscles**
Lymph node	Yes	**Yes**	Yes	Cortex/paracortex/medulla; HEV
Spleen	Yes	**None**	Yes	**White and red pulp**
Tonsil	No	None	Yes	Crypts

CLINICAL CORRELATION

DiGeorge syndrome is thymic aplasia (third and fourth pharyngeal pouches) with T-cell deficiency and hypocalcaemia. **Severe combined immunodeficiency** is failure of both arms. **Hodgkin lymphoma** spreads contiguously through nodes and shows Reed–Sternberg cells. **Splenectomy** increases susceptibility to encapsulated organisms — pneumococcus, meningococcus, Haemophilus influenzae — because the spleen's marginal zone is the first line against polysaccharide capsules.

UNDER THE MICROSCOPE

Ask **"what is at the center of the nodule?"** Thymus: lobules with dark cortex and pale medulla plus pink whorls (Hassall corpuscles). Lymph node: capsule, subcapsular sinus, hilum. Spleen: **white pulp islands inside a red sea** of sinusoids and cords.

MEMORY HOOK

"Thymus Teaches T cells; Nodes Filter lymph; Spleen Filters blood." And **"CD4 sees MHC II (2 × 2); CD8 sees MHC I (8 × 1)."**

EXAM TIP

Two negative facts are worth more than most positive ones: the **thymus has no follicles**, and the **spleen has no afferent lymphatics**.

Chapter 14 — one-line answers

- Innate immunity = present at birth, granulocytes and soluble proteins; adaptive = antigen presentation to lymphocytes.
- MHC I is on **all nucleated cells** and presents endogenous peptide to **$CD8^+$** cells.
- MHC II is on **APCs** (mostly monocyte-derived) and presents endocytosed peptide to **$CD4^+$** cells.
- BCR = surface **IgM or IgD**; TCR is usually **αβ** with a CD co-receptor.
- T cells: CD4 helper, CD8 cytotoxic, $CD4^+CD25^+$ regulatory, γδ in epithelia.
- B cells make antibodies (humoral); T cells mediate cell-mediated immunity.
- Plasma cells secrete antibody; memory B and T cells give the rapid second response.
- Primary lymphoid organs: **bone marrow (B)** and **thymus (T)**.
- Secondary: lymph nodes, MALT, spleen — where lymphocytes are activated.
- FDCs display antigen–antibody complexes; activated B cells form germinal centers.
- Thymus: cortex = positive selection, medulla = negative selection; **Aire** displays self-antigens.
- Thymus has **no follicles** and Hassall corpuscles in the medulla.
- Lymph node: cortex B cells, paracortex T cells and HEVs, medulla cords and sinuses.
- Spleen: white pulp (PALS = T zone, follicles = B zone) and red pulp (cords + sinusoids).
- Spleen filters blood; **open circulation** lets macrophages remove stiff 120-day erythrocytes.
- Iron is stored as **ferritin**, carried by **transferrin**, and heme iron binds **hemopexin**; the rest becomes bilirubin.

PART IV — DIGESTION AND RESPIRATION

Chapters 15–18 · gut, liver, pancreas, lung, and skin

15 · Digestive Tract

BOOK PAGES 295–328 | Part 3, pages 23–70 · General Structure of the Digestive Tract · Oral Cavity · Esophagus · Stomach · Small Intestine · Large Intestine

The core idea

The gut is one tube with four concentric layers, and every region is a modification of those four layers for a specific job: **grinding, acidifying, absorbing, or dehydrating**. Learn the four layers and their regional variations and you can describe any segment from esophagus to anus.

General Structure of the Digestive Tract

Layer	Contents	Function
Mucosa	Epithelium + **lamina propria** + **muscularis mucosae**	Digestion, absorption, secretion, barrier
Submucosa	Loose connective tissue, vessels, lymphatics, **Meissner (submucosal) plexus**, glands where present	Supply and control of secretion
Muscularis	Inner circular + outer longitudinal smooth muscle, **Auerbach (myenteric) plexus** between them	Peristalsis
Serosa or adventitia	Mesothelium + connective tissue (intraperitoneal) or connective tissue only (retroperitoneal)	Suspension and protection

This is the skeleton of every answer about the gut. Two rules make it exam-proof:

- The **mucosa always has three components** — a lining epithelium, a lamina propria of loose connective tissue, and smooth muscle from the **muscularis mucosae**. The epithelium changes; the other two do not.
- **Serosa = suspended in the abdomen; adventitia = embedded in surrounding tissue.** The esophagus develops within the mediastinum and therefore has an adventitia along almost its whole length.

Both plexuses belong to the **enteric nervous system**: Meissner's controls secretion and blood flow, Auerbach's controls motility. The lymphocytes and nodules scattered through mucosa and submucosa are **MALT** (Chapter 14).

Oral Cavity

The lining is **stratified squamous** everywhere, but the degree of keratinization tracks wear: **keratinized on the hard palate and gingiva**, partly keratinized on the lips, and **nonkeratinized** over most of the soft oral mucosa.

Lips — three zones: cutaneous (skin with hair and sebaceous glands), **vermilion** (thin keratinized epithelium over vascular connective tissue — the red of the lip), and mucosal (nonkeratinized, wet). A core of **striated muscle** makes the lips mobile.

Tongue — skeletal muscle in bundles, covered by stratified squamous epithelium with four papilla types:

Papilla	Shape	Taste buds	Notes
Filiform	Cone-shaped, **heavily keratinized**	**No**	Most numerous; mechanical; gives the whitish dorsal appearance
Fungiform	Mushroom-shaped, lightly keratinized	Yes	Scattered among filiform
Foliate	Leaf-like folds	Yes	Lateral margins

Papilla	Shape	Taste buds	Notes
Vallate (circumvallate)	Largest, **1–3 mm** wide, in a trench	**Yes, ~250 per papilla**	**8–12** in a V in front of the terminal sulcus

Taste buds are ovoid intraepithelial organs of **50–100 cells**, about half of them elongated **gustatory cells** with a **7–10-day** turnover. Their microvilli face a **2-µm taste pore**; support cells and basal stem cells complete the bud. **Filiform papillae never carry taste buds** — a standard trap.

Teeth. An adult has **32 permanent teeth**. Each has a crown, a constricted neck, and roots in dental alveoli.

- **Enamel** — hardest material in the body: **96% calcium hydroxyapatite**, only **2–3% organic**, **no collagen**. It is built of **enamel rods ~5 µm wide** running the full **~2 mm** thickness. **Ameloblasts** secrete a collagen-free matrix rich in **amelogenin (20 kDa)**, which initiates and guides crystal growth. Enamel is **acellular**; ameloblasts disappear at eruption, so enamel cannot regenerate, and it is lost from decalcified sections.
- **Dentin** — **70% hydroxyapatite** in a **type I collagen** matrix, harder than bone. **Odontoblasts** (cranial neural crest) secrete **predentin**, which mineralizes within about a day. Odontoblasts persist for life, and their processes occupy **dentinal tubules** across the full thickness of the dentin, accompanied by unmyelinated nerves — the basis of dentin sensitivity.
- **Cementum** — bone-like covering of the roots. **Pulp** — vascular, innervated loose connective tissue entering through apical foramina.
- **Periodontium = cementum + periodontal ligament + alveolar bone + gingiva**. The ligament's collagen bundles (**Sharpey fibers**) bind cementum to bone; it is **150–350 µm** thick.

Esophagus

A muscular tube about **25 cm** long. **Nonkeratinized stratified squamous** epithelium, a thick muscularis mucosae, **esophageal glands in the submucosa** (mucus-secreting, lubricating), and small **cardiac glands** in the lamina propria near the junction.

The muscularis is the examinable progression: **skeletal muscle at the superior end → mixed in the middle → exclusively smooth in the lower third**. The outer layer is **adventitia** except for the distal **1–2 cm**, which lies in the peritoneal cavity and has a serosa. At the **esophagogastric junction** the epithelium changes abruptly from stratified squamous to simple columnar, and the adventitia gives way to serosa.

Stomach

Four regions: the superior **cardia** and inferior **pylorus** resemble each other histologically (branching mucous glands, few parietal or chief cells), while the intervening **fundus and body** contain the main gastric glands. The surface is perforated by **gastric pits**, which are lined by the same **surface mucous cells** as the stomach lumen and lead into branching glands. **No goblet cells, no villi.**

The gland epithelium contains **five functional cell types plus pluripotent stem cells**:

Cell	Location	Product	Function
Parietal (oxyntic)	Upper half of gland	**HCl + intrinsic factor**	Large cell, many mitochondria, **intracellular canaliculi**; intrinsic factor is required for **vitamin B_{12}** uptake; **eosinophilic**
Chief (zymogenic)	Deeper half of gland	**Pepsinogen**	Activated to **pepsin** by luminal acid; **basophilic** with apical zymogen granules
Mucous neck	Neck	Less alkaline mucus	Immature precursors that migrate up into the pits
Surface mucous	Surface and pits	**Viscous mucus + bicarbonate**	The barrier that protects the mucosa

Cell	Location	Product	Function
Enteroendocrine	Throughout	**Gastrin, histamine**, others	Paracrine and endocrine control of secretion
Stem cells	Narrow neck region	—	Pluripotent source of all the above

Parietal cells are controlled both by **parasympathetic innervation** and by paracrine **histamine** and **gastrin** from enteroendocrine cells — the pharmacological basis of proton-pump inhibitors and H_2 blockers. The **muscularis has three layers** (oblique, circular, longitudinal), an exception to the four-layer template that allows churning.

Small Intestine

The absorptive surface is amplified three times: **plicae circulares** (mucosa + submucosa), **villi** (mucosa), and **microvilli** (cell surface) — the clearest "structure follows function" example in the book.

Each **villus** is a finger of lamina propria covered by simple columnar epithelium, containing a **lacteal** and smooth muscle. Between villi, simple tubular **crypts (of Lieberkühn)** are the proliferative compartment. Smooth muscle of the lamina propria and muscularis mucosae, controlled by **Meissner's plexus**, moves the villi and propels lymph; the muscularis, controlled by **Auerbach's plexus**, produces peristalsis.

Cell types and what each absorbs or secretes:

- **Enterocytes** — brush border; **sugars and amino acids** produced by final digestion in the glycocalyx are taken up by **transcytosis** and pass to capillaries.
- **Goblet cells** — mucus.
- **Paneth cells** — base of crypts, **eosinophilic** granules, secrete **defensins** (and lysozyme).
- **Enteroendocrine cells** — CCK, secretin, GIP, motilin.
- **Stem cells** — crypt base, replacing the epithelium every few days.

Lipid absorption is different and examinable: lipid products associate with **bile salts**, enter enterocytes, are re-esterified to **triglycerides** and packaged as **chylomicrons**, and leave through the **lacteal** rather than the capillary.

Region	Defining feature
Duodenum	**Duodenal (Brunner) glands in the submucosa**; leaf-like villi
Jejunum	Tallest villi, no Brunner glands, no Peyer's patches
Ileum	**Peyer's patches** (mucosal and submucosal); shortest villi

Large Intestine

Three regions: the short **cecum** with the **appendix**; the long **colon** (ascending, transverse, descending, **sigmoid**); and the **rectum**.

No villi, straight tubular crypts lined by **colonocytes** (absorptive, water and electrolyte uptake) and **abundant goblet cells**. The outer longitudinal muscle is gathered into three **teniae coli**, which produce the bulging **haustra**; the serosa carries hanging fat masses, the **omental appendages** (appendices epiploicae). Function: water and electrolyte absorption, mucus secretion, bacterial fermentation. The **appendix** is a blind diverticulum of the cecum with abundant lymphoid tissue in its lamina propria and submucosa — part of MALT.

Anal canal — the mucosa is highly vascular, folded into **anal columns (of Morgagni)** with intervening anal sinuses; it ends as **stratified squamous** epithelium. Swollen mucosal or submucosal veins here are **hemorrhoids**.

CLINICAL CORRELATION

Peptic ulcer — H. pylori breaches the mucous barrier; acid and pepsin digest the wall, classically in the duodenal bulb or gastric antrum. **Pernicious anemia** — loss of parietal cells means no intrinsic factor, so B_{12} cannot be absorbed in the terminal ileum. **Celiac disease** — villous atrophy, crypt hyperplasia, and intraepithelial lymphocytes in the proximal small intestine, driven by gluten. **Barrett esophagus** — chronic acid reflux causes metaplasia from stratified squamous to **intestinal-type columnar epithelium with goblet cells**, a premalignant change. **Diverticulosis** — mucosa and submucosa herniate between the teniae coli; trapped feces cause diverticulitis.

UNDER THE MICROSCOPE

Identify the segment by **the surface, not the glands**. **Villi + crypts = small intestine. No villi + deep pits + thick gland mass = stomach. No villi + straight crypts + many goblet cells = colon.** Then find the regional marker: Brunner glands = duodenum; Peyer's patches = ileum; three muscle layers = stomach.

MEMORY HOOK

"Duodenum has Brunner Below; Ileum has Peyer's Patches." And **"Filiform = Feel only; Vallate = Carry most taste."**

HIGH-YIELD PEARL

The numbers that appear in questions: enamel **96%** mineral and **2 mm** thick; dentin **70%**; esophageal length **25 cm**; **8–12** vallate papillae with about **250** taste buds each. The stomach has **no goblet cells** and **no villi**; the intestine has **both**.

Chapter 15 — one-line answers

- Four gut layers, outside in: mucosa, submucosa, muscularis, serosa or adventitia.
- Mucosa always = epithelium + lamina propria + muscularis mucosae.
- Serosa where the gut is suspended; adventitia where it is embedded.
- Auerbach lies between the muscle layers (motility); Meissner lies in the submucosa (secretion, blood flow).
- Keratinized oral epithelium: hard palate and gingiva. Everywhere else, mostly nonkeratinized.
- Filiform papillae are the only type without taste buds.
- Vallate papillae: 8–12, largest, ~250 taste buds each.
- Enamel is acellular and cannot regenerate; ameloblasts disappear at eruption.
- Dentin is 70% hydroxyapatite in type I collagen; odontoblast processes stay in dentinal tubules.
- Esophageal muscle: skeletal above, mixed in the middle, smooth below.
- Esophagogastric junction: squamous → columnar, and adventitia → serosa.
- Parietal cells: HCl + intrinsic factor, intracellular canaliculi. Chief cells: pepsinogen.
- Stomach has three muscle layers and no goblet cells; the colon has teniae coli and no villi.
- Duodenum = Brunner glands; ileum = Peyer's patches; colon = teniae coli.
- Lipids leave the villus as chylomicrons in the lacteal; sugars and amino acids enter capillaries.

16 · Organs Associated with the Digestive Tract

BOOK PAGES 329–348 | Part 3, pages 71–99 · Salivary Glands · Pancreas · Liver · Biliary Tract & Gallbladder

The core idea

Four gland systems share one duct system and one embryological origin: the salivary glands **lubricate and begin digestion**, the pancreas **digests**, the liver **processes**, and the gallbladder **stores**. Each is built around its job, and each has one histologic signature that identifies it.

Salivary Glands

Three major pairs produce about **90%** of saliva; minor glands scattered through the oral mucosa produce the other **10%**. Total output is **0.75–1.5 L/day**.

Gland	Secretory units	Secretion
Parotid	**Serous acini only**	Watery, enzyme-rich
Submandibular	Mixed, **mainly serous** acini	Mixed
Sublingual	Mixed, **mainly mucous tubules**, some with **serous demilunes**	Viscous

Serous cells are protein-secreting and form round or oval acini; **mucous cells** secrete mucin and form elongated tubules. **Myoepithelial cells** sit between the secretory cells and the basal lamina and squeeze the secretion onward.

The duct system is a three-step sequence, and each step has a different epithelium:

1. **Intercalated ducts** — simple cuboidal; the smallest.
2. **Striated ducts** — simple columnar, with **mitochondria-lined basolateral folds** that reabsorb **Na^+ and Cl^-** and add **HCO_3^-**. This is why saliva is **hypotonic** and why the striated duct is the histologic signature of a salivary gland.
3. **Interlobular (excretory) ducts** — unusually, lined by **stratified cuboidal or columnar** epithelium.

Pancreas

A compound acinar gland, entirely **exocrine** except for the scattered **islets of Langerhans**. It secretes about **1.5 L/day** of alkaline juice.

- **Acinar cells** — **pyramidal**, with a basal nucleus, abundant **rough ER and Golgi** in the basal cytoplasm, and **eosinophilic zymogen granules** apically. The cell is polarized for export: make protein at the base, package it at the apex.
- **Centroacinar cells** — pale cells inserted into the acinar lumen; they begin the duct system inside the acinus.
- **Ducts** — intercalated → intralobular → interlobular → main pancreatic duct. **Centroacinar and intercalated duct cells secrete HCO_3^-** under **secretin**, neutralizing chyme and establishing the pH at which pancreatic enzymes work. The high pH also keeps the enzymes inactive until they reach the duodenum.
- **Endocrine islets** — pale, round clusters between acini (details in Chapter 20).

The pancreas has only a **very thin capsule**, whose septa carry the larger vessels and ducts and separate the parenchyma into lobules; each acinus is surrounded by a basal lamina and a delicate reticular sheath — a useful contrast with the parotid, which has striated ducts and no islets.

Liver

The liver is both an **endocrine** gland (plasma proteins into blood) and an **exocrine** gland (bile into ducts). That dual output explains its unusual architecture: two fluid streams run in opposite directions through the same lobule.

Blood supply. About **75%** of hepatic blood is nutrient-rich, O_2-poor **portal vein** blood; **25%** is oxygenated **hepatic artery** blood. Both enter at the portal areas and flow **toward the central vein**.

- **Classic lobule** — a polygon of **plates of hepatocytes radiating toward a central vein**, with **portal areas** at the corners. Connective tissue between lobules is sparse, becoming more abundant in the portal areas.
- **Portal area (triad + one)** — a **portal venule**, a **hepatic arteriole**, a **bile ductule**, and a small **lymphatic**.
- **Hepatocytes** — large polygonal epithelial cells with large central nuclei; about **50% are polyploid** and many are **binucleated**. They contain abundant **rough and smooth ER**, many small **Golgi complexes**, glycogen, and peroxisomes.
- **Sinusoids** — discontinuous, **fenestrated** endothelium. Between endothelium and hepatocytes is the **perisinusoidal space (of Disse)**, where plasma exchanges with hepatocytes.
- **Kupffer cells** — stellate macrophages in the sinusoidal lining; remove effete erythrocytes and release iron and bilirubin.
- **Hepatic stellate (Ito) cells** — in the perisinusoidal space, storing **vitamin A** and other fat-soluble vitamins in lipid droplets; they become myofibroblasts in fibrosis.
- **Bile canaliculi** — grooves **between adjacent hepatocytes**, sealed by tight junctions. Bile flows **toward** the portal area while blood flows **away** from it.
- **Canals of Hering** — where canaliculi join the bile ductules; all ducts eventually form the left and right hepatic ducts.

Functions: carbohydrate, lipid, and protein metabolism; **albumin** and clotting factor synthesis; detoxification by SER and peroxisomes; bile production; bilirubin conjugation; storage of glycogen, iron, and vitamins A, D, and B_{12}.

Zonation. Zone 1 (periportal) receives the most oxygen and performs oxidative metabolism; **zone 3** (pericentral) receives the least and performs biotransformation. Zone 3 dies first in ischemia and is the site of alcoholic injury.

Biliary Tract & Gallbladder

Every bile-conducting structure after the canaliculi is lined by **cholangiocytes** — simple cuboidal or columnar epithelium.

Route: canaliculi → canals of Hering → bile ductules → interlobular ducts → left and right hepatic ducts → **common hepatic duct** → **cystic duct** → gallbladder → **common bile duct** → **hepatopancreatic ampulla (of Vater)** → duodenum.

The **gallbladder** is a sac with a **highly folded mucosa**, a well-vascularized lamina propria, **no muscularis mucosae and no submucosa**, and a single muscular layer. Its columnar cholangiocytes **absorb water**, concentrating bile; the muscle contracts under **cholecystokinin (CCK)** released by duodenal enteroendocrine cells when food arrives.

CLINICAL CORRELATION

Cirrhosis is the end result of hepatocyte injury: fibrosis with regenerative nodules, portal hypertension, and synthetic failure. **Acute pancreatitis** is autodigestion by prematurely activated enzymes; **chronic pancreatitis** destroys acini and islets, producing diabetes and steatorrhea. **Carcinoma of the head of the pancreas** obstructs the common bile duct and produces painless obstructive jaundice. **Gallstones** form when bile is supersaturated with cholesterol or pigment; impaction in the cystic or common bile duct produces biliary colic and obstructive jaundice.

UNDER THE MICROSCOPE

The liver is recognized by **cords of eosinophilic cells with central veins and portal areas**. The pancreas by **dark purple acini with pink apical granules and pale islets**. Pancreas versus parotid: the pancreas has **islets** and **no striated ducts**; the parotid has **striated ducts** and **no islets**.

EXAM TIP

Bile flows **toward** the portal triad; blood flows **away** from it. Students who memorize "everything flows to the center" lose this mark every year.

Chapter 16 — one-line answers

- Major salivary glands make ~90% of saliva; minor glands ~10%.
- Parotid = serous only. Submandibular = mixed, mainly serous. Sublingual = mixed, mainly mucous with demilunes.
- Intercalated duct = simple cuboidal; striated duct = simple columnar with basal folds; excretory duct = stratified cuboidal or columnar.
- Striated ducts reabsorb Na^+ and Cl^- and add HCO_3^-, making saliva hypotonic.
- Myoepithelial cells contract to expel secretion.
- Pancreatic acinar cell: basal RER and Golgi, apical zymogen granules — polarized for export.
- Centroacinar cells sit inside the acinus and begin the duct.
- Secretin drives HCO_3^- secretion from centroacinar and intercalated duct cells.
- Liver blood: 75% portal vein, 25% hepatic artery.
- Portal area = portal venule + hepatic arteriole + bile ductule + lymphatic.
- About half of hepatocytes are polyploid; many are binucleated.
- Kupffer cells = sinusoidal macrophages. Ito cells = vitamin A storage.
- Bile canaliculi lie between hepatocytes, sealed by tight junctions, and drain to the canals of Hering.
- Hepatic zone 1 = periportal, most oxygen; zone 3 = pericentral, most biotransformation and first to die in ischemia.
- Gallbladder: folded mucosa, no muscularis mucosae, no submucosa; cholangiocytes absorb water; CCK contracts the muscle.

17 · The Respiratory System

BOOK PAGES 349–370 | Part 3, pages 100–131 · Nasal Cavities · Pharynx · Larynx · Trachea · Bronchial Tree & Lung · Lung Vasculature & Nerves · Pleural Membranes · Respiratory Movements

The core idea

The respiratory tract is a **conducting** series (nose to terminal bronchiole) followed by a **respiratory** series (respiratory bronchiole to alveolus). The transition is marked by the disappearance of cartilage and the appearance of alveoli. Walking down the tree, each level loses one structural support and gains a thinner epithelium.

Nasal Cavities

Each nasal cavity has a **vestibule** where air enters, and **three conchae** projecting from the medial wall that create turbulence and force air over a large mucosal surface.

- **Vestibule** — stratified squamous epithelium with **vibrissae** (stiff hairs) that filter large particles.
- **Respiratory region** — **respiratory epithelium**: pseudostratified ciliated columnar with goblet cells, over a highly vascular lamina propria containing seromucous glands. It warms, humidifies, and cleans air; the cilia sweep mucus toward the pharynx.
- **Olfactory region** — the roof and part of the superior concha: **olfactory epithelium**, about **10 cm²** and up to **100 µm** thick, containing **bipolar olfactory neurons**, supporting cells, and basal stem cells. Each neuron has about **20 long nonmotile cilia** bearing odorant receptors. Olfactory neurons are replaced every **1–2 months** — unusual for neurons. **No goblet cells**, but **Bowman glands** in the lamina propria.

The respiratory epithelium is a mixed population, and the proportions are examinable:

Cell	Share	Function
Ciliated columnar cells	Majority	Move the mucus blanket
Goblet cells	—	Secrete mucus
Brush cells	~3%	Chemosensory; short blunt microvilli
Small granule (Kulchitsky) cells	~3%	**DNES**; dense-core granules 100–300 nm
Basal cells	~30%	Stem cells

Pharynx

Nasopharynx — respiratory epithelium with **lymphoid tissue** (pharyngeal tonsil) in the lamina propria. **Oropharynx** and **laryngopharynx** — stratified squamous epithelium, continuous with the oral cavity and esophagus. The change from respiratory to stratified squamous epithelium marks where the airway is exposed to abrasion from food.

Larynx

A short air passage between pharynx and trachea, with skeletal cartilages and intrinsic muscles.

- **Epiglottis** — elastic cartilage, with stratified squamous epithelium on its lingual surface and respiratory epithelium on its laryngeal surface; the switch is abrupt.
- **Vocal folds (true cords)** — **stratified squamous** epithelium, avascular, with the **vocal ligament** and the **vocalis muscle**. Tension and vibration produce sound.
- **False cords (vestibular folds)** — respiratory epithelium and glands.

Trachea

- **Epithelium** — respiratory epithelium (pseudostratified ciliated columnar with goblet cells).

- **Lamina propria** — elastic fibers, lymphocytes, seromucous glands.
- **Cartilage** — **C-shaped hyaline rings**, incomplete posteriorly.
- **Trachealis muscle** — smooth muscle bridging the posterior gap, allowing the esophagus to expand.

Bronchial Tree & Lung

Bronchi — the primary bronchus enters each lung and branches repeatedly. Cartilage breaks into **irregular plates** rather than rings, a complete ring of smooth muscle appears, and elastic fibers and glands persist. Each **tertiary (segmental) bronchus** with its branches forms a **bronchopulmonary segment** — about **10–12%** of a lung, with its own connective tissue capsule and blood supply.

Bronchioles — airways of **1 mm or less**, formed after about the **tenth generation** of branching. They have **no cartilage and no glands**, a simple ciliated columnar or cuboidal epithelium, circular smooth muscle, and elastic fibers.

- **Club cells** (formerly Clara cells) replace goblet cells as the main secretory cell; they secrete **surfactant-like material**, help innate immunity, and their **smooth ER P450 system detoxifies inhaled compounds**. They are the progenitor cells of the bronchiolar epithelium.
- **Terminal bronchiole** — the last conducting airway, lined mainly by club cells, with **no alveoli**.

Respiratory region: terminal bronchiole → **respiratory bronchiole** (simple cuboidal, interrupted by alveoli) → **alveolar duct** (a linear series of alveoli) → **alveolar sac** (a cluster). Gas exchange begins at the respiratory bronchiole.

Cell	Shape	Function	Share of alveolar lining
Type I pneumocyte	Squamous, extremely thin	**Gas exchange**; forms the blood–air barrier	**~95%**
Type II pneumocyte	Cuboidal, **lamellar bodies**	**Surfactant**; innate immunity; progenitor for type I	~5%
Alveolar macrophage	Large, free ("dust cell")	Phagocytosis	—

Numbers to know:

- Each alveolus is about **200 µm** in diameter.
- Each adult lung has about **200 million alveoli**, with a total surface area of about **75 m²**.
- The **blood–air barrier** = surfactant film + **type I pneumocyte** + fused basal laminae + capillary endothelium. Its total thickness is **0.1–1.5 µm**, and the attenuated type I cytoplasm can be as thin as **25 nm**.
- **Surfactant** is an oily mixture of **cholesterol, phospholipids, and surfactant proteins** secreted by **type II cells and club cells**. It lowers surface tension and prevents alveolar collapse at end-expiration.
- **Pores of Kohn** connect adjacent alveoli.

Lung Vasculature & Nerves

The lung has a **double blood supply**: the **pulmonary artery** carries deoxygenated blood for gas exchange and forms a dense capillary network around alveoli, while the **bronchial arteries** (from the aorta) supply the conducting airways and pleura. Pulmonary arteries are unusual in carrying **deoxygenated** blood and in having thin walls suited to low pressure. Autonomic nerves control bronchial smooth muscle and gland secretion — parasympathetic (vagal) constriction, sympathetic relaxation.

Pleural Membranes

A **serous membrane** of simple squamous mesothelium over connective tissue: **visceral pleura** covers the lung and is continuous with **parietal pleura** lining the thoracic wall. A thin fluid film lets the layers slide while surface tension holds them together, so the lung follows the chest wall.

Respiratory Movements

Ventilation is pressure-driven: the **diaphragm** and **external intercostal muscles** enlarge the thoracic cavity, pleural pressure falls, the lung expands passively, and air enters. **Elastic recoil** of the lung and surface tension at the alveolar air–liquid interface drive expiration — precisely why surfactant deficiency causes collapse.

CLINICAL CORRELATION

Respiratory distress syndrome of the newborn — immature type II pneumocytes produce insufficient surfactant, so alveoli collapse; treatment is exogenous surfactant and antenatal steroids. **Emphysema** — destruction of alveolar walls enlarges air spaces and reduces surface area. **Asthma** — bronchoconstriction, mucosal edema, eosinophilic inflammation, thickened basement membrane, and smooth muscle hypertrophy. **Chronic bronchitis** — mucus hypersecretion and gland hypertrophy. **Cystic fibrosis** — defective CFTR thickens secretions, causing bronchiectasis and recurrent infection. **Lung cancer** — the commonest sites are the major bronchi; squamous cell carcinoma, adenocarcinoma, small cell carcinoma, and large cell carcinoma are the four main types.

UNDER THE MICROSCOPE

Walk down the tree and count what disappears. **Cartilage rings + glands = trachea. Cartilage plates + no glands = bronchus. No cartilage, no glands, muscular wall = bronchiole.** Then look for alveoli in the wall: if present, it is a respiratory bronchiole.

MEMORY HOOK

"Rings, Plates, None" for cartilage from trachea → bronchus → bronchiole. And **"Type I for gas, Type II for surfactant, macrophage for garbage."**

EXAM TIP

Club cells are found in **bronchioles**, not alveoli, and are the progenitor cells of the bronchiolar epithelium. A question that places them in the alveolus is testing whether you know the level of the airway.

18 · Skin

BOOK PAGES 371–392 | Part 3, pages 132–160 · Epidermis · Dermis · Subcutaneous Tissue · Sensory Receptors · Hair · Nails · Skin Glands · Skin Repair

The core idea

Skin is a **stratified, keratinizing, self-renewing barrier** with appendages embedded in it. The epidermis is a factory that moves cells outward and turns them into keratin; the dermis provides mechanical and nutritional support; the appendages provide temperature control, sensation, and protection. Every fact in the chapter is a consequence of one of those three jobs.

Epidermis

Thickness: 75–150 µm in thin skin, **400–600 µm** in thick skin. Full skin (epidermis + dermis) is about **4 mm** on the back and **1.5 mm** on the scalp. The epidermis is **avascular** — nutrients and O_2 arrive by diffusion from the dermal papillae. It turns over about every **15–30 days**.

Layer	Cells	Key feature
Stratum basale	Mitotic stem cells, melanocytes, Merkel cells	Single layer on the basement membrane, attached by **hemidesmosomes and integrins**; **keratin filaments ~10 nm** first appear here
Stratum spinosum	Polyhedral cells with **desmosomes**	"Spiny" because shrinkage pulls apart desmosome-bound bundles (**tonofibrils**); **Langerhans cells** live here
Stratum granulosum	Flattened cells with **keratohyalin granules** and **lamellar granules**	Keratohyalin contains **filaggrin**, which aggregates keratin; lamellar granules discharge lipid into the intercellular space
Stratum lucidum	Clear, anucleate, refractile	**Only in thick skin** of palms and soles
Stratum corneum	Anucleate **squames**	The barrier; keratin packed in an electron-dense matrix with a lipid envelope

Keratin filaments contain at least **six polypeptides of 40–70 kDa**, synthesized as the cell differentiates. The **lipid envelope** between squames is what makes epidermis impermeable to water.

Cell	Origin	Function	Identifying feature
Keratinocyte	Ectoderm	Barrier, keratin production	90% of epidermal cells; desmosomes
Melanocyte	**Neural crest**	**Melanin** via tyrosinase; transferred in melanosomes	Dark, dendritic; basal layer
Langerhans cell	**Monocyte/bone marrow**	Antigen presentation	Dendritic, stratum spinosum; **2–8%** of epidermal cells
Merkel cell	Neural crest	**Light touch** (mechanoreception)	Basal, with a sensory nerve ending

Melanocytes — about **one per five or six keratinocytes** (**600–1200/mm²** of thin skin), with processes reaching **10–30 keratinocytes**. **Tyrosinase** converts tyrosine to DOPA and then to **eumelanin** (brown-black) or **pheomelanin** (red-yellow). Mature melanosomes are ellipsoid, **0.5 × 1 µm**, and are transferred to keratinocytes, which hold more pigment than the cells that make it. The pigment sits over the nucleus and **protects nuclear DNA from UV damage** — the reason tanning exists and the reason melanoma risk tracks sun exposure. One melanocyte plus its keratinocytes is an **epidermal-melanin unit**. Melanocytes are about **four times more numerous and active in thin skin** than in thick palmar and plantar skin.

The race-and-melanin rule: densities of epidermal-melanin units are **similar in all individuals**; what differs is how rapidly melanin is made and how abundantly it accumulates in keratinocytes.

Dark skin is not skin with more melanocytes.

Dermis

- **Papillary dermis** — loose connective tissue forming **dermal papillae** that interdigitate with the epidermis; contains the **subpapillary microvascular plexus**, which sends capillary loops into the papillae.
- **Reticular dermis** — dense irregular connective tissue with thick collagen bundles and elastic fibers; up to **4 mm** thick on the back.

Dermal vasculature has two jobs: **nutrition** (two plexuses, subpapillary and deep) and **thermoregulation** through **arteriovenous anastomoses** between them.

Thick versus thin skin. Thick skin (palms, soles) has a thick stratum corneum, a **stratum lucidum**, and **no hair follicles**; it has many sweat glands. Thin skin covers most of the body, has hair follicles and sebaceous glands, and a thinner epidermis.

Subcutaneous Tissue

The **hypodermis** — loose connective tissue and adipose tissue (**panniculus adiposus**) binding skin loosely to underlying organs, so the skin can slide. It insulates, cushions, stores energy, and determines how a wound behaves and how an injection is absorbed.

Cutaneous Sensory Receptors

Receptor	Location	Structure	Modality
Free nerve endings	Epidermis and dermis	Unencapsulated	Pain, temperature extremes, mechanical movement
Merkel cell	Basal epidermis	Cell + disc-like nerve ending	**Light touch**, sustained pressure
Meissner corpuscle	Dermal papilla	Elliptical, **~150 µm**, stacked Schwann cells	**Light touch**, fine discrimination
Pacinian corpuscle	Deep dermis and hypodermis	Oval, **~1 mm**, **15–50 concentric lamellae**	**Pressure and high-frequency vibration**
Krause end bulb	Penis and clitoris mainly	Thin collagenous capsule	Low-frequency vibration
Ruffini corpuscle	Dermis	Fusiform capsule anchored to collagen	**Stretch (tension) and twisting**

Hair

Hairs are keratinized structures formed in **hair follicles**, epidermal invaginations extending into the dermis.

- The **hair bulb** at the base contains the **matrix** (rapidly proliferating keratinocytes) and is invaginated by a vascular **dermal papilla** that supplies it.
- The hair root has three concentric layers — **medulla, cortex, cuticle** — surrounded by **internal and external root sheaths** continuous with the epidermis, a **glassy membrane** (partly basal lamina), and a connective tissue sheath.
- The **arrector pili** smooth muscle raises the hair; a **root plexus** provides sensation.
- Density: about **600 hairs/cm²** on the face and **60/cm²** elsewhere.
- Growth cycles: **anagen** (growth), **catagen** (involution), **telogen** (rest).

Nails

Hard keratin plates on the dorsal surface of the distal phalanx. The **nail root** lies under a skin fold whose cornified layer forms the **cuticle (eponychium)**; keratinocytes proliferate in the **matrix** and differentiate into the **nail plate**, which slides forward over the **nail bed** (which does not contribute).

Growth is about **3 mm/month** for fingernails and **1 mm/month** for toenails. The **lunula** is the visible part of the matrix.

Skin Glands

Sebaceous glands — **holocrine**: sebocytes accumulate lipid, die, and discharge **sebum** into the hair follicle (pilosebaceous unit). Absent from palms and soles.

Eccrine sweat glands — **merocrine**, coiled tubular, with a secretory portion of cuboidal cells and a duct of **stratified cuboidal** cells; abundant on palms and soles; watery sweat for thermoregulation.

Apocrine sweat glands — **axillae and perineum**, much **wider lumens**, develop after **puberty**, and secrete protein-rich sweat onto hair follicles; odor follows bacterial action.

Skin Repair

First intention — clean, apposed edges; a clot forms, neutrophils and macrophages arrive, keratinocytes migrate across the wound, and minimal granulation tissue is needed. **Second intention** — open wounds; abundant **granulation tissue**, wound contraction by myofibroblasts, and a larger scar. **Burns** are classified by depth: first-degree (epidermis), second-degree (into dermis, with blistering), third-degree (full thickness, including appendages, so it cannot re-epithelialize).

CLINICAL CORRELATION

Psoriasis — accelerated keratinocyte turnover with parakeratosis, acanthosis, and Munro microabscesses. **Melanoma** — malignant melanocytes; **Breslow thickness** (depth of invasion) predicts survival, which is why the dermis matters. **Vitiligo** — autoimmune destruction of melanocytes. **Albinism** — defective tyrosinase or another melanin-pathway component. **Basal cell carcinoma** — from basal keratinocytes, locally invasive but rarely metastasizes. **Squamous cell carcinoma** — from spinous cells, can metastasize. **Merkel cell carcinoma** — rare but about **twice as lethal** as melanoma. **Pemphigus** and **bullous pemphigoid** are the blistering diseases of desmosomes and hemidesmosomes described in Chapter 4.

UNDER THE MICROSCOPE

Count layers and look for appendages. **Thick skin: five layers, thick keratin, no hair. Thin skin: four layers, hair follicles and sebaceous glands.** A dark condensed nucleus at the basal layer with a clear halo suggests a melanocyte.

EXAM TIP

"Melanocytes are more numerous in dark skin" is false. **Unit density is similar in all people; melanin production and transfer differ.** Also remember that the **nail bed makes no contribution** to the nail plate — the matrix does.

MEMORY HOOK

"Come Let's Get Some Beer" — **C**orneum, **L**ucidum, **G**ranulosum, **S**pinosum, **B**asale (superficial to deep) — and remember **lucidum exists only in thick skin**.

PART V — EXCRETION AND REPRODUCTION

Chapters 19–23 · kidney, endocrine glands, and the reproductive systems

19 · The Urinary System

BOOK PAGES 393–412 | Part 3, pages 161–190 · Kidneys · Blood Circulation · Renal Function: Filtration, Secretion, & Reabsorption · Ureters, Bladder, & Urethra

The core idea

The kidney is a **countercurrent filtration and reabsorption machine** built from millions of identical tubes. Each segment of the tubule has a distinct epithelium matched to a distinct transport job, so identifying the segment from its histology is the same as naming its function.

Kidneys

Each kidney is about **12 cm long, 6 cm wide, and 2.5 cm thick**. The **hilum** on the medial border admits nerves and vessels and lets the ureter and lymphatics exit. Inside the hilum the ureter expands as the **renal pelvis**, which divides into **two or three major calyces**, then **minor calyces**.

- **Capsule → cortex** (renal corpuscles + convoluted tubules) → **medulla** (loops of Henle, collecting ducts, vasa recta) organized into **8–15 pyramids** (commonly quoted as 8–12).
- **Renal columns (of Bertin)** — cortical tissue between pyramids.
- **Lobe** = pyramid + overlying cortex; **lobule** = a medullary ray + the cortical tissue around it.
- **Papilla** → minor calyx → major calyx → renal pelvis → ureter.
- The cortex receives **more than 10 times** the blood flow of the medulla.

The nephron — about **1.25 million per kidney** — begins with the **renal corpuscle**: the **glomerulus** (a tuft of fenestrated capillaries) inside the **glomerular (Bowman) capsule**.

- **Parietal layer** — simple squamous epithelium, continuous with the proximal tubule.
- **Visceral layer** — **podocytes** with primary processes and interdigitating **pedicels** that embrace the capillaries. Between pedicels are **filtration slit pores 25–30 nm wide**, bridged by a **slit diaphragm** of P-cadherin, **nephrin**, and the negatively charged glycoprotein **podocalyxin**.
- **Mesangial cells** — between capillaries, covering surfaces not covered by podocytes; **contractile** (like pericytes) and produce external lamina.

The filtration barrier has three layers: fenestrated endothelium → glomerular basement membrane (GBM), about **330 nm** thick, of **type IV collagen** and other proteins made by endothelial cells and podocytes → **podocyte slit diaphragms**. The barrier is both a size and a charge filter.

Functional numbers that examiners love:

- About **20%** of the plasma entering a glomerulus is filtered.
- Total filtration area ≈ **500 cm²**.
- **GFR ≈ 125 mL/min ≈ 180 L/day**.
- Plasma volume ≈ **3 L**, so the kidneys filter the entire plasma volume about **60 times a day**.

Juxtaglomerular apparatus (JGA) — at the vascular pole: **juxtaglomerular (JG) cells** in the afferent arteriole (modified smooth muscle, contain **renin**), the **macula densa** (a plaque of tall, closely packed cells in the distal straight tubule sensing NaCl), and **extraglomerular mesangial (lacis) cells**.

Blood Circulation

Renal artery → segmental → interlobar (between pyramids) → arcuate (at the corticomedullary junction) → interlobular (radiating into the cortex) → **afferent arteriole** → glomerulus → **efferent arteriole** → peritubular capillaries in the cortex and **vasa recta** in the medulla. The **efferent arteriole**, not a venule, drains the glomerulus — an arrangement that keeps glomerular hydrostatic pressure high. Efferent arterioles of **cortical** glomeruli branch into a diffuse peritubular network;

those of **juxtamedullary** glomeruli form the long hairpin **vasa recta** that run alongside the loops of Henle and are the anatomical basis of the **countercurrent multiplier**.

Renal Function: Filtration, Secretion, & Reabsorption

Segment	Epithelium	Key features	Function
Proximal convoluted tubule (PCT)	Simple cuboidal, **eosinophilic**	**Long microvilli (brush border)**, abundant mitochondria, large interdigitating **basolateral folds**	**Major reabsorption**: all glucose and organic nutrients, all small proteins and peptides (degraded to amino acids), much water and electrolyte
Thin descending limb	Simple squamous	Thin	Water permeability
Thin ascending limb	Simple squamous	Thin	—
Thick ascending limb (TAL = distal straight tubule)	Simple cuboidal, low brush border	Ends as the **macula densa** at the vascular pole	**NaCl reabsorption, impermeable to water** — the diluting segment and loop-diuretic site
Distal convoluted tubule (DCT)	Simple cuboidal, **pale**, no brush border	Many mitochondria, basal infoldings	Further adjustment of electrolytes; thiazide site; leads to **connecting tubules**
Collecting duct	Simple cuboidal → columnar in medulla	**Principal cells** (aquaporins, ADH) and **intercalated cells** (type A secrete H^+, type B secrete HCO_3^-)	Final water and acid–base regulation

Size contrast: the straight proximal tubule is about **60 µm** in outer diameter and narrows abruptly to about **30 µm** in the thin limbs of the loop.

The visual contrast that answers most questions: the **proximal tubule is eosinophilic with a ragged brush border and a star-shaped lumen; the distal tubule is pale with a clean lumen and no brush border.**

Collecting ducts. Connecting tubules from several nephrons join to form cortical collecting ducts, which enter the medulla parallel to the loops and vasa recta and become larger with more columnar cells. The largest (papillary) ducts — **ducts of Bellini** — open into a minor calyx, where filtrate undergoes no further modification and is called **urine**.

Ureters, Bladder, & Urethra

Ureter — **urothelium (transitional epithelium)**, a folded mucosa, and a **thick muscularis** that moves urine by peristalsis; inner longitudinal and outer circular layers, with a third layer added near the bladder. **Bladder** — urothelium, a highly folded mucosa, and the **detrusor** muscle in three poorly delineated interlacing layers. It holds **400–600 mL**, with the urge to void at about **150–200 mL**.

Urothelium is a specialized barrier, not just a stratified epithelium:

- **Umbrella cells** — large superficial cells up to **100 µm** in diameter, with **uroplakin plaques ~16 nm** across, hinged to allow unfolding as the bladder fills.
- The empty bladder has **5–7 layers**; the full bladder has **2–3**, because intermediate cells are stretched laterally.
- Tight junctions plus plaques make it an osmotic barrier against hypertonic urine.

Urethra — in both sexes lined initially by urothelium. In men there are three parts: **prostatic** (3–4 cm, urothelium), **membranous** (short, through the urogenital diaphragm; stratified or pseudostratified columnar), and **penile** (long; stratified columnar/pseudostratified proximally, stratified squamous distally). In women the urethra is **3–5 cm**, urothelium proximally, then nonkeratinized stratified squamous; the middle part is surrounded by the **external striated sphincter**. The short female urethra explains the frequency of cystitis.

CLINICAL CORRELATION

Diabetic nephropathy thickens the glomerular basement membrane and later produces mesangial expansion (Kimmelstiel–Wilson nodules), with proteinuria. **Nephrotic syndrome** — heavy proteinuria, hypoalbuminemia, edema, hyperlipidemia; podocyte injury is central. **Acute tubular necrosis** — ischemic or toxic injury to the proximal tubule, the most metabolically demanding segment. **Polycystic kidney disease** — the primary cilium and its signaling fail. **Renal cell carcinoma** arises from proximal tubule cells; **transitional cell carcinoma** from the urothelium. **Kidney stones** form in the pelvis or calyces and cause extreme pain when they move into the ureter.

UNDER THE MICROSCOPE

Start at the renal corpuscle: the **glomerulus** is a tuft of capillaries inside a capsule. Then compare tubules: **eosinophilic + brush border = proximal; pale + clean = distal; large pale cells in a row = collecting duct.** The **macula densa** is the patch of tall cells on the distal tubule facing the vascular pole.

EXAM TIP

The proximal tubule is the **most eosinophilic** structure in the renal cortex, packed with mitochondria for active transport. Students often call it "basophilic" because the brush border looks fuzzy — the cytoplasm is pink.

Chapter 19 — one-line answers

- Kidney: 12 × 6 × 2.5 cm; **8–15 pyramids**; about **1.25 million nephrons**.
- Lobe = pyramid + cortex; lobule = medullary ray + surrounding cortex.
- Cortex receives **more than 10×** the blood flow of the medulla.
- Afferent arteriole → glomerulus → **efferent arteriole** (not a venule), keeping glomerular pressure high.
- About **20%** of entering plasma is filtered; GFR ≈ **125 mL/min ≈ 180 L/day**.
- Filtration barrier = fenestrated endothelium + **GBM (type IV collagen, ~330 nm)** + slit diaphragms (nephrin, podocalyxin).
- Mesangial cells are contractile and support the capillary tuft.
- **PCT**: brush border, eosinophilic, reabsorbs all glucose and small proteins.
- **TAL**: water-impermeable, reabsorbs NaCl, ends as the **macula densa**.
- **JG cells** in the afferent arteriole secrete **renin**; macula densa senses NaCl.
- Collecting duct: **principal cells** (aquaporins, ADH) and **intercalated cells** (type A H^+, type B HCO_3^-).
- Urothelium: **umbrella cells** with uroplakin plaques; 5–7 layers empty, 2–3 layers full.
- Bladder holds **400–600 mL**; urge at **150–200 mL**.
- Male urethra: prostatic, membranous, penile; female urethra 3–5 cm — short, hence cystitis.

20 · Endocrine Glands

BOOK PAGES 413–438 | Part 3, pages 191–217 (to the thyroid) · Pituitary Gland (Hypophysis) · Adrenal Glands · Pancreatic Islets · Diffuse Neuroendocrine System · Thyroid Gland · Parathyroid Glands · Pineal Gland

The core idea

Endocrine glands release hormone into the blood, so their histology is organized around **storage and release** rather than ducts. The three patterns to recognize are **cords** (pituitary, adrenal, parathyroid), **follicles** (thyroid), and **clusters** (pancreatic islets). Each pattern is dictated by how the hormone is stored.

Pituitary Gland (Hypophysis)

Two embryologically distinct parts in one sella turcica.

Adenohypophysis (oral ectoderm — **Rathke's pouch**). The **pars distalis accounts for 75%** of the adenohypophysis and is covered by a thin fibrous capsule:

Region	Cells	Hormones
Pars distalis	**Acidophils**	**GH, prolactin**
	Basophils	**ACTH, TSH, FSH, LH**
	Chromophobes	Inactive or degranulated
Pars intermedia	Basophils, colloid cysts	MSH (rudimentary in humans)
Pars tuberalis	—	Wraps the infundibulum

Reading the stain. Routine H&E divides the pars distalis into **acidophils** (pink), **basophils** (blue), and **chromophobes** (poorly stained, the least numerous). The modern names describe the hormone, not the color: **somatotrophs** (GH), **lactotrophs** (prolactin), **corticotrophs** (ACTH), **thyrotrophs** (TSH), and **gonadotrophs** (FSH, LH). Acidophils are the somatotrophs and lactotrophs; basophils are the corticotrophs, thyrotrophs, and gonadotrophs.

Neurohypophysis (neuroectoderm): the **pars nervosa** stores **oxytocin and ADH (arginine vasopressin)** — both **9-amino-acid peptides** — synthesized in the **supraoptic and paraventricular nuclei** of the hypothalamus and carried down axons. In the axon terminals they are bound to **10-kDa carrier proteins, neurophysin I and II**, and accumulate in dilatations called **neurosecretory (Herring) bodies**, visible as faintly eosinophilic structures. **Pituicytes** are the glial cells of the lobe. The neurohypophysis **makes no hormone of its own** — it stores and releases hypothalamic hormones.

Control: the hypothalamus releases **releasing and inhibiting hormones** into the **hypophyseal portal system**, carrying them directly to the pars distalis — which is why a tiny hypothalamic signal controls a large pituitary output.

Adrenal Glands

Two glands in one organ, with entirely different embryology and function:

Region	Cells	Hormone	Control
Zona glomerulosa	Rounded or arched cords under the capsule, **~15% of the cortex**	**Aldosterone**	**Angiotensin II, K^+**
Zona fasciculata	Long cords of pale, foamy cells	**Cortisol**	**ACTH**
Zona reticularis	Irregular cords, pigmented cells	**Androgens (DHEA)**	ACTH
Medulla	**Chromaffin cells** (neural crest)	**Epinephrine, norepinephrine**	Sympathetic splanchnic nerve

Each adrenal gland is a flattened half-moon about **4–6 cm long, 1–2 cm wide, and 4–6 mm thick**, weighing about **8 g** in total. The **fasciculata** is the **thickest** zone, and its cells are pale and vacuolated because they are full of **lipid droplets (cholesterol esters)**, **tubular mitochondrial cristae**, and **smooth ER** for steroid synthesis. The **medulla** is functionally a sympathetic ganglion without axons: **chromaffin cells** contain electron-dense granules **150–350 nm** in diameter, storing catecholamines with Ca^{2+}, ATP, and **49-kDa chromogranins**. About **80%** of the catecholamine released is **epinephrine**. Mnemonic for the zones: **"GFR — Salt, Sugar, Sex"** from outside in.

Pancreatic Islets

Pale, round clusters between the acini of the exocrine pancreas (Chapter 16). The pancreas has **more than 1 million islets**, mostly in the tail, but they occupy only **1–2%** of the organ's volume. Most are **100–200 µm** in diameter and contain several hundred cells. The book recognizes **four major cell types**:

Cell	Product	Notes
β (beta)	**Insulin**	About **twice as abundant** as α cells; central in the islet
α (alpha)	**Glucagon**	Peripheral
δ (delta)	**Somatostatin**	Paracrine inhibition
PP	Pancreatic polypeptide	—

Islets are richly vascularized, and their hormones pass directly into the bloodstream rather than into ducts.

Diffuse Neuroendocrine System

Scattered hormone-secreting cells throughout the body — gut enteroendocrine cells, respiratory neuroendocrine cells, and similar — sharing the ability to take up amine precursors (**APUD** cells). They are the reason a "pure" endocrine organ is not the only source of hormones: the gut alone produces more regulatory peptides than any single gland.

Thyroid Gland

- **Follicles** of varying size filled with **colloid** (thyroglobulin), lined by a simple epithelium whose height ranges from **squamous to low columnar**.
- **Thyrocytes (follicular cells)** — synthesize **T4 and T3**. Activity is read from the epithelium: **active gland = more follicles of tall columnar cells with scalloped colloid; hypoactive gland = squamous cells with dense colloid.**
- **Parafollicular (C) cells** — large, pale cells in the follicular epithelium or between follicles, of **neural crest** origin, secreting **calcitonin**.
- **Folliculogenesis:** thyroglobulin is synthesized, iodinated at the apical surface, stored as colloid, taken back by endocytosis, and digested to release T3/T4.

Parathyroid Glands

- **Chief cells** — small, pale, secretory cells producing **PTH**.
- **Oxyphil cells** — larger, eosinophilic, abundant mitochondria, function uncertain; increase with age.
- Arranged in cords with fat and capillaries; **no follicles**.

Pineal Gland

Pinealocytes (modified neurons) in cords, with **corpora arenacea (brain sand)** — calcified concretions visible on X-ray. Secretes **melatonin** in darkness under sympathetic control; a biological

clock component, and a favourite "identify this organ" image because of the concretions.

CLINICAL CORRELATION

Graves disease — diffuse hyperthyroidism with tall columnar follicular cells, scalloped colloid, and lymphoid infiltration. **Hashimoto thyroiditis** — autoimmune destruction with **Hürthle cells** and germinal centers. **Cushing syndrome** — excess cortisol, with fasciculata hyperplasia or an adrenal adenoma. **Conn syndrome** — aldosterone excess from a glomerulosa adenoma. **Pheochromocytoma** — a chromaffin-cell tumor producing episodic hypertension. **Diabetes mellitus** — progressive β-cell loss. **Pituitary adenoma** — bitemporal hemianopia from compression of the optic chiasm plus a hormone syndrome.

UNDER THE MICROSCOPE

Three patterns, three organs. **Follicles of colloid = thyroid. Cords of pale foamy cells under a capsule = adrenal cortex; the medulla is paler and central. Small pale chief cells in dense cords = parathyroid.** Pituitary: **acidophils and basophils in sheets (anterior); pale fibrillary tissue without epithelial cells (posterior).**

EXAM TIP

The adrenal cortex has **three** zones and the medulla has **none** of the cortex's function. Catecholamines are **medullary**.

Chapter 20 — one-line answers

- Pars distalis = **75%** of the adenohypophysis; acidophils, basophils, chromophobes.
- Acidophils = somatotrophs (GH) and lactotrophs (prolactin); basophils = corticotrophs, thyrotrophs, gonadotrophs.
- Hypothalamic releasing and inhibiting hormones reach the pars distalis through the **hypophyseal portal system**.
- Neurohypophysis stores **oxytocin and ADH** made in the supraoptic and paraventricular nuclei.
- ADH and oxytocin are **9-amino-acid peptides** carried by **neurophysin I and II**; Herring bodies are their axonal dilatations.
- Pituicytes are the glial cells of the pars nervosa.
- Adrenal cortex: glomerulosa (**aldosterone**) → fasciculata (**cortisol**) → reticularis (**androgens**).
- Zona fasciculata is the **thickest** zone; glomerulosa is ~15% of the cortex.
- Adrenal medulla = **chromaffin cells**, neural crest, **80% epinephrine**.
- Pancreatic islets: **β insulin, α glucagon, δ somatostatin, PP pancreatic polypeptide**; β about twice as abundant as α.
- Thyroid: follicles of **colloid (thyroglobulin)**; active epithelium is columnar with scalloped colloid, inactive is squamous.
- **Parafollicular C cells** secrete **calcitonin**.
- Parathyroid: **chief cells** (PTH) and **oxyphil cells** (function uncertain).
- Pineal: **pinealocytes** and **corpora arenacea**; melatonin under sympathetic control.

21 · The Male Reproductive System

BOOK PAGES 439–459 | Testes · Intratesticular Ducts · Excretory Genital Ducts · Accessory Glands · Penis

The core idea

The testis does two jobs that require opposite conditions: **producing sperm**, which needs a temperature below core (hence the scrotum), and **producing testosterone**, which needs a rich blood supply. The architecture keeps them apart by separating the **seminiferous tubules** from the **interstitium**.

Testes

- **Tunica albuginea** — thick capsule; the **mediastinum testis** sends septa dividing the gland into lobules.
- **Seminiferous tubules** — lined by a **stratified epithelium of germ cells** plus **Sertoli cells**, on a basal lamina, surrounded by **peritubular myoid cells**.
- **Interstitium** — **Leydig cells**, blood vessels, connective tissue.

Sertoli cells extend from base to lumen; their nuclei are pale, oval, and have a prominent nucleolus. They are **not** germ-line cells. Functions:

- **Physical support and nutrition** of developing germ cells.
- **Blood–testis barrier** — tight junctions between adjacent Sertoli cells near the base, creating basal and adluminal compartments. This is why the immune system does not reject sperm.
- **Phagocytosis** of residual bodies.
- **Secretion** of **androgen-binding protein (ABP)**, **inhibin**, anti-Müllerian hormone, transferrin, and plasminogen activator.
- **Response to FSH** — which acts on Sertoli cells, not on germ cells.

Spermatogenesis: spermatogonia (type A dark/pale → type B) → primary spermatocyte → secondary spermatocyte → spermatid → spermatozoon. Meiosis I halves the chromosome number and **meiosis II** separates chromatids, giving four spermatids per primary spermatocyte. **Spermiogenesis** is the metamorphosis into a sperm: **acrosome** from the Golgi, nuclear condensation, flagellum growth, shedding of residual cytoplasm. **Spermiation** releases the sperm into the lumen.

Because the barrier makes the adluminal compartment immunologically privileged, sperm antigens are never tolerized — which is why vasectomy can produce anti-sperm antibodies.

Leydig cells — large, polygonal, **eosinophilic** cells with abundant SER and lipid droplets, sometimes containing **Reinke crystals**; they secrete **testosterone** under **LH** stimulation.

Intratesticular Ducts

Rete testis (simple cuboidal) collects sperm from the seminiferous tubules and delivers them through the **efferent ductules** (ciliated and absorptive columnar) to the epididymis.

Excretory Genital Ducts

Segment	Epithelium	Function
Epididymis	**Pseudostratified columnar with stereocilia**	**Maturation and storage of sperm**
Ductus (vas) deferens	Pseudostratified columnar	Transport; **very thick muscular wall** (inner longitudinal, middle circular, outer longitudinal)
Ejaculatory duct	Columnar	Delivers into the prostatic urethra

Accessory Glands

- **Seminal vesicles** — folded mucosa, secretory cells rich in **fructose** and prostaglandins; about 60% of semen volume.
- **Prostate** — tubuloalveolar glands in a **fibromuscular stroma**; **corpora amylacea** are characteristic. Glands are arranged in zones: the **peripheral zone** is where most carcinomas arise; the **transition zone** is where BPH arises.
- **Bulbourethral (Cowper) glands** — mucin for lubrication.

Penis

Three erectile bodies: two **corpora cavernosa** and one **corpus spongiosum** containing the urethra. Each is a sponge of **endothelium-lined cavernous spaces** surrounded by smooth muscle and a **tunica albuginea**. Erection is a vascular event: parasympathetic nitric oxide dilates the helicine arteries, filling the spaces and compressing venous outflow.

CLINICAL CORRELATION

Cryptorchidism — an undescended testis is exposed to core temperature, so spermatogenesis fails and malignancy risk rises; orchidopexy is performed early. **Varicocele** — dilation of the pampiniform plexus raises testicular temperature and impairs spermatogenesis. **Prostatic carcinoma** — arises in the peripheral zone, is palpable on rectal examination, and metastasizes to bone. **BPH** — nodular hyperplasia of the transition zone, compressing the urethra. **Testicular tumors** — germ-cell tumors in young men; a painless mass is malignant until proven otherwise.

UNDER THE MICROSCOPE

Identify the two compartments instantly. **Inside the tubule: Sertoli cells (pale, tall) and germ cells at various stages. Outside the tubule: Leydig cells (eosinophilic, polygonal, in clusters).** If asked for the cell that forms the blood–testis barrier, the answer is Sertoli.

EXAM TIP

FSH acts on **Sertoli** cells; LH acts on **Leydig** cells. One sentence, and it accounts for many marks.

22 · The Female Reproductive System

BOOK PAGES 460–489 | Ovaries · Uterine Tubes · Major Events of Fertilization · Uterus · Embryonic Implantation, Decidua, & the Placenta · Cervix · Vagina · External Genitalia · Mammary Glands

The core idea

The ovary is the only adult structure that must **rebuild itself through a monthly cycle**. Every histological feature of the ovary, tube, uterus, and vagina is a stage in one of two cycles — **ovarian** (follicle to corpus luteum) and **endometrial** (proliferative to secretory) — locked together by the same four hormones.

Ovaries

- **Germinal epithelium** — simple cuboidal (a misnomer; it does not produce germ cells).
- **Tunica albuginea** — dense connective tissue.
- **Cortex** — follicles at all stages, plus corpus luteum and corpus albicans.
- **Medulla** — loose connective tissue with vessels and nerves.

Stage	Structure	Hormonal influence
Primordial follicle	Oocyte + single layer of flat granulosa cells	None; present from birth
Primary follicle	Oocyte enlarges; granulosa cells become **cuboidal**; **zona pellucida** appears	FSH-dependent from this stage
Secondary (antral) follicle	Antrum forms; **theca interna** and **theca externa** differentiate	FSH and LH
Graafian (mature) follicle	Large antrum; oocyte on the **cumulus oophorus**; **corona radiata**	LH surge triggers ovulation
Atretic follicle	Degenerating; **glassy membrane** remains	—
Corpus luteum	Granulosa-lutein cells (progesterone) and theca-lutein cells	LH maintains it
Corpus albicans	White fibrous scar	—

Two-cell, two-gonadotropin theory: LH stimulates **theca interna** cells to produce androgens; **FSH** stimulates **granulosa** cells to aromatize them into **estrogen**. This division of labour explains why both gonadotropins are needed for normal estrogen production.

Uterine Tubes

Four parts — **infundibulum with fimbriae, ampulla, isthmus, intramural portion**. The mucosa is folded and lined by **ciliated columnar cells** (move the ovum) and **peg (secretory) cells**; the muscular wall is thin.

Major Events of Fertilization

Fertilization normally occurs in the **ampulla**. The sperm penetrates the corona radiata and zona pellucida, the membranes fuse, the oocyte completes meiosis II, and the zygote forms. The zygote divides as it travels toward the uterus, reaching the blastocyst stage by the time it arrives — about five days.

Uterus

- **Perimetrium (serosa)**, **myometrium** (thick smooth muscle in interlacing bundles), **endometrium**.
- **Endometrium** — **stratum functionalis** (shed each month) and **stratum basalis** (retained, regenerates it). **Spiral arteries** supply the functionalis and constrict to cause shedding.

Phase	Days	Hormone	Histology
Menstrual	1–4	Falling progesterone	Shedding, hemorrhage, neutrophilic infiltrate
Proliferative	5–14	**Estrogen**	Glands straight and narrow, mitoses, dense stroma
Secretory	15–28	**Progesterone**	Glands **tortuous and dilated with secretion**, oedematous stroma, coiled spiral arteries

The single best discriminator: **straight, narrow glands = proliferative; saw-toothed, dilated, secreting glands = secretory.**

Embryonic Implantation, Decidua, & the Placenta

The blastocyst implants around day 6–7. The **trophoblast** differentiates into **cytotrophoblast** (mitotic, inner) and **syncytiotrophoblast** (invasive, outer). **Chorionic villi** project into maternal blood in the **intervillous space**.

Decidua — the modified endometrium of pregnancy: **decidua basalis** (under the embryo, forms the maternal placenta), **decidua capsularis** (over it), **decidua parietalis** (the rest).

Placental hormones: hCG (maintains the corpus luteum early; the basis of pregnancy tests), **human placental lactogen**, progesterone, and estrogen.

Cervix

Endocervical canal lined by **simple columnar mucus-secreting** cells; the ectocervix by **stratified squamous** epithelium. They meet at the **squamocolumnar junction**, and the zone between original and current junction is the **transformation zone** — the site of HPV-related dysplasia and the target of the **Pap smear**.

Vagina

Stratified squamous non-keratinized epithelium rich in **glycogen**, no glands (lubrication comes from cervical and Bartholin glands), and a lamina propria with elastic fibers and venous plexuses. **Lactobacillus** metabolises glycogen to lactic acid, maintaining an acidic pH.

External Genitalia

The **vulva**: the **labia majora** (skin with hair and sebaceous glands, containing the homologue of the scrotum), **labia minora** (thin skin without hair, with sebaceous glands), the **clitoris** (erectile tissue homologous to the penis), and the **vestibule** containing the urethral and vaginal openings and the openings of the **greater vestibular (Bartholin) glands**, which provide lubrication.

Mammary Glands

A modified **apocrine sweat gland**. In the non-lactating state the gland is mostly **dense connective tissue and adipose tissue with small ducts**; during pregnancy lobules proliferate; during lactation **alveoli are distended with secretion**. The **lactiferous ducts** converge on the nipple, with **lactiferous sinuses** beneath it.

Hormonal control: **estrogen** for ductal growth, **progesterone** for lobuloalveolar development, **prolactin** for milk synthesis, **oxytocin** for milk ejection.

CLINICAL CORRELATION

Polycystic ovary syndrome — multiple arrested antral follicles, anovulation, hyperandrogenism. **Endometriosis** — functional endometrial tissue outside the uterus, bleeding cyclically. **Leiomyoma (fibroid)** — a benign smooth-muscle tumor of the myometrium. **Endometrial carcinoma** — unopposed estrogen, with atypical hyperplasia. **Cervical carcinoma** — HPV in the transformation zone. **Ectopic pregnancy** — implantation outside the uterine cavity, usually in the ampulla. **Breast carcinoma** — arises from ductal or lobular epithelium; the **terminal duct lobular unit** is the functional unit to remember.

UNDER THE MICROSCOPE

Place the ovary in the cycle by the **dominant structure**: many small primordial follicles = inactive/early; a large antral follicle = late follicular; a corpus luteum with folded, pale, luteinized cells = secretory phase; a hyaline scar = post-ovulatory. In the uterus, look at the **glands**, not the stroma.

MEMORY HOOK

"Proliferative = Pencil-thin glands; Secretory = Saw-tooth and swollen."

EXAM TIP

The corpus luteum secretes **progesterone**, and it is maintained by **LH** (and, in pregnancy, by **hCG**). Do not say estrogen maintains the corpus luteum.

Chapter 22 — one-line answers

- Ovary has three regions: cortex with follicles, medulla with vessels, and a surface of **germinal epithelium** over a tunica albuginea.
- Follicle sequence: primordial → primary → secondary (antral) → mature (Graafian) → ovulation.
- **Zona pellucida** = glycoprotein coat around the oocyte; **corona radiata** = granulosa cells carried with it at ovulation.
- **Theca interna** makes androgens; **granulosa cells** aromatize them to estrogen under FSH.
- **Cumulus oophorus** = granulosa mound carrying the oocyte into the antrum.
- After ovulation: **corpus luteum** (progesterone) → **corpus albicans** (hyaline scar).
- Uterine tube: **infundibulum with fimbriae → ampulla → isthmus**; fertilization in the ampulla.
- Endometrium has a **functional layer** that sheds and a **basal layer** that regenerates it.
- Proliferative phase = estrogen, straight narrow glands, mitoses; secretory phase = progesterone, saw-toothed glands, edema.
- Menstrual phase = spiral artery constriction and functionalis shedding.
- Cervix: columnar endocervix, stratified squamous ectocervix, squamocolumnar junction = transformation zone.
- Vagina = stratified squamous nonkeratinized epithelium with glycogen, no glands; acidity comes from lactobacilli.
- Mammary gland: **TDLU** is the functional unit; estrogen drives ducts, progesterone drives lobuloalveolar growth, prolactin drives milk synthesis.

23 · The Eye & Ear: Special Sense Organs

BOOK PAGES 490–524 | Eyes: The Photoreceptor System · Ears: The Vestibuloauditory System

The core idea

Both organs convert a physical signal into a nerve impulse through a **specialized epithelium** — the retina for light, the organ of Corti for sound. Their histology is therefore about **transduction**, not merely structure.

Eyes: The Photoreceptor System

Three tunics:

Tunic	Components	Function
Fibrous	**Cornea** and **sclera**	Protection and support; the cornea also refracts
Vascular (uvea)	**Choroid, ciliary body, iris**	Nutrition, accommodation, pupil control
Nervous	**Retina**	Phototransduction

The cornea has five layers, and the order is a standard question:

1. **Corneal epithelium** — stratified squamous, non-keratinized.
2. **Bowman's membrane** — acellular collagen.
3. **Stroma** — regular lamellae of type I collagen and keratocytes; **avascular**.
4. **Descemet's membrane** — thick basal lamina.
5. **Corneal endothelium** — simple squamous; pumps water out to maintain transparency.

The cornea is **transparent because it is avascular and its collagen is regularly arranged**. When it becomes oedematous or scarred, it loses transparency — the anatomical explanation for corneal blindness.

- **Choroid** — pigmented, vascular; nourishes the outer retina.
- **Ciliary body** — **ciliary muscle** (accommodation) and **ciliary processes** (aqueous humour secretion) with **zonular fibers** suspending the lens.
- **Iris** — pigment epithelium and a stroma with **sphincter pupillae** (constricts, parasympathetic) and **dilator pupillae** (dilates, sympathetic).

Aqueous humour flows from the posterior chamber through the pupil into the anterior chamber and drains through the **trabecular meshwork** into the **canal of Schlemm**. Blockage raises intraocular pressure — glaucoma.

The lens: capsule → subcapsular epithelium → lens fibers (elongated cells packed with **crystallins**, without organelles). New fibers are added at the equator throughout life, so the lens grows and its nucleus becomes denser. Loss of transparency is **cataract**.

The retina — five of its ten layers matter for exams:

Layer	Contents
Retinal pigment epithelium (RPE)	Pigmented; absorbs stray light, phagocytoses outer segments, regenerates visual pigment
Photoreceptor layer	**Rods** (rhodopsin, dim light, ~120 million) and **cones** (iodopsin, color, ~6 million)
Outer nuclear layer	Photoreceptor nuclei
Inner nuclear layer	Bipolar, horizontal, amacrine, and Müller cell nuclei
Ganglion cell layer	Ganglion cells whose axons form the optic nerve

The retina is **inverted**: light passes through the ganglion cell and inner layers before reaching the photoreceptors, which sit against the pigment epithelium. The **macula lutea** — and its center, the **fovea centralis** — contains only **cones**, has the highest acuity, and is the site of central vision. The **optic disc** is the **blind spot**, because photoreceptors are absent there. **Müller cells** are the principal glia.

Ears: The Vestibuloauditory System

External ear — auricle of **elastic cartilage** covered by skin; external acoustic meatus lined by skin with **ceruminous glands** and hairs; **tympanic membrane** with a fibrous core.

Middle ear — air-filled tympanic cavity with the **ossicles** (malleus, incus, stapes); the **auditory tube** equalises pressure; the **oval window** and **round window** transmit and relieve pressure waves.

Inner ear — **bony labyrinth** containing the membranous labyrinth:

- **Cochlea** — the **organ of Corti** on the basilar membrane, with **inner and outer hair cells** (the transducers), a **tectorial membrane** that deflects their stereocilia, supporting cells, and the **spiral ganglion**. The **stria vascularis** secretes endolymph.
- **Vestibule** — **utricle** and **saccule**, each with a **macula** containing hair cells and otoliths (gravity and linear acceleration).
- **Semicircular canals** — **cristae ampullares** with hair cells in a cupula (rotational acceleration).

Endolymph (high K^+) fills the membranous labyrinth; **perilymph** (high Na^+) fills the bony labyrinth. The ionic difference is the battery that drives transduction.

CLINICAL CORRELATION

Glaucoma — raised intraocular pressure from impaired drainage through the trabecular meshwork, damaging the optic nerve. **Cataract** — lens opacity from crystallin denaturation. **Age-related macular degeneration** — degeneration of the RPE and photoreceptors at the macula. **Retinal detachment** — separation of the neural retina from the RPE, which starves the photoreceptors. **Diabetic retinopathy** — microvascular damage with hemorrhage and neovascularization. **Otosclerosis** — ossification of the stapes footplate, producing conductive deafness. **Presbycusis** — age-related loss of hair cells and neurons. **Ménière disease** — endolymphatic hydrops with vertigo, tinnitus, and fluctuating deafness.

UNDER THE MICROSCOPE

The **cornea** is identified by five layers and a regular, avascular stroma. The **retina** by its ten layers with a pigmented layer at the outer border and ganglion cells at the inner border. The **organ of Corti** by the tunnel of Corti between inner and outer hair cells on the basilar membrane, with the tectorial membrane above.

EXAM TIP

Rods detect **dim light**, cones detect **color and detail**; the **fovea has only cones**, which is why acuity is highest there and color vision is lost first in the periphery.

PART VI — RAPID REVIEW AND REFERENCE

Chapter 24, the master tables, and the reference appendices

24 · The Last-24-Hours Rapid Review

HIGH-YIELD PEARL
This chapter is designed to be read in one sitting, twice. Read it once slowly, then again out loud. Everything in it is either a rule that generates other facts or a fact that is asked almost every year.

The 60-second organ identification algorithm

Step 1 — Is there a lumen?

- **No lumen, solid cords or follicles** → thyroid (follicles), adrenal (cords), liver (cords + central veins), parathyroid (cords), pituitary (cell sheets), lymphoid organ (nodules).
- **Yes, a lumen** → go to Step 2.

Step 2 — What is the epithelium?

- **Stratified squamous** → esophagus, oral cavity, vagina, anal canal, epidermis (no lumen).
- **Simple columnar with crypts** → stomach, intestine, gallbladder, uterus.
- **Pseudostratified ciliated** → respiratory tract, epididymis, vas deferens.
- **Transitional** → ureter, bladder, renal pelvis.
- **Simple cuboidal/columnar tubules** → kidney, salivary ducts, epididymis.

Step 3 — What surrounds it?

- **Cartilage rings** → trachea. **Cartilage plates** → bronchus. **No cartilage** → bronchiole.
- **Three muscle layers** → stomach. **Taeniae coli** → colon. **Thick muscle wall** → vas deferens, ureter.
- **Serosa** → intraperitoneal; **adventitia** → retroperitoneal.

The ten master tables

1. Epithelium by location

Epithelium	Where
Simple squamous	Alveoli, endothelium, mesothelium, Bowman capsule
Simple cuboidal	Kidney tubules, thyroid follicles, small ducts, ovary surface
Simple columnar	Stomach, intestine, gallbladder, uterine tube
Pseudostratified ciliated	Trachea, bronchi, nasal cavity, epididymis, vas deferens
Stratified squamous keratinized	Epidermis
Stratified squamous non-keratinized	Oral cavity, esophagus, vagina, cornea, anal canal
Transitional	Renal pelvis, ureter, bladder, proximal urethra
Epithelioid	Adrenal cortex, islets, liver

2. Collagen types

Type	Where	Disease
I	Bone, skin, tendon, cornea	Osteogenesis imperfecta
II	Cartilage, vitreous	Chondrodysplasia
III	Reticulin: lymphoid organs, marrow, liver	Ehlers–Danlos vascular
IV	Basement membrane	Alport syndrome

Type	Where	Disease
V	Cornea, bone, placenta	Ehlers–Danlos

3. Muscle

Feature	Skeletal	Cardiac	Smooth
Striations	Yes	Yes	**No**
Nuclei	Many, **peripheral**	1–2, **central**	1, **central**
Junctions	—	**Intercalated discs**	Gap junctions
T-tubules	Triads at A–I	Diads at Z	Caveolae
Ca^{2+} sensor	Troponin C	Troponin C	**Calmodulin**
Regeneration	Limited (satellite)	**None**	Hyperplasia

4. Cartilage

Feature	Hyaline	Elastic	Fibrocartilage
Collagen	II	II + elastic	**I + II**
Perichondrium	Yes	Yes	**No**
Where	Trachea, ribs, joints, nose	Epiglottis, pinna	Disc, symphysis, menisci
Cells	Isogenous groups	Numerous	**Rows**

5. Bone cells

Cell	Origin	Job
Osteoprogenitor	Mesenchyme	Precursor
Osteoblast	Mesenchyme	**Make osteoid**, RANKL
Osteocyte	Osteoblast	Maintain, sense load
Osteoclast	**Monocyte**	**Resorb**, ruffled border

6. Blood cells

Cell	%	Nucleus	Signature
Neutrophil	60–70	3–5 lobes	Pale granules
Eosinophil	2–4	**Bilobed**	**Orange-red** granules
Basophil	0.5–1	S-shaped, hidden	Dark blue granules
Lymphocyte	20–30	Round, dense	Scant cytoplasm
Monocyte	3–8	**Kidney-shaped**	Largest cell

7. Lymphoid organs

Organ	Follicles	Afferent lymph	Unique
Thymus	**No**	No	**Hassall corpuscles**
Lymph node	Yes	**Yes**	Cortex/paracortex/medulla
Spleen	Yes	**No**	White + red pulp
Tonsil	Yes	No	Crypts, no capsule

8. Gut segments

Segment	Diagnostic feature
Esophagus	Stratified squamous, skeletal muscle above
Stomach	**Three muscle layers**, parietal + chief cells, no goblet
Duodenum	**Brunner glands (submucosa)**
Jejunum	Tallest villi, no Brunner, no Peyer
Ileum	**Peyer's patches**
Colon	**No villi**, crypts, many goblet, taeniae
Appendix	Lymphoid tissue all round, blind tube

9. Endocrine

Gland	Pattern	Hormone
Thyroid	**Follicles with colloid**	T3, T4; C cells → calcitonin
Parathyroid	Cords of chief cells	PTH
Adrenal cortex G/F/R	Cords	Aldosterone / cortisol / androgens
Adrenal medulla	Chromaffin	Epinephrine
Pituitary anterior	Acidophils + basophils	GH, PRL / ACTH, TSH, FSH, LH
Pituitary posterior	Fibrillary, Herring bodies	Oxytocin, ADH
Islets	Clusters	Insulin, glucagon, somatostatin

10. Airway

Level	Cartilage	Glands	Muscle
Trachea	**Rings**	Yes	Trachealis
Bronchus	**Plates**	Yes	Complete ring
Bronchiole	**None**	**None**	Prominent
Respiratory bronchiole	None	None	+ alveoli in wall

One hundred facts in one hundred lines

1. Basophilic = binds hematoxylin = nucleic acid.
2. Eosinophilic = binds eosin = protein.
3. PAS stains glycogen and basement membrane.
4. Silver stains reticulin (type III collagen).
5. Light microscope resolution = 0.2 µm.
6. Frozen section = rapid diagnosis + lipid preservation.
7. Plasma membrane thickness = 7–10 nm.
8. Na^+/K^+-ATPase pumps 3 Na^+ out, 2 K^+ in.
9. Clathrin mediates receptor-mediated endocytosis.
10. Mitochondria are maternally inherited.
11. Steroid-secreting cells have **tubular** cristae.
12. Nissl bodies = rough ER + polysomes.

13. Golgi **modifies and sorts**; RER **synthesises**.
14. Lysosomes are acidic (pH ≈ 5).
15. Lipofuscin = residual body pigment.
16. Peroxisomes contain catalase.
17. Proteasomes degrade ubiquitin-tagged protein.
18. Kinesin = anterograde; dynein = retrograde.
19. Cilia = 9 + 2; basal body = 9 + 0.
20. Keratin = epithelium; vimentin = mesenchyme; desmin = muscle; GFAP = astrocyte.
21. Lamin A mutation = progeria.
22. Nucleolus makes rRNA; NORs on 13, 14, 15, 21, 22.
23. Barr body = inactivated X.
24. p53 = guardian of the genome.
25. Meiosis = four haploid cells; crossing over in pachytene.
26. Apoptosis = no inflammation.
27. Labile tissue = epidermis, gut, marrow; permanent = neurons, cardiac muscle.
28. Epithelium is avascular.
29. Basement membrane = lamina lucida, densa, reticularis.
30. Type IV collagen is in the basal lamina.
31. Desmosomes link to keratin; hemidesmosomes to laminin via integrin.
32. E-cadherin is calcium-dependent.
33. Gap junctions = connexins.
34. Microvilli core = actin.
35. Stereocilia are long microvilli (epididymis).
36. Endothelium and mesothelium are simple squamous.
37. Pseudostratified epithelium is truly one layer.
38. Transitional epithelium has umbrella cells and uroplakins.
39. Goblet cell = unicellular mucous gland.
40. Sebaceous gland = holocrine.
41. Mammary gland = apocrine.
42. Pancreas = merocrine.
43. Myoepithelial cells compress acini.
44. Collagen synthesis needs **vitamin C** and **copper**.
45. Scurvy = defective hydroxylation.
46. Marfan = fibrillin.
47. Ehlers–Danlos = collagen processing.
48. Plasma cell nucleus = clock-face, eccentric.
49. Mast cell = histamine + heparin.
50. Brown fat = multilocular + UCP1 + central nucleus.
51. Leptin reduces appetite; adiponectin increases insulin sensitivity.
52. Hyaline cartilage = type II collagen.
53. Fibrocartilage = **no perichondrium**.
54. Interstitial growth = from within lacunae.
55. Osteoclast = monocyte lineage, TRAP-positive.
56. Osteoblast = RANKL; OPG is the decoy.
57. Osteon = concentric lamellae + Haversian canal.

58. Epiphyseal zones: reserve → proliferate → hypertrophy → calcify → ossify.
59. Intramembranous ossification = flat bones.
60. PTH raises calcium; calcitonin lowers it.
61. Oligodendrocyte = CNS, many axons; Schwann = PNS, one segment.
62. Microglia are mesodermal.
63. Wallerian degeneration occurs distal to injury.
64. Regeneration ≈ 1 mm/day.
65. A band never shortens.
66. Skeletal T-tubules form triads at the A–I junction.
67. Cardiac T-tubules form diads at the Z line.
68. Smooth muscle uses calmodulin, not troponin.
69. Intercalated discs = fascia adherens + desmosome + gap junction.
70. Purkinje fibers conduct; they are not neurons.
71. Elastic artery = elastic lamellae in media.
72. Muscular artery = prominent internal elastic lamina.
73. Sinusoidal capillaries = liver, spleen, marrow.
74. Blood–brain barrier = continuous capillaries + tight junctions.
75. Lymphatic capillaries are blind-ended.
76. Erythrocyte lifespan = 120 days.
77. Neutrophil = 3–5 lobes, acute inflammation.
78. Eosinophil = parasites and allergy.
79. Basophil = histamine, IgE.
80. Monocyte = largest leukocyte.
81. Platelets = 2–3 µm, from megakaryocytes.
82. EPO from kidney; TPO from liver.
83. M:E ratio = 2:1 to 4:1.
84. CD4 ↔ MHC II; CD8 ↔ MHC I.
85. Thymus has no follicles; spleen has no afferent lymphatics.
86. Hassall corpuscles = thymus.
87. PALS = T-cell zone of spleen.
88. Germinal center = B-cell proliferation.
89. Enteric plexuses: Meissner (submucosa), Auerbach (myenteric).
90. Parietal cell = HCl + intrinsic factor.
91. Chief cell = pepsinogen.
92. Paneth cell = lysozyme and defensins.
93. Brunner glands = duodenum.
94. Peyer's patches = ileum.
95. Portal triad = artery, portal vein, bile duct.
96. Bile flows toward the triad; blood flows away.
97. Type I pneumocyte = gas exchange; type II = surfactant.
98. Club (formerly Clara) cells = bronchioles; they secrete surfactant-like material and detoxify inhaled compounds.
99. Podocytes form the filtration slit diaphragm (nephrin).
100. Macula densa = distal tubule; JG cells = afferent arteriole.

Twenty-five distinctions that decide marks

Do not confuse	With	The difference
Basophilic	Eosinophilic	DNA/RNA versus protein
Mucous	Mucosa	Secretion versus lining layer
Endothelium	Mesothelium	Vessels versus body cavities
Pseudostratified	Stratified	One layer versus many
Hyaline cartilage	Fibrocartilage	Type II + perichondrium versus I + II + none
Osteoblast	Osteoclast	Builds versus resorbs; mesenchymal versus monocyte
Astrocyte	Oligodendrocyte	Support/BBB versus myelination
Schwann cell	Oligodendrocyte	One segment versus many axons
Neutrophil	Eosinophil	3–5 lobes versus bilobed
Monocyte	Lymphocyte	Kidney-shaped versus round
Proximal tubule	Distal tubule	Brush border, eosinophilic versus clean, pale
Type I pneumocyte	Type II	Thin for gas versus cuboidal for surfactant
Parietal cell	Chief cell	HCl versus pepsinogen
Brunner gland	Peyer's patch	Duodenum submucosa versus ileum lymphoid
Zona glomerulosa	Zona fasciculata	Aldosterone versus cortisol
Adrenal cortex	Adrenal medulla	Steroid versus catecholamine
Anterior pituitary	Posterior pituitary	Glandular versus neural
Sertoli cell	Leydig cell	Barrier/nurture versus testosterone
Granulosa cell	Theca cell	Estrogen (aromatase) versus androgen
Proliferative	Secretory endometrium	Straight glands versus tortuous secreting glands
Follicular	Luteal phase	Estrogen-dominant versus progesterone-dominant
Rod	Cone	Dim light versus color and acuity
Endolymph	Perilymph	High K^+ versus high Na^+
Apoptosis	Necrosis	No inflammation versus inflammation
Labile	Permanent tissue	Continuous turnover versus no division

Confessions of the examiner — the eight most common errors

1. Saying the osteoclast is mesenchymal. It is **hematopoietic**.
2. Saying the A band shortens. It **does not**.
3. Saying the thymus has follicles. It **does not**.
4. Saying the spleen has afferent lymphatics. It **does not**.
5. Saying the adrenal medulla makes steroids. It makes **catecholamines**.
6. Saying FSH acts on Leydig cells. It acts on **Sertoli** cells; LH acts on Leydig.
7. Saying rods are at the fovea. The fovea has **cones only**.
8. Saying the stomach has goblet cells. It has **surface mucous cells**, not goblet cells.

Appendix A · Stains and What They Prove

Stain	Target	Color	Classic exam use
Hematoxylin	Nucleic acids	Blue-purple	Nuclei, rough ER
Eosin	Proteins	Pink-red	Cytoplasm, collagen, RBC
PAS	Glycogen, mucin, glycoproteins, basement membrane	Magenta	Goblet cells, liver glycogen, fungal walls
Masson trichrome	Collagen versus muscle	Blue/green versus red	Liver fibrosis, kidney biopsy
Silver (reticulin)	Type III collagen, nerve fibers	Black	Lymph node, spleen, marrow
Orcein / resorcin–fuchsin	Elastin	Dark brown-black	Aorta, lung, skin
Sudan black / oil red O	Lipid	Black/red	Steatosis, adipocytes (frozen)
Osmium tetroxide	Lipid, myelin	Black	Fat, myelin
Cresyl violet (Nissl)	Rough ER	Violet	Neurons
Feulgen	DNA	Magenta	DNA quantification
Prussian blue	Iron (hemosiderin)	Blue	Iron overload, old hemorrhage
Congo red	Amyloid	Apple-green birefringence	Amyloidosis
Gram	Bacterial wall	Purple (positive) / pink (negative)	Infection
Ziehl–Neelsen	Acid-fast bacilli	Red	Tuberculosis
Immunohistochemistry	Specific antigen	Brown (DAB) or fluorescent	Tumor typing, cell identification

Appendix B · The Vocabulary Examiners Expect

Term	Meaning
Parenchyma	The functional cells of an organ
Stroma	The supportive connective tissue framework
Capsule	Dense connective tissue enclosing an organ
Hilum	The entry/exit point for vessels, ducts, and nerves
Cortex / medulla	Outer versus inner region
Trabecula	A partition or supporting beam of connective tissue
Sinusoid	A wide, discontinuous capillary
Acinus	A berry-like cluster of secretory cells
Tubuloacinar	A gland with both tubular and acinar secretory units
Metaplasia	Change from one mature cell type to another
Dysplasia	Disordered growth with atypical cells
Neoplasia	New, autonomous growth
Hypertrophy	Increase in cell size
Hyperplasia	Increase in cell number
Atrophy	Decrease in size or number
Apoptosis	Programmed cell death
Necrosis	Pathological cell death with inflammation
Autolysis	Self-digestion after death
Fixation	Chemical preservation of tissue
Artefact	A structure produced by processing, not present in life

PART VII — ASSESS YOUR KNOWLEDGE

One hundred examination questions with explained answers

100 Examination Questions

EXAM TIP
Answer all 100 before turning to the key. Mark every guess with a dot; guesses, not errors, are what you must revise. A score of 80 or more means you are exam-ready for histology. Below 65 means go back to the tables in Part VI.

Cells and techniques (1–12)

1. A tissue section is treated with a basic dye. Which structure will stain most intensely?

A. Collagen bundles
B. Nuclei and rough endoplasmic reticulum
C. Erythrocytes
D. Elastic fibers

2. Which stain would you choose to demonstrate a basement membrane?

A. Eosin
B. Orcein
C. PAS
D. Sudan black

3. A surgeon requires an immediate diagnosis during an operation. The correct specimen preparation is:

A. Routine paraffin section
B. Frozen section
C. Resin embedding
D. Whole-mount preparation

4. The resolving power of the light microscope is approximately:

A. 0.2 nm
B. 2 µm
C. 0.2 µm
D. 20 µm

5. Which technique localizes a specific protein in a tissue section using an antibody?

A. Autoradiography
B. In situ hybridization
C. Immunohistochemistry
D. Polarizing microscopy

6. A cell specialized for steroid hormone synthesis would show abundant:

A. Rough ER and Golgi
B. Smooth ER and lipid droplets
C. Lysosomes and residual bodies
D. Free ribosomes and proteasomes

7. Which organelle contains catalase and performs β-oxidation of very long-chain fatty acids?

A. Lysosome
B. Peroxisome
C. Proteasome
D. Mitochondrion

8. Retrograde axonal transport of viruses toward the neuronal soma is mediated by:

A. Kinesin
B. Dynein
C. Myosin

D. Dynein and kinesin equally

9. Which intermediate filament is characteristic of astrocytes?

A. Desmin
B. Vimentin
C. GFAP
D. Neurofilament

10. A cell with a 9 + 0 arrangement of microtubules is:

A. A motile cilium
B. A basal body
C. A sperm flagellum
D. A microvillus

11. A mutation in lamin A produces:

A. Progeria
B. Duchenne dystrophy
C. Marfan syndrome
D. Alport syndrome

12. The nucleolar organizer regions are located on which chromosomes?

A. 1, 2, 3, 4, 5
B. 13, 14, 15, 21, 22
C. 16, 17, 18, 19, 20
D. X and Y

Epithelium, connective tissue, cartilage and bone (13–32)

13. Which junctional complex is principally responsible for sealing the intercellular space?

A. Desmosome
B. Gap junction
C. Tight junction
D. Hemidesmosome

14. Pemphigus vulgaris results from autoantibodies against:

A. Desmoglein
B. Integrin
C. Connexin
D. Laminin

15. Which epithelium lines the normal ureter?

A. Stratified squamous
B. Transitional
C. Simple columnar
D. Pseudostratified ciliated

16. Stereocilia are found in the:

A. Trachea
B. Epididymis
C. Fallopian tube
D. Intestinal villi

17. The basement membrane's lamina reticularis is composed mainly of:

A. Type I collagen
B. Type II collagen
C. Type III collagen
D. Type IV collagen

18. Which collagen type is defective in osteogenesis imperfecta?

A. Type I
B. Type II
C. Type III
D. Type IV

19. Hydroxylation of proline and lysine during collagen synthesis requires:

A. Vitamin A
B. Vitamin C
C. Vitamin D
D. Copper

20. A cell with an eccentric nucleus showing a clock-face chromatin pattern and a pale perinuclear halo is a:

A. Mast cell
B. Plasma cell
C. Macrophage
D. Eosinophil

21. The main function of UCP1 in brown adipose tissue is:

A. ATP synthesis
B. Uncoupling of oxidative phosphorylation to produce heat
C. Fatty acid storage
D. Leptin secretion

22. Adiponectin, secreted by adipocytes, acts to:

A. Reduce appetite
B. Increase insulin sensitivity
C. Increase blood pressure
D. Stimulate lipolysis in brown fat

23. Which cartilage type lacks a perichondrium?

A. Hyaline
B. Elastic
C. Fibrocartilage
D. Articular hyaline

24. The intervertebral disc's annulus fibrosus is composed of:

A. Hyaline cartilage
B. Elastic cartilage
C. Fibrocartilage
D. Dense regular connective tissue

25. Which cell is responsible for bone resorption?

A. Osteoblast
B. Osteocyte
C. Osteoclast
D. Osteoprogenitor cell

26. Osteoclasts are derived from:

A. Mesenchymal cells
B. Monocyte–macrophage lineage
C. Osteoblasts
D. Neural crest

27. In endochondral ossification, the zone in which chondrocytes enlarge and the matrix calcifies is the:

A. Reserve zone
B. Proliferative zone
C. Hypertrophic zone
D. Ossification zone

28. Which hormone lowers serum calcium?

A. PTH
B. Calcitriol
C. Calcitonin
D. Growth hormone

29. The primary ossification center of a long bone appears in the:

A. Epiphysis
B. Diaphysis
C. Metaphysis
D. Periosteum

30. A multinucleated cell lying in a scalloped resorption pit is:

A. An osteoblast
B. An osteocyte
C. An osteoclast
D. A macrophage

31. Dense regular connective tissue is characterized by:

A. Randomly oriented collagen bundles
B. Parallel collagen bundles
C. Abundant elastic fibers
D. A reticular network

32. Which tissue is the site of hematopoiesis in the adult?

A. Liver
B. Spleen
C. Red bone marrow
D. Yellow bone marrow

Nerve and muscle (33–44)

33. Nissl bodies are composed of:

A. Smooth ER and Golgi
B. Rough ER and polysomes
C. Neurofilaments
D. Synaptic vesicles

34. Which glial cell myelinates several axons in the central nervous system?

A. Astrocyte
B. Oligodendrocyte
C. Schwann cell
D. Microglia

35. Microglia are derived from:

A. Neural crest
B. Neural tube
C. Mesoderm (monocyte lineage)
D. Endoderm

36. The blood–brain barrier is formed principally by:

A. Astrocyte end-feet alone
B. Tight junctions between capillary endothelial cells

C. Ependymal cells
D. Pericytes alone

37. In a peripheral nerve, the connective tissue sheath surrounding a single axon is the:

A. Epineurium
B. Perineurium
C. Endoneurium
D. Mesoneurium

38. During muscle contraction, which band does not change in length?

A. I band
B. H band
C. A band
D. None of these

39. Calcium binds which molecule in skeletal muscle to initiate contraction?

A. Calmodulin
B. Troponin C
C. Myosin light chain
D. Actinin

40. Intercalated discs contain all of the following except:

A. Fascia adherens
B. Desmosomes
C. Gap junctions
D. Tight junctions

41. Which feature distinguishes cardiac from skeletal muscle?

A. Striations
B. Central nucleus
C. Myofibrils
D. Mitochondria

42. Smooth muscle contraction differs from striated muscle because it uses:

A. Troponin C
B. Calmodulin and myosin light-chain kinase
C. Tropomyosin only
D. No calcium

43. T-tubules and terminal cisternae form triads at the:

A. Z line
B. A–I junction
C. M line
D. H band

44. A muscle fiber that is rich in mitochondria and myoglobin and resistant to fatigue is:

A. Type I
B. Type IIa
C. Type IIx
D. All equally

Blood, hemopoiesis and immunity (45–60)

45. The most abundant leukocyte in normal peripheral blood is the:

A. Lymphocyte
B. Neutrophil
C. Monocyte

D. Eosinophil

46. A bilobed nucleus with bright orange-red granules identifies the:

A. Basophil
B. Neutrophil
C. Eosinophil
D. Monocyte

47. The largest leukocyte, with a kidney-shaped nucleus, is the:

A. Lymphocyte
B. Monocyte
C. Neutrophil
D. Basophil

48. Platelets are derived from:

A. Erythroblasts
B. Megakaryocytes
C. Monocytes
D. Lymphocytes

49. Erythropoietin is produced mainly by the:

A. Liver
B. Kidney
C. Spleen
D. Bone marrow

50. The normal myeloid-to-erythroid ratio in bone marrow is:

A. 1:1
B. 2:1 to 4:1
C. 1:2
D. 10:1

51. Which immunoglobulin is the main secretory antibody at mucosal surfaces?

A. IgG
B. IgM
C. IgA
D. IgE

52. CD4-positive T cells recognize antigen presented by:

A. MHC class I
B. MHC class II
C. MHC class III
D. Antibody alone

53. Hassall corpuscles are found in the:

A. Spleen
B. Lymph node
C. Thymus
D. Tonsil

54. Which organ lacks afferent lymphatic vessels?

A. Lymph node
B. Thymus and spleen
C. Tonsil
D. Appendix

55. The periarteriolar lymphoid sheath of the spleen contains mainly:

A. B cells
B. T cells
C. Macrophages
D. Plasma cells

56. A germinal center is a site of:

A. T-cell maturation
B. B-cell proliferation
C. Erythropoiesis
D. Platelet production

57. The red pulp of the spleen functions mainly to:

A. Produce T cells
B. Filter blood and remove aged erythrocytes
C. Store lymphocytes
D. Secrete antibodies

58. The lifespan of a normal erythrocyte is approximately:

A. 7 days
B. 30 days
C. 120 days
D. 1 year

59. Neonatal hemopoiesis occurs mainly in the:

A. Yolk sac
B. Liver and spleen
C. Bone marrow
D. Thymus

60. Which cell is the principal antigen-presenting cell of the epidermis?

A. Keratinocyte
B. Melanocyte
C. Langerhans cell
D. Merkel cell

Digestive system (61–72)

61. Which layer of the gut wall contains the myenteric (Auerbach) plexus?

A. Mucosa
B. Submucosa
C. Between the circular and longitudinal muscle layers
D. Serosa

62. Parietal cells of the gastric glands secrete:

A. Pepsinogen
B. HCl and intrinsic factor
C. Mucus
D. Gastrin

63. Brunner glands are found in the:

A. Duodenum
B. Jejunum
C. Ileum
D. Colon

64. Peyer's patches are characteristic of the:

A. Duodenum
B. Jejunum

C. Ileum
D. Stomach

65. Paneth cells are located at the base of intestinal crypts and secrete:

A. Mucus
B. Lysozyme and defensins
C. Gastrin
D. Intrinsic factor

66. Which feature is absent in the colon but present in the small intestine?

A. Goblet cells
B. Crypts
C. Villi
D. Enteroendocrine cells

67. The portal triad contains all of the following except:

A. Hepatic artery
B. Portal vein
C. Bile duct
D. Central vein

68. Kupffer cells are:

A. Hepatocytes
B. Sinusoidal macrophages
C. Fat-storing cells
D. Endothelial cells

69. Bile flows in which direction relative to the portal triad?

A. Away from the triad
B. Toward the triad
C. Parallel to blood flow
D. Into the central vein

70. The exocrine pancreas is identified by:

A. Islets and striated ducts
B. Basophilic acini and zymogen granules
C. Mucous acini only
D. Follicles of colloid

71. Which cells secrete insulin?

A. Alpha cells
B. Beta cells
C. Delta cells
D. PP cells

72. The esophagus is lined by:

A. Simple columnar epithelium
B. Stratified squamous non-keratinized epithelium
C. Transitional epithelium
D. Pseudostratified ciliated epithelium

Respiratory system and skin (73–82)

73. Type II pneumocytes are characterized by:

A. A squamous shape and no organelles
B. Lamellar bodies and surfactant production
C. Cilia and goblet granules

D. Phagocytic vacuoles

74. Clara (club) cells are found in the:

A. Alveoli
B. Trachea
C. Bronchioles
D. Pleura

75. Cartilage in the bronchial tree changes in which sequence?

A. None → plates → rings
B. Rings → plates → none
C. Plates → rings → none
D. Rings → none → plates

76. The blood–air barrier does not include:

A. Surfactant
B. Type I pneumocyte
C. Basement membrane
D. Type II pneumocyte

77. The layer of the epidermis present only in thick skin is the:

A. Basale
B. Spinosum
C. Granulosum
D. Lucidum

78. Melanocytes are derived from:

A. Ectoderm
B. Neural crest
C. Mesoderm
D. Endoderm

79. Sebaceous glands secrete by which mechanism?

A. Merocrine
B. Apocrine
C. Holocrine
D. Endocrine

80. Pacinian corpuscles are located in the:

A. Basal epidermis
B. Dermal papillae
C. Deep dermis and hypodermis
D. Stratum corneum

81. Which cell type is most numerous in the epidermis?

A. Melanocyte
B. Langerhans cell
C. Keratinocyte
D. Merkel cell

82. Which receptor is responsible for fine tactile discrimination in the fingertips?

A. Pacinian corpuscle
B. Meissner corpuscle
C. Ruffini ending
D. Free nerve ending

Urinary and endocrine (83–92)

83. The filtration slit diaphragm between podocyte foot processes contains:

A. Nephrin
B. Desmoglein
C. Connexin
D. Laminin only

84. The macula densa is located in the:

A. Proximal tubule
B. Thin limb
C. Distal tubule
D. Collecting duct

85. Renin is secreted by:

A. Macula densa cells
B. Juxtaglomerular cells
C. Podocytes
D. Mesangial cells

86. Which tubule segment is most eosinophilic because of abundant mitochondria?

A. Proximal convoluted tubule
B. Thin descending limb
C. Distal convoluted tubule
D. Collecting duct

87. The epithelium of the ureter is:

A. Simple columnar
B. Transitional
C. Stratified squamous
D. Pseudostratified

88. Thyroid follicles are lined by:

A. Simple squamous epithelium
B. Simple cuboidal epithelium
C. Stratified cuboidal epithelium
D. Pseudostratified epithelium

89. Calcitonin is secreted by:

A. Follicular cells
B. Parafollicular C cells
C. Chief cells
D. Oxyphil cells

90. The zona fasciculata of the adrenal cortex secretes:

A. Aldosterone
B. Cortisol
C. Androgens
D. Epinephrine

91. Which is derived from neural crest?

A. Adrenal cortex
B. Adrenal medulla
C. Thyroid follicle
D. Parathyroid chief cells

92. Herring bodies are found in the:

A. Anterior pituitary
B. Posterior pituitary

C. Pineal gland
D. Thyroid

Reproductive system, eye and ear (93–100)

93. The blood–testis barrier is formed by tight junctions between:

A. Spermatogonia
B. Sertoli cells
C. Leydig cells
D. Peritubular myoid cells

94. LH acts primarily on:

A. Sertoli cells
B. Leydig cells
C. Spermatogonia
D. Epididymal cells

95. The usual site of fertilization is the:

A. Uterine cavity
B. Ampulla of the uterine tube
C. Isthmus
D. Cervix

96. Straight, narrow endometrial glands with numerous mitoses indicate which phase?

A. Menstrual
B. Proliferative
C. Secretory
D. Gestational

97. The corpus luteum secretes mainly:

A. Estrogen
B. Progesterone
C. hCG
D. FSH

98. The fovea centralis contains:

A. Rods only
B. Cones only
C. Both equally
D. Neither

99. Aqueous humor drains into the:

A. Vitreous body
B. Canal of Schlemm
C. Ciliary body
D. Choroid

100. The sensory cells of the organ of Corti are:

A. Hair cells
B. Supporting cells
C. Spiral ganglion cells
D. Tectorial cells

Answer Key with Explanations

1. B — Basic dyes such as hematoxylin bind acidic structures, and nucleic acids (nucleus, rough ER) are the most acidic components of the cell.

2. C — PAS stains the glycoprotein-rich basement membrane magenta; eosin stains it only weakly and orcein stains elastin.

3. B — A frozen section gives a diagnosis within minutes, at the cost of some morphological distortion.

4. C — The light microscope resolves about 0.2 µm; electron microscopy is required for anything smaller.

5. C — Immunohistochemistry uses a specific antibody, visualized by an enzyme reaction or a fluorophore.

6. B — Steroid synthesis requires smooth ER and lipid (cholesterol) stores; such cells appear pale and foamy.

7. B — Peroxisomes contain catalase and carry out β-oxidation of very long-chain fatty acids.

8. B — Dynein moves cargo toward the minus end, i.e. toward the cell body (retrograde).

9. C — GFAP is the characteristic intermediate filament of astrocytes.

10. B — The basal body is 9 + 0; the motile cilium and flagellum are 9 + 2.

11. A — Lamin A mutations destabilize the nuclear envelope and cause Hutchinson–Gilford progeria.

12. B — The acrocentric chromosomes 13, 14, 15, 21, and 22 carry the nucleolar organizer regions.

13. C — The tight junction (zonula occludens) seals the intercellular space and creates polarity.

14. A — Anti-desmoglein antibodies disrupt desmosomes and cause intraepithelial blistering.

15. B — The ureter is lined by transitional epithelium, which allows distension.

16. B — Long, stiff microvilli (stereocilia) are found in the epididymis and inner ear.

17. C — The lamina reticularis contains type III (reticular) collagen, produced by fibroblasts.

18. A — Osteogenesis imperfecta results from defective type I collagen.

19. B — Vitamin C is required for prolyl and lysyl hydroxylase; deficiency causes scurvy.

20. B — The plasma cell has an eccentric clock-face nucleus and a pale perinuclear Golgi zone.

21. B — UCP1 uncouples the proton gradient so that energy is released as heat.

22. B — Adiponectin increases insulin sensitivity; it is reduced in obesity.

23. C — Fibrocartilage has no perichondrium, which contributes to its poor healing.

24. C — The annulus fibrosus is fibrocartilage, with type I and type II collagen.

25. C — The osteoclast resorbs bone using a ruffled border and acid secretion.

26. B — Osteoclasts arise from the monocyte–macrophage lineage, not from mesenchyme.

27. C — In the hypertrophic zone chondrocytes enlarge and the matrix begins to calcify.

28. C — Calcitonin from thyroid C cells inhibits osteoclasts and lowers serum calcium.

29. B — The primary center appears in the diaphysis; secondary centers appear in the epiphyses.

30. C — A multinucleated cell in a Howship lacuna is an osteoclast.

31. B — Dense regular connective tissue has parallel collagen bundles, as in tendon.

32. C — Red bone marrow is the adult hematopoietic site.

33. B — Nissl bodies are stacked rough ER and polysomes.

34. B — One oligodendrocyte myelinates segments of several CNS axons.

35. C — Microglia are mesodermal, derived from monocytes — the exception among glia.

36. B — Tight junctions between endothelial cells are the anatomical basis; astrocyte feet and pericytes reinforce it.

37. C — Endoneurium surrounds individual nerve fibers; perineurium surrounds fascicles.

38. C — The A band is the length of the thick filaments and does not change during contraction.

39. B — Calcium binds troponin C, which moves tropomyosin and exposes the myosin-binding site.

40. D — Tight junctions are not a component of intercalated discs.

41. B — Cardiac muscle has one or two central nuclei; skeletal muscle nuclei are peripheral and multiple.

42. B — Smooth muscle uses calmodulin and myosin light-chain kinase; it has no troponin.

43. B — In mammalian skeletal muscle, triads lie at the A–I junction; in cardiac muscle, diads lie at the Z line.

44. A — Type I fibers are slow, oxidative, and fatigue-resistant.

45. B — Neutrophils constitute 60–70% of circulating leukocytes.

46. C — The eosinophil has a bilobed nucleus and bright orange-red granules.

47. B — The monocyte is the largest leukocyte and has a kidney-shaped nucleus.

48. B — Platelets are fragments of megakaryocyte cytoplasm.

49. B — Peritubular interstitial cells of the kidney produce erythropoietin.

50. B — The normal M:E ratio is 2:1 to 4:1.

51. C — IgA is the main secretory immunoglobulin; it exists as a dimer with a secretory component.

52. B — $CD4^+$ helper T cells recognize antigen on MHC class II.

53. C — Hassall corpuscles are concentric epithelial whorls unique to the thymic medulla.

54. B — Neither the thymus nor the spleen has afferent lymphatics.

55. B — The PALS is the T-cell zone of the spleen; follicles are the B-cell zone.

56. B — Germinal centers are sites of B-cell proliferation and affinity maturation.

57. B — The red pulp filters blood and removes aged or damaged erythrocytes.

58. C — Erythrocytes circulate for about 120 days.

59. B — In the second trimester, the liver and spleen are the main hemopoietic organs.

60. C — Langerhans cells are the epidermal antigen-presenting cells; they originate in bone marrow.

61. C — Auerbach's plexus lies between the circular and longitudinal layers of the muscularis externa.

62. B — Parietal cells secrete hydrochloric acid and intrinsic factor.

63. A — Brunner glands lie in the submucosa of the duodenum.

64. C — Peyer's patches are characteristic of the ileum.

65. B — Paneth cells secrete lysozyme and defensins, contributing to innate immunity.

66. C — Villi are present in the small intestine and absent in the colon.

67. D — The central vein is at the center of the lobule, not in the portal triad.

68. B — Kupffer cells are the resident macrophages of the hepatic sinusoids.

69. B — Bile flows toward the portal triad while blood flows away from it toward the central vein.

70. B — The exocrine pancreas has basophilic acini with eosinophilic zymogen granules.

71. B — Beta cells secrete insulin and are about twice as abundant as alpha cells.

72. B — The esophagus is lined by stratified squamous non-keratinized epithelium.

73. B — Type II pneumocytes contain lamellar bodies and secrete surfactant.

74. C — Clara (club) cells line bronchioles.

75. B — Cartilage forms rings in the trachea, plates in bronchi, and is absent in bronchioles.

76. D — Type II pneumocytes do not form part of the thin blood–air barrier.

77. D — The stratum lucidum is present only in thick skin of palms and soles.

78. B — Melanocytes derive from the neural crest.

79. C — Sebaceous glands are holocrine; the whole cell disintegrates into the secretion.

80. C — Pacinian corpuscles lie deep in the dermis and hypodermis and sense vibration.

81. C — Keratinocytes make up about 90% of epidermal cells.

82. B — Meissner corpuscles in the dermal papillae mediate fine tactile discrimination.

83. A — Nephrin is the key protein of the slit diaphragm.

84. C — The macula densa is the thickened plaque of the distal straight tubule (thick ascending limb) where it touches the vascular pole.

85. B — Juxtaglomerular cells of the afferent arteriole secrete renin.

86. A — The proximal tubule is intensely eosinophilic because of abundant mitochondria and brush border.

87. B — The ureter is lined by transitional epithelium.

88. B — Thyroid follicles are lined by a simple epithelium whose cells range from squamous (inactive) to low columnar (active), classically described as cuboidal.

89. B — Parafollicular C cells secrete calcitonin.

90. B — The zona fasciculata secretes cortisol under ACTH control.

91. B — The adrenal medulla is neural crest derived; the cortex is mesodermal.

92. B — Herring bodies are neurosecretory dilatations in the posterior pituitary.

93. B — Tight junctions between adjacent Sertoli cells form the blood–testis barrier.

94. B — LH stimulates Leydig cells; FSH acts on Sertoli cells.

95. B — Fertilization normally occurs in the ampulla of the uterine tube.

96. B — Straight, narrow glands with mitoses are typical of the proliferative phase.

97. B — The corpus luteum secretes mainly progesterone.

98. B — The fovea contains cones only, giving maximal acuity and color vision.

99. B — Aqueous humor drains through the trabecular meshwork into the canal of Schlemm.

100. A — The inner and outer hair cells of the organ of Corti are the transducing receptors.

Scoring and What to Do Next

Score	Interpretation	Action
90–100	Excellent	Skim the rapid review once daily; do slide identification only
80–89	Exam-ready	Reread the twenty-five distinctions table; revisit missed chapters
65–79	Adequate	Redo the master tables in Part VI, then retake the questions
50–64	At risk	Read each chapter's "core idea" and tables, not the prose
Below 50	Start again from Part II	Build the six core tables from memory before anything else

HIGH-YIELD PEARL

The final revision loop is not reading — it is **recall**. Close the book and write the ten master tables from memory. Whatever you cannot reproduce is exactly what you must study. Students who do this for three days typically gain ten to fifteen marks.